Title Page

SBT-DF204 Computer Forensics Case Studies

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Course Name: Investigating DJI Mavic Air Flight Evidence

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Introduction

This investigation examined digital evidence linked to a DJI Mavic Air using three main evidence sources: an Android logical acquisition containing DJI GO 4 application data, a raw internal microSD image, and an Expert Witness Format (EWF) image of microSD media. The examination was carried out in Kali Linux using verified working copies so that the original evidence files remained unchanged.

The Android evidence was reviewed for the main DJI GO 4 flight record, cached flight video, detailed DAT flight record, and cached picture artefacts. The raw microSD image was then examined through a read-only forensic workflow to identify recorded videos, thumbnails, timestamps, hashes, filesystem information, and the volume identifier. The EWF image was independently verified and examined so its filesystem, media files, timestamps, hashes, and volume information could be compared with the raw microSD evidence.

Throughout the investigation, SHA-256 values, file metadata, source paths, filesystem information, and controlled media previews were preserved before conclusions were drawn. Particular care was taken with timestamps because some ZIP and FAT-derived values did not contain a reliable timezone and could not safely be treated as UTC without supporting evidence.

Objectives

- Preserve the supplied Android, raw microSD, and EWF evidence without altering the original files.
- Create verified working copies and confirm their integrity using cryptographic hashes and byte-for-byte checks.
- Locate the main DJI GO 4 artefacts, including the flight record, cached video, detailed DAT records, and cached images.
- Record file type, size, timestamps, metadata, and SHA-256 values before opening or previewing relevant media.
- Examine the raw internal microSD through a read-only workflow and document its FAT32 structure, videos, thumbnails, and volume identifier.
- Verify and examine the EWF image with the approved EWF and Sleuth Kit tools.
- Compare Android, raw microSD, and EWF findings while separating direct observations from interpretation.
- Build the most defensible UTC timeline possible and clearly mark timestamps whose timezone could not be established.
- State the evidential limitations and avoid assigning drone ownership or operator identity where the evidence does not support it.

Laboratory / Tools Used

The examination was completed in a Kali Linux virtual machine hosted in VMware Workstation. The tools below were used only for the functions supported by the evidence examined in this case.

Evidence handling: ls, find, tree, file, stat, cp --preserve=all, cmp, sha256sum, and md5sum were used to inventory evidence, create working copies, and verify integrity.

Archive examination: unzip, zipinfo, and unzip -Z -v were used to extract the Android working copy and review source ZIP metadata.

Binary/text inspection: xxd and strings were used to view binary content and recover readable identifiers such as Mavic Air strings and the microSD volume GUID.

Media metadata: MediaInfo and ExifTool were used to record video and image properties before controlled preview.

Media preview: VLC and Thunar were used only after metadata and hashes had been preserved.

Raw microSD access: losetup, mount, findmnt, fdisk, and read-only options were used to expose and document the raw FAT32 image without writing to it.

EWF examination: ewfinfo, ewfverify, and ewfmount were used to identify, verify, and expose DF048.E01.

Filesystem forensics: fsstat, fls, istat, and icat were used to examine the FAT32 EWF data, file metadata, inode references, and file hashes directly.

Lab Folder Structure and Evidence Preparation

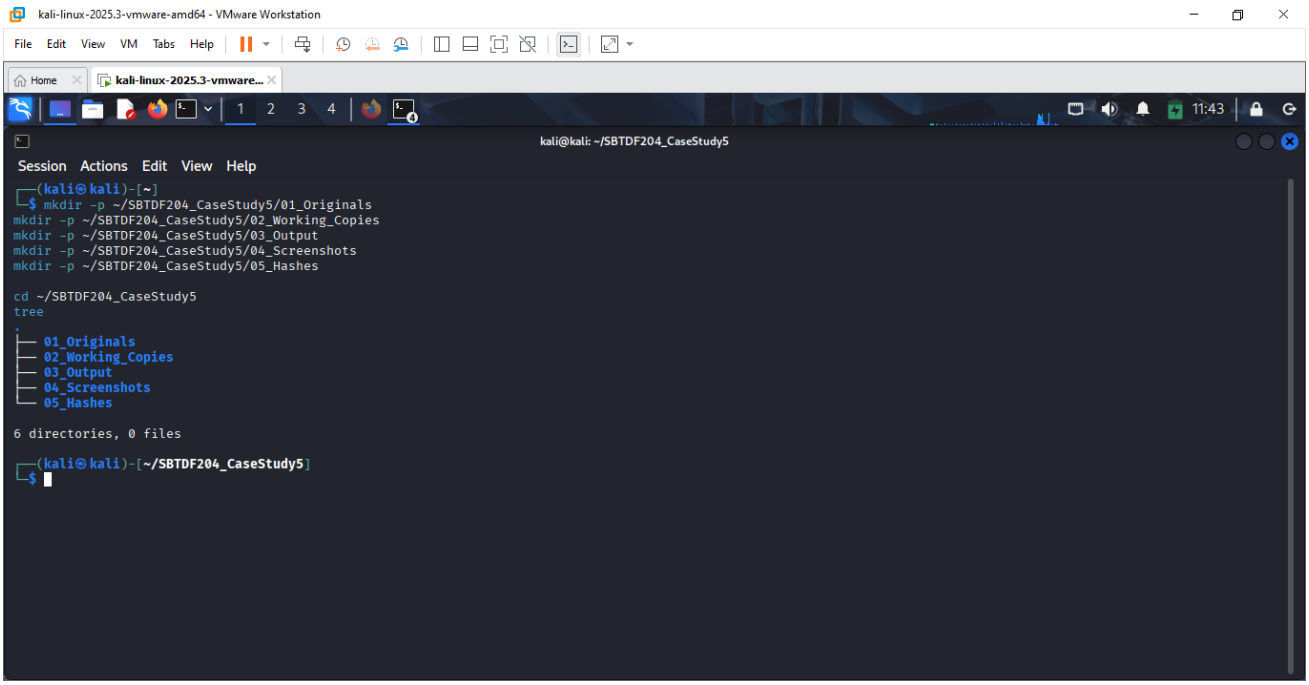


Fig 1.0 Forensic Investigation Directory Structure.

Explanation

The case folders separate originals, working copies, outputs, screenshots, and hash records. This structure kept the examination organised and reduced the risk of accidentally analysing the preserved originals.

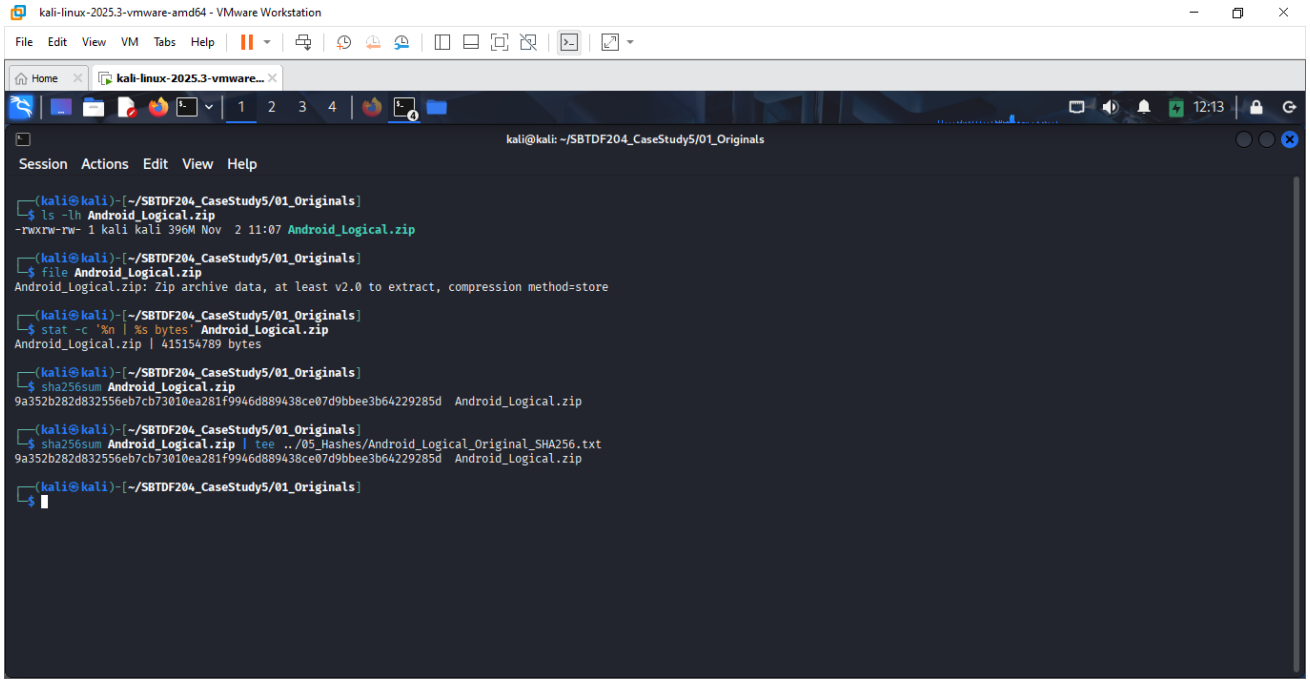


Fig 1.1 Android Logical Original File Integrity Verification.

Explanation

The original Android_Logical.zip was identified and its exact size and SHA-256 value were recorded before

analysis. This created the starting integrity record for the Android evidence.

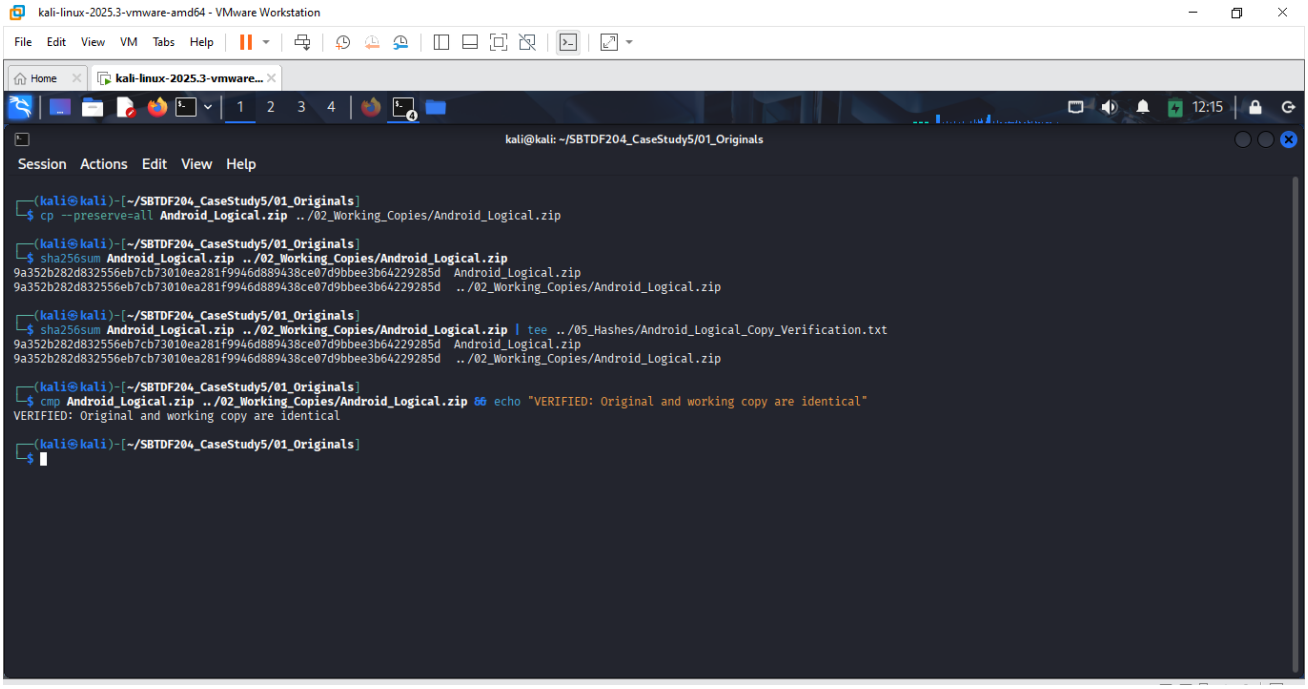


Fig 1.2 Android Logical Working Copy Integrity Verification.

Explanation

The original and working Android archives produced the same SHA-256 value and passed the byte-for-byte comparison. This confirmed that the working copy accurately reproduced the preserved source.

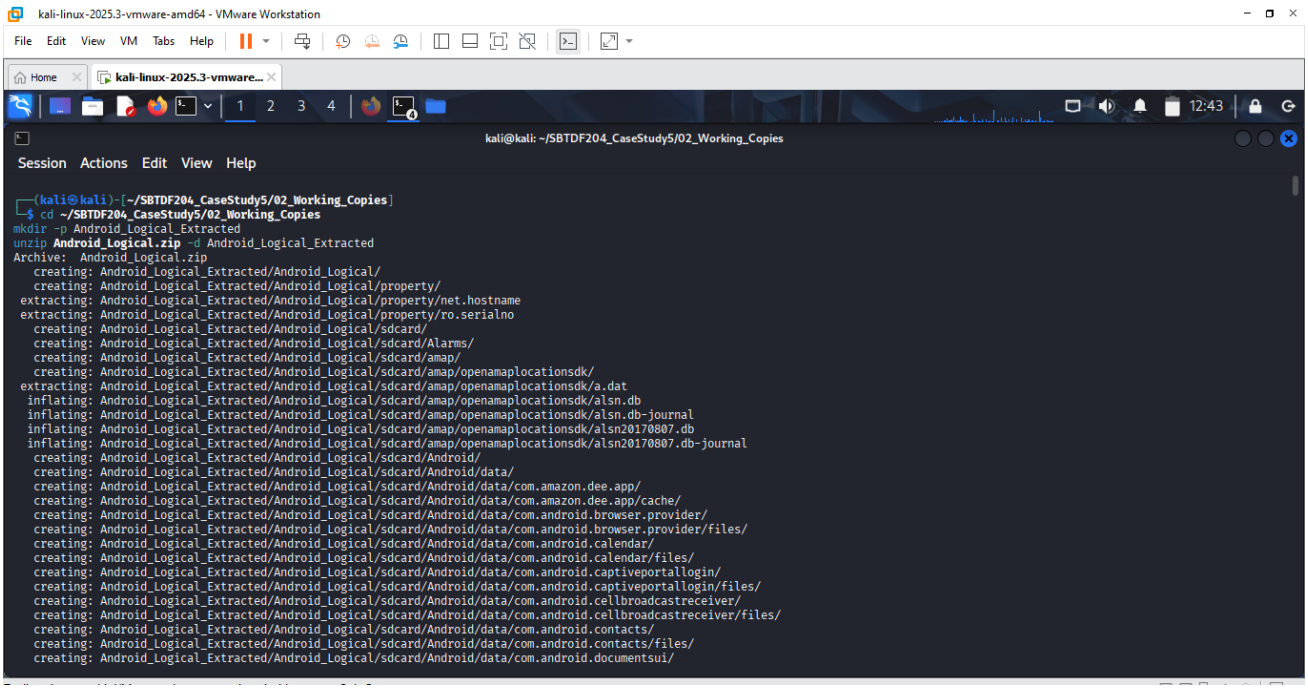


Fig 1.3 Android Logical Working Copy Extraction.

Explanation

The verified Android working archive was extracted into the case workspace rather than examining the original directly. The terminal output records the beginning of that controlled extraction.

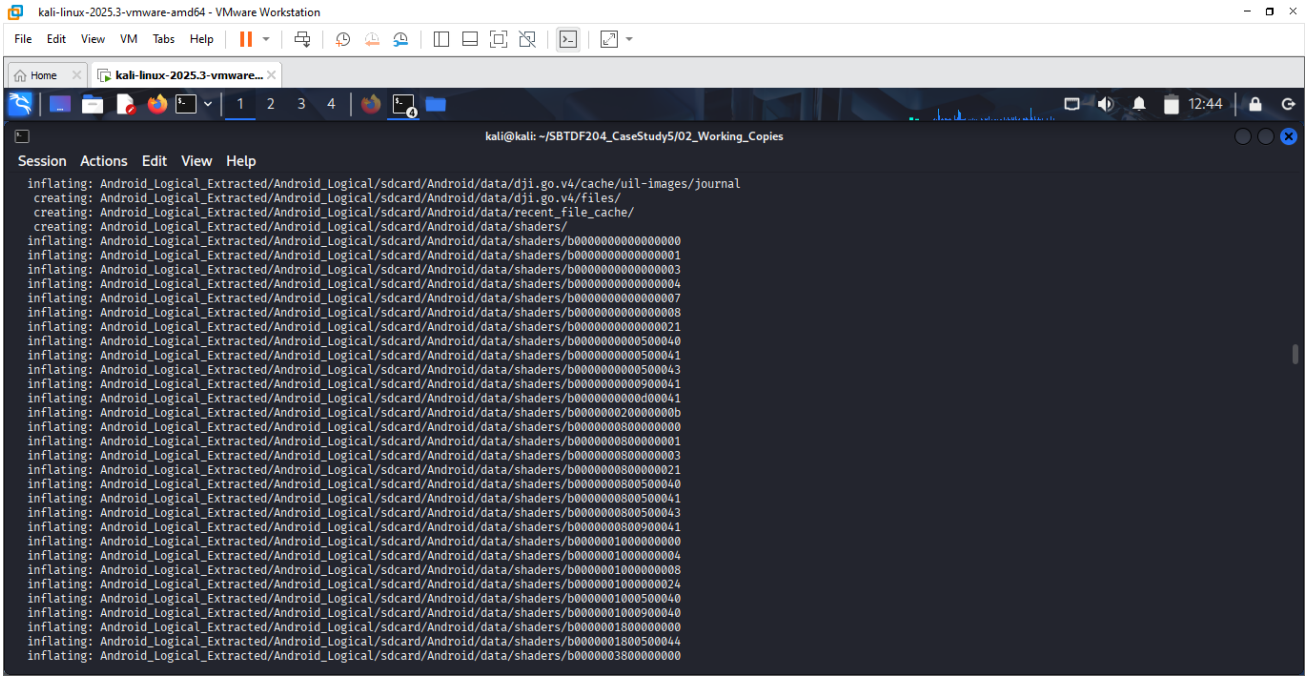


Fig 1.4 Android Logical Working Copy Extraction page 1.

Explanation

The extraction continued through Android application and storage artefacts. These recovered directories provided the files later searched for DJI GO 4 evidence.

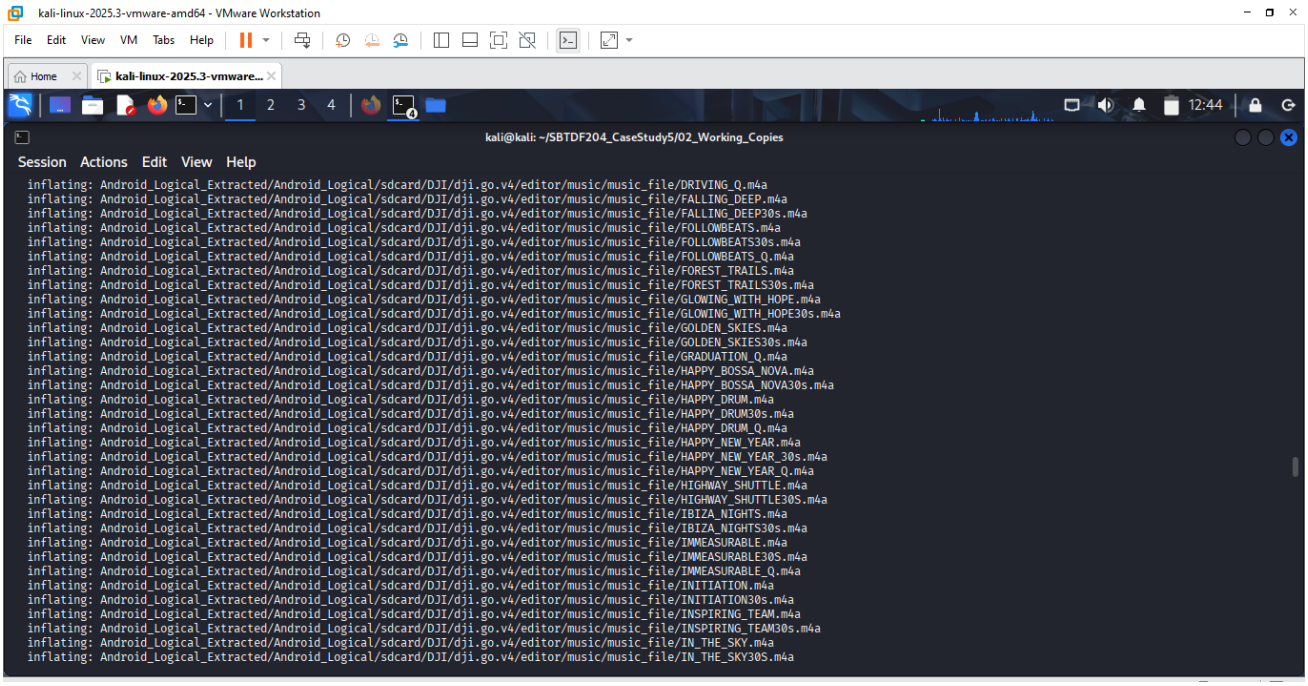


Fig 1.5 Android Logical Working Copy Extraction page 2.

Explanation

Additional application files were extracted from the logical package. The continued output shows that the working copy contained a broad set of Android user and application data.

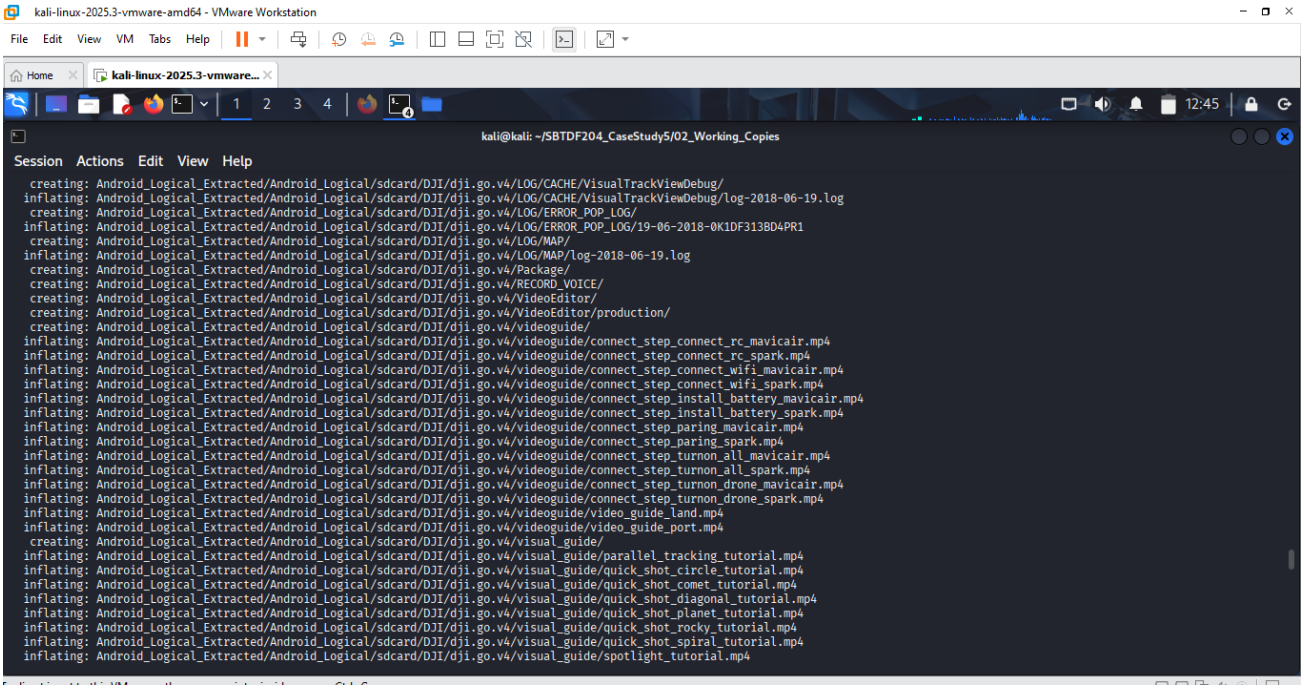


Fig 1.6 Android Logical Working Copy Extraction page 3.

Explanation

The extraction reached DJI-related folders and supporting application resources. This confirmed that DJI GO 4 content was present in the logical acquisition.

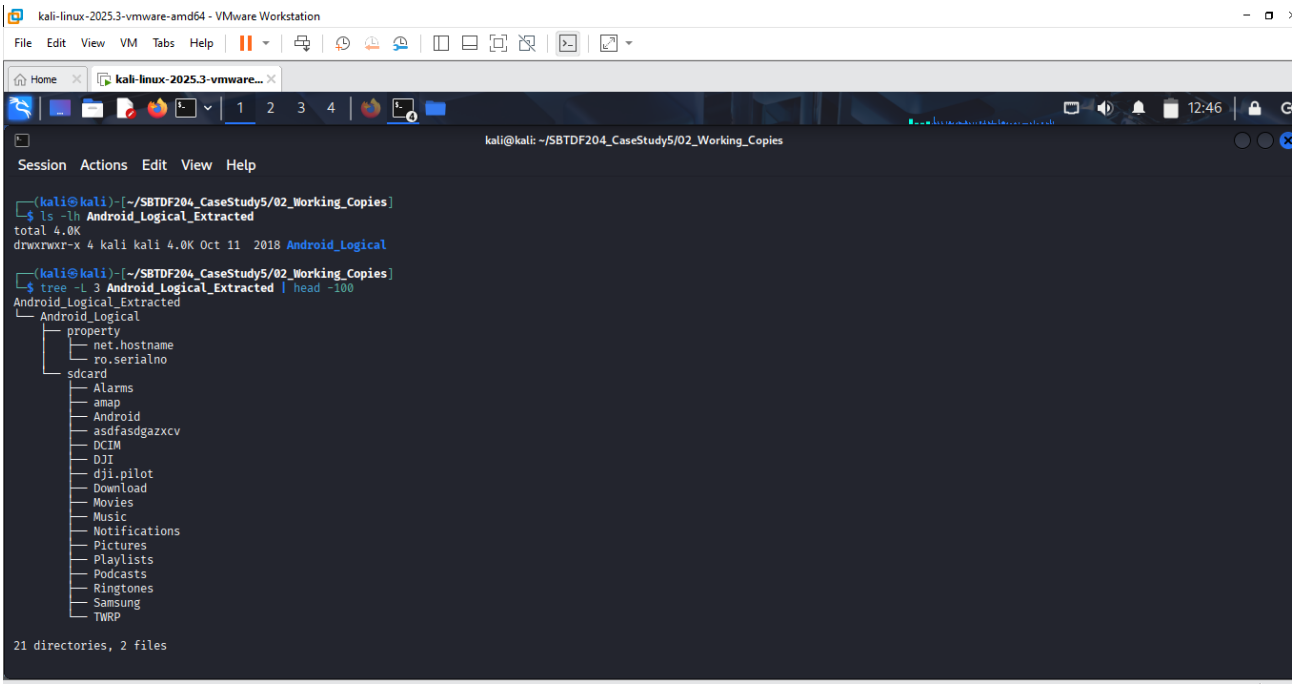


Fig 1.7 Android Logical Working Copy Extraction page 4.

Explanation

The completed extraction was reviewed as a directory tree. The resulting structure provided the starting point for locating the dji.go.v4 evidence folders.

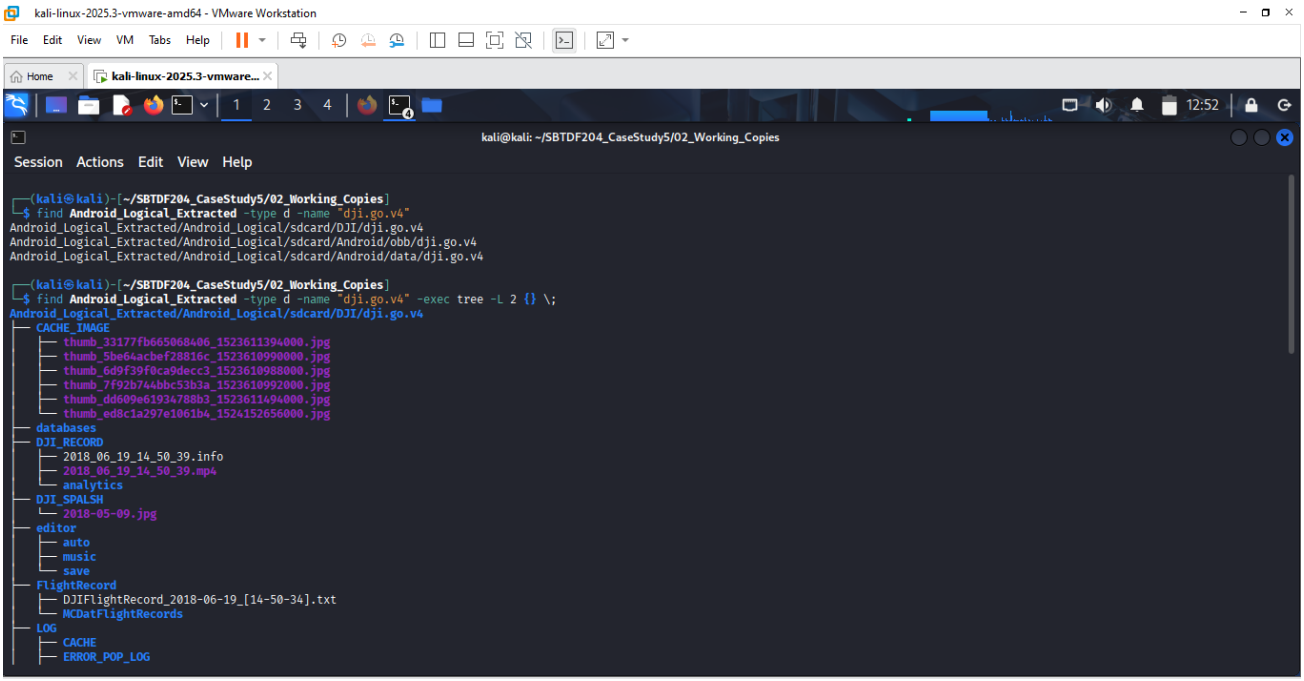


Fig 1.8 DJI GO 4 Application Directory Structure.

Explanation

The dji.go.v4 directory shows important locations such as FlightRecord, DJI_RECORD, CACHE_IMAGE, and supporting folders. These paths became the main focus of the Android examination.

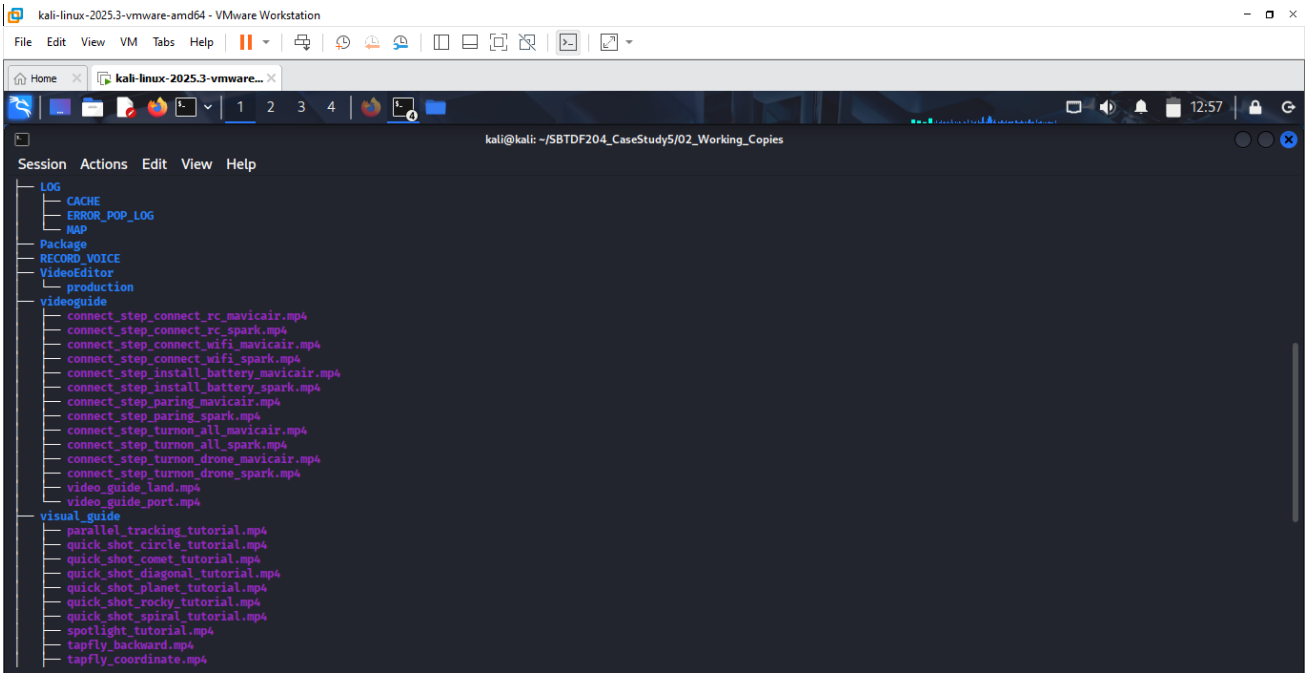


Fig 1.9 DJI GO 4 Application Directory Structure page 1.

Explanation

This continuation shows video-guide and other internal DJI GO 4 resources. It helped separate general application content from artefacts created during DJI GO 4 use.

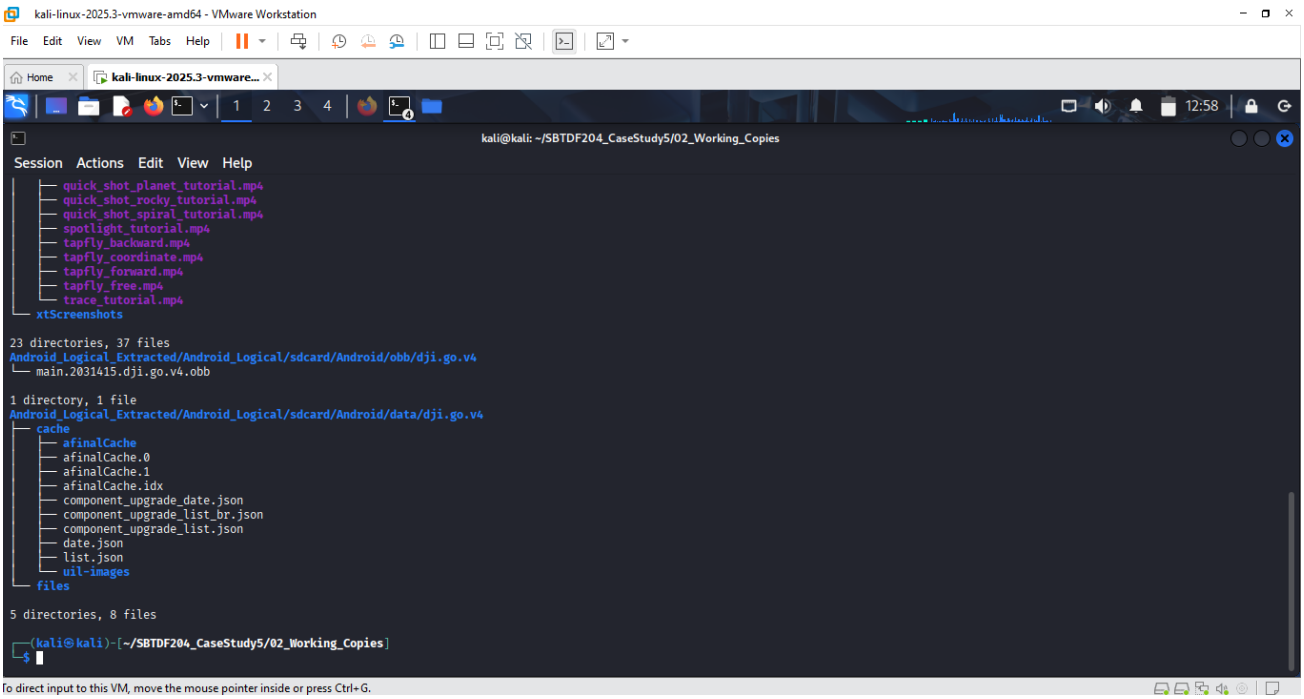


Fig 2.0 DJI GO 4 Application Directory Structure page 2.

Explanation

The remaining DJI GO 4 folders and configuration files complete the initial application inventory. This confirmed the wider structure before individual evidence files were examined.

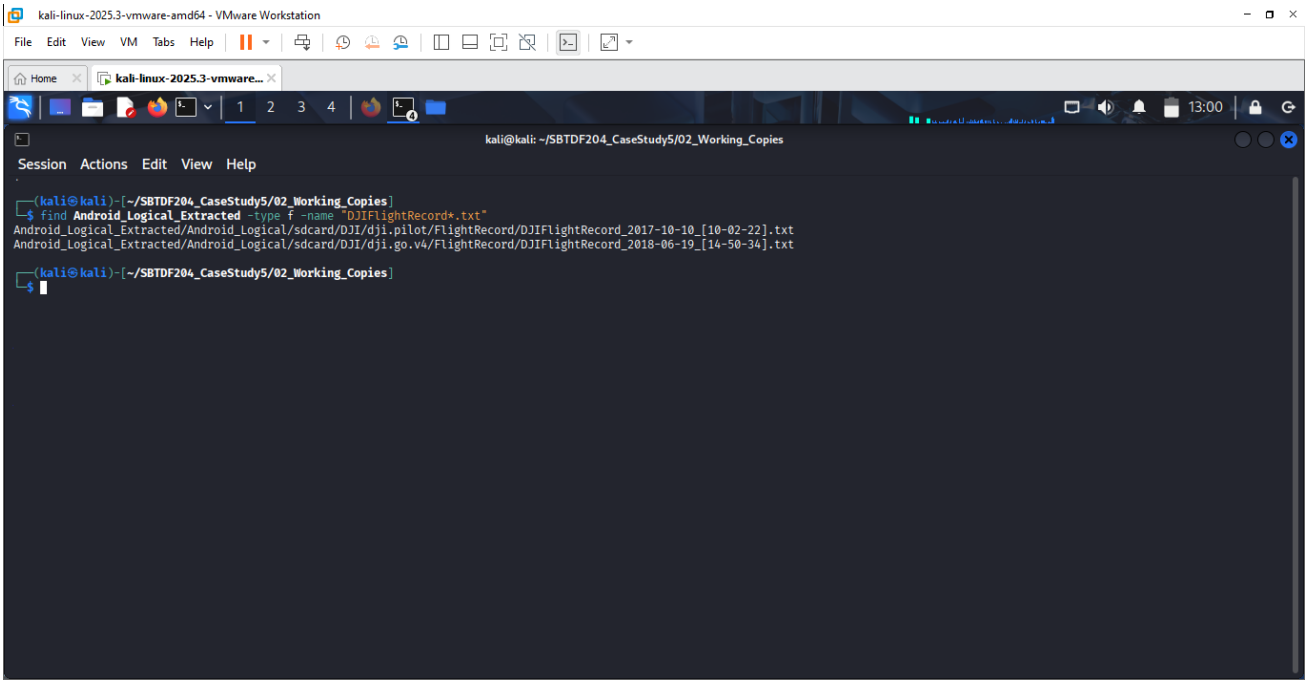


Fig 2.1 DJI GO 4 Main Flight Record Located.

Explanation

The main DJIFlightRecord_2018-06-19_[14-50-34].txt file was located in the FlightRecord directory. Its exact path was preserved for later metadata, hash, and timestamp checks.

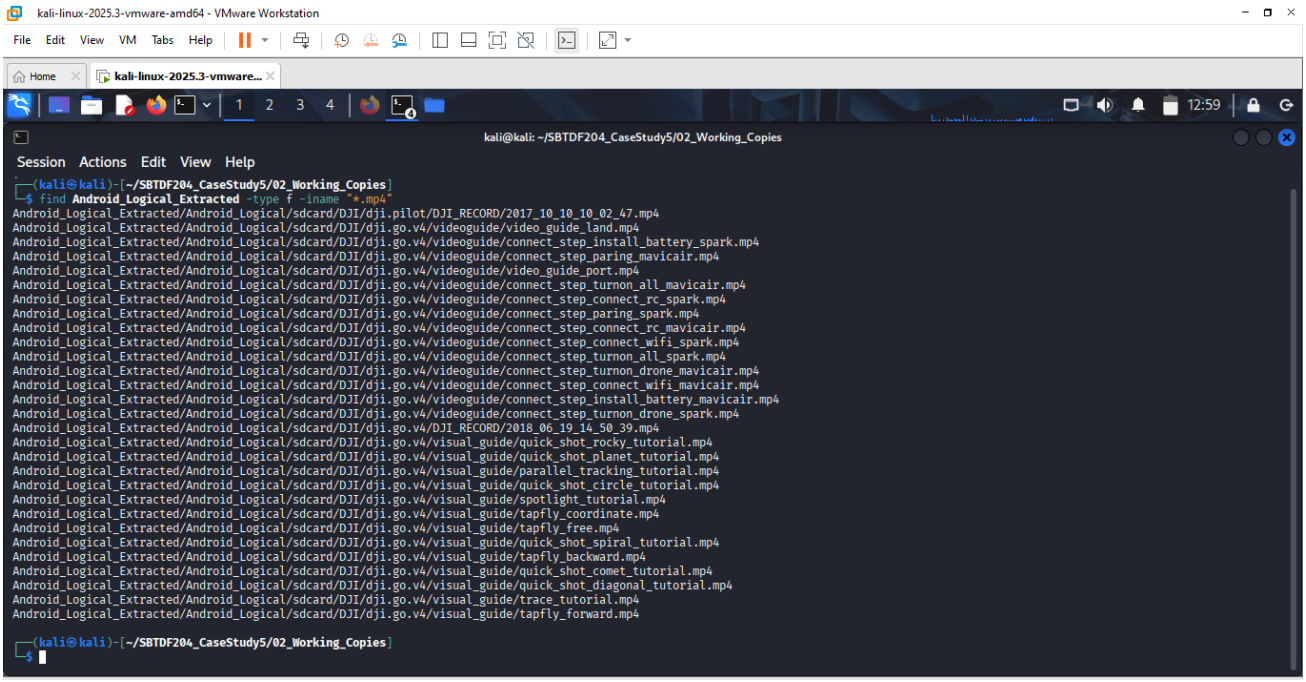


Fig 2.2 DJI GO 4 Cached Flight Video Located.

Explanation

The search identified the cached flight video in the DJI_RECORD folder and distinguished it from the application's tutorial videos. This established the exact MP4 used for later examination.

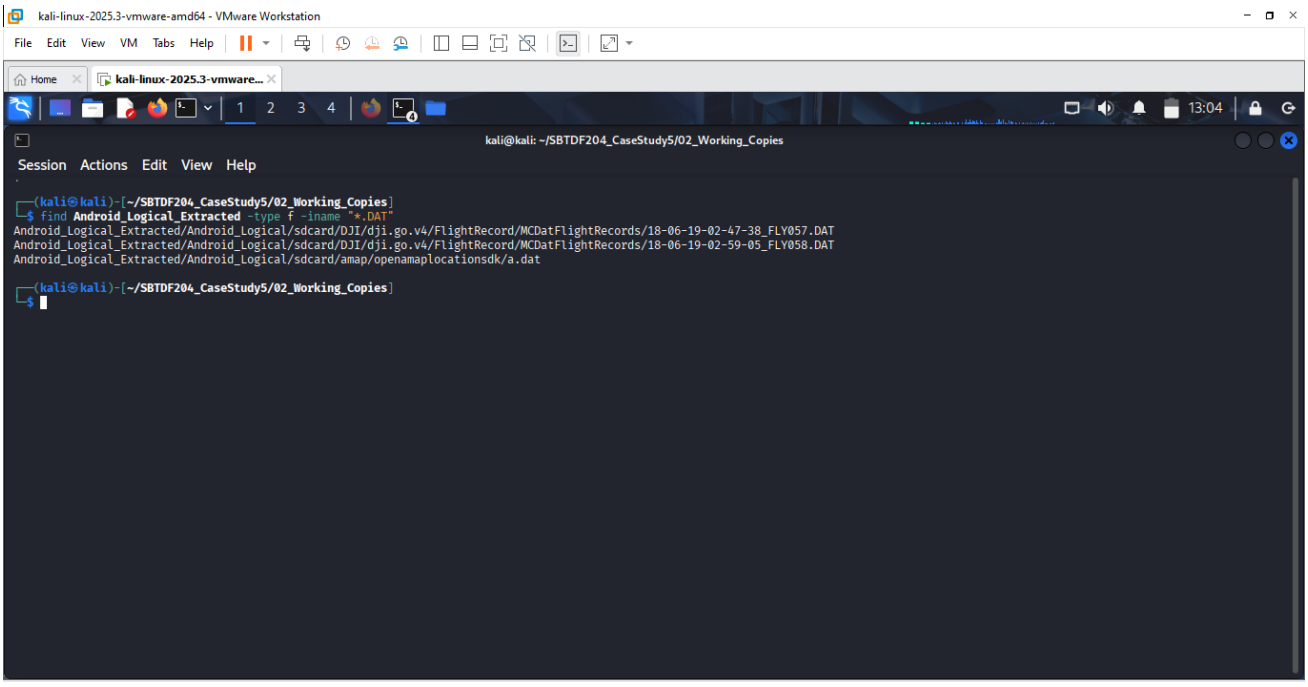


Fig 2.3 DJI GO 4 Detailed DAT Flight Records Located.

Explanation

Detailed DAT flight records were located in the MCDatFlightRecords directory. The investigation later focused on 18-06-19-02-47-38_FLY057.DAT.

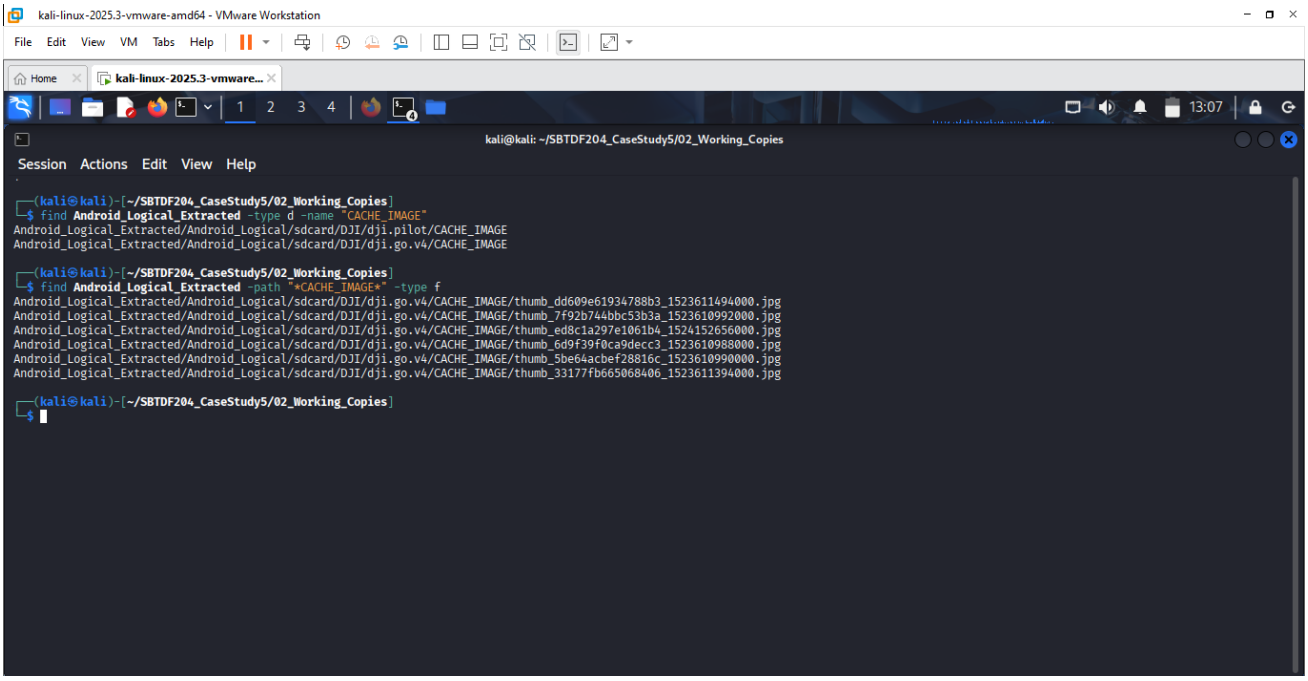


Fig 2.4 DJI GO 4 Cached Picture Artefacts Located.

Explanation

Six cached image files were located in CACHE_IMAGE. Their paths and filenames were recorded before metadata or visual inspection was performed.

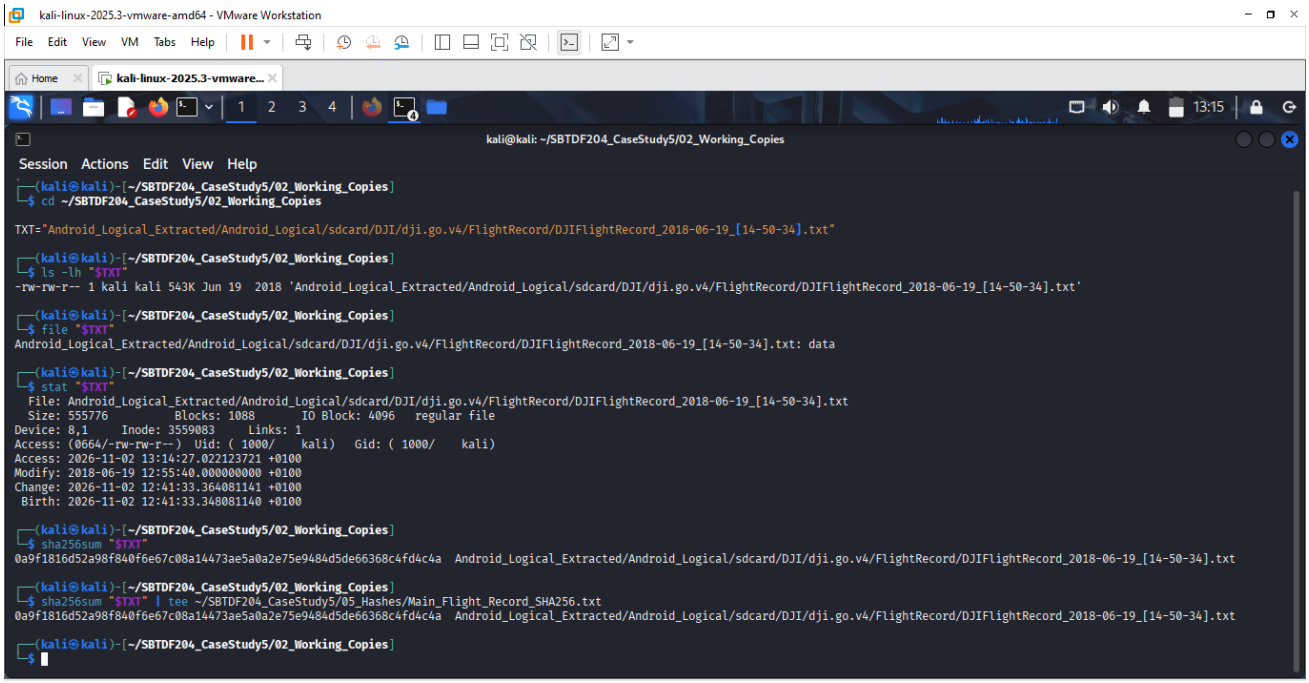


Fig 2.5 DJI GO 4 Main Flight Record Metadata and Integrity.

Explanation

The main flight record was checked with file, stat, and SHA-256 hashing. Its size and integrity value were preserved before deeper examination.

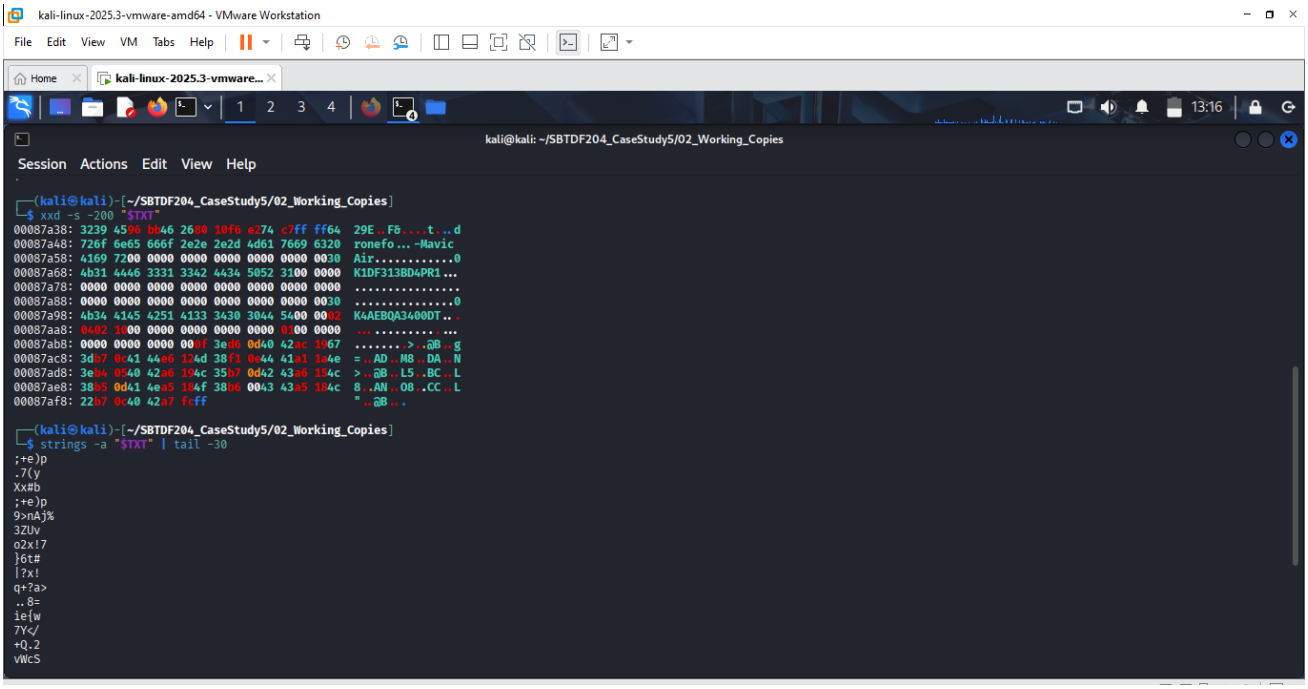


Fig 2.6 Main Flight Record Aircraft Identification.

Explanation

Manual binary and readable-text inspection exposed Mavic Air-related strings in the main flight record. This supported the record's association with a DJI Mavic Air aircraft.

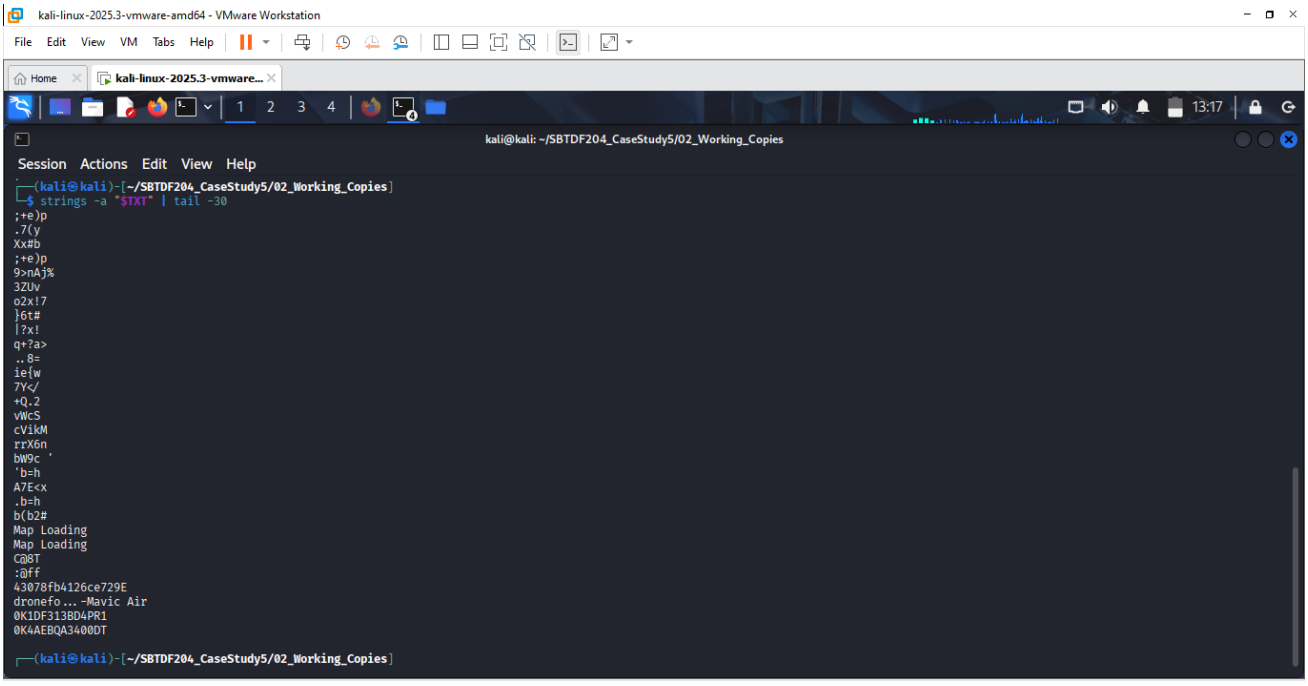


Fig 2.7 Main Flight Record Aircraft Identification page 1.

Explanation

Additional readable strings provided application and aircraft context from the same record. These observations were recorded without changing the binary evidence.

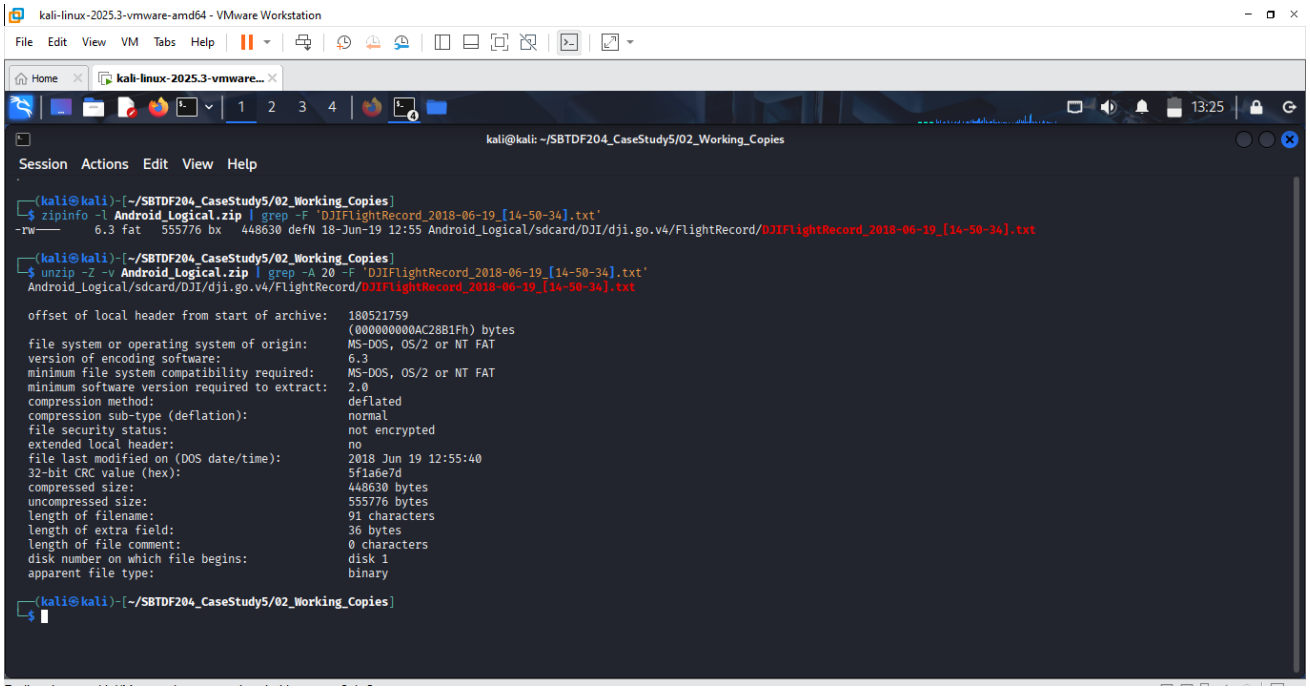


Fig 2.8 Main Flight Record ZIP Timestamp Verification.

Explanation

The source ZIP entry was examined to recover the timestamp stored in the original archive. Because the ZIP metadata did not preserve a reliable timezone, the value was treated cautiously during normalisation.

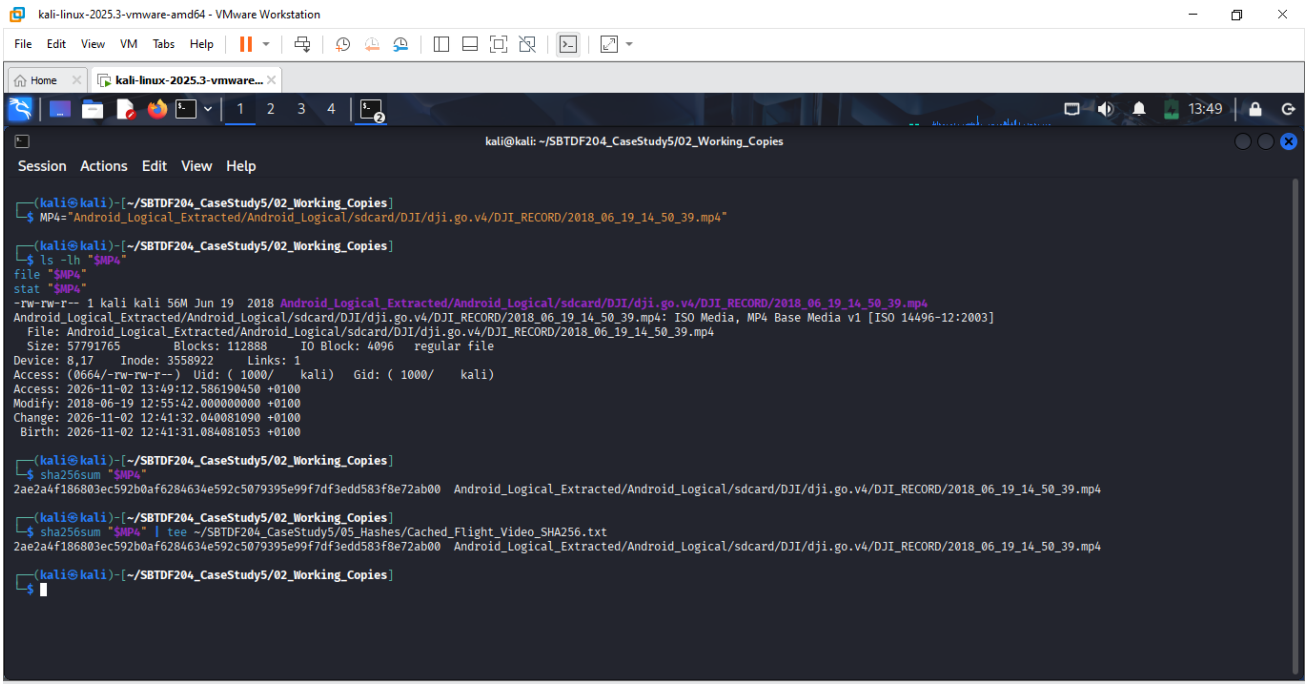


Fig 2.9 DJI GO 4 Cached Flight Video Metadata and Integrity.

Explanation

The cached MP4 was checked for file type, size, timestamps, and SHA-256 before playback. This preserved its forensic state before visual examination.

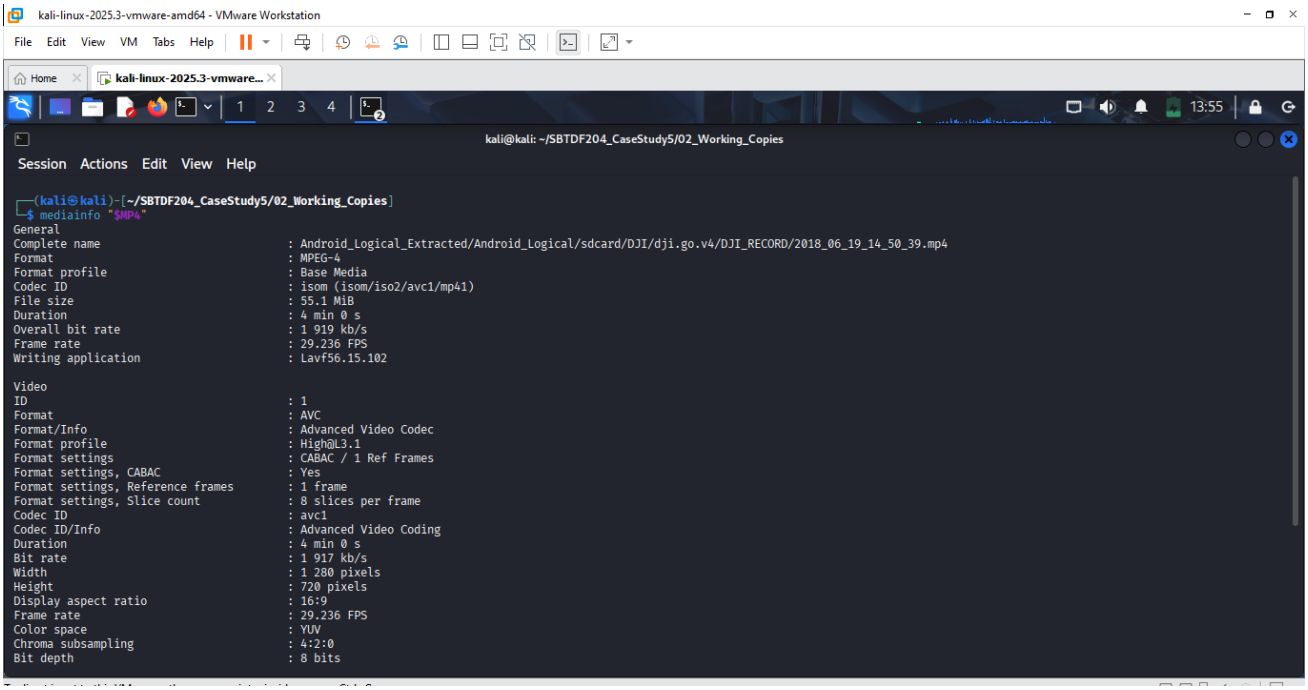


Fig 3.0 Cached Flight Video MediaInfo Metadata.

Explanation

MediaInfo identified the cached evidence as an MPEG-4 video using AVC/H.264 compression. The output also recorded its approximate four-minute duration and other technical properties.

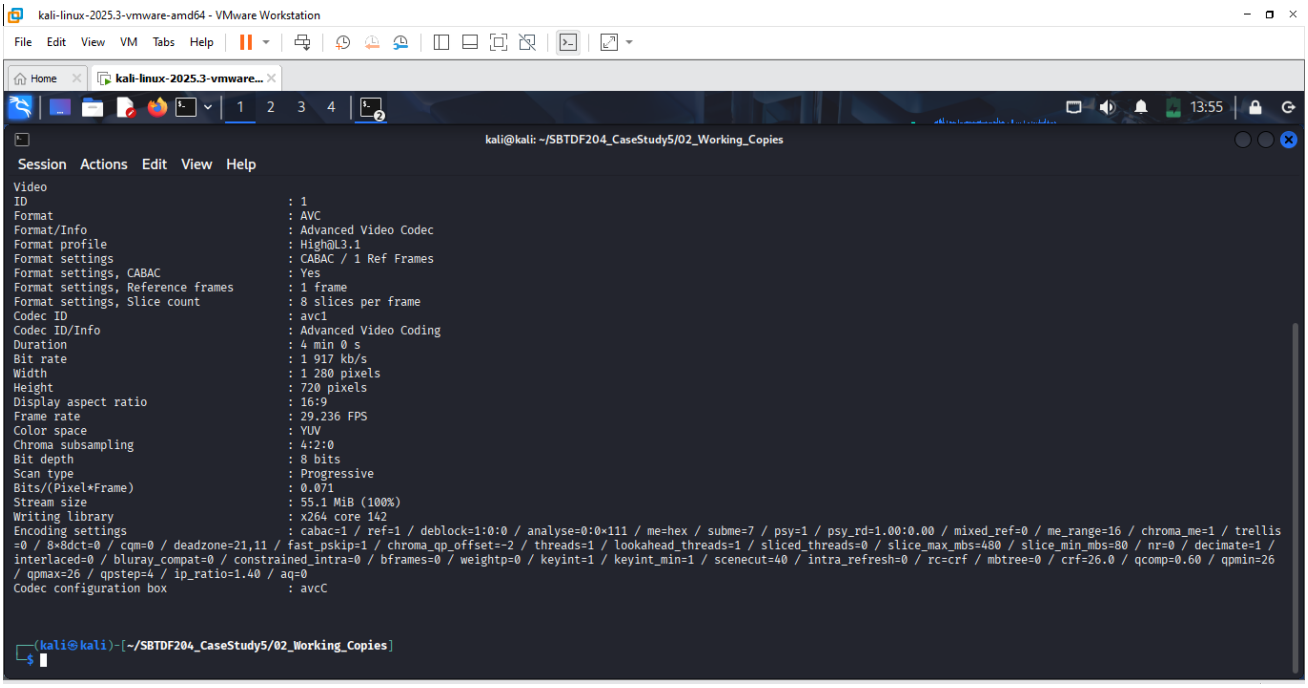


Fig 3.1 Cached Flight Video MediaInfo Metadata page 1.

Explanation

The continuation records the video's detailed codec, frame-rate, resolution, bitrate, and encoding information. These fields describe the technical characteristics of the cached recording.

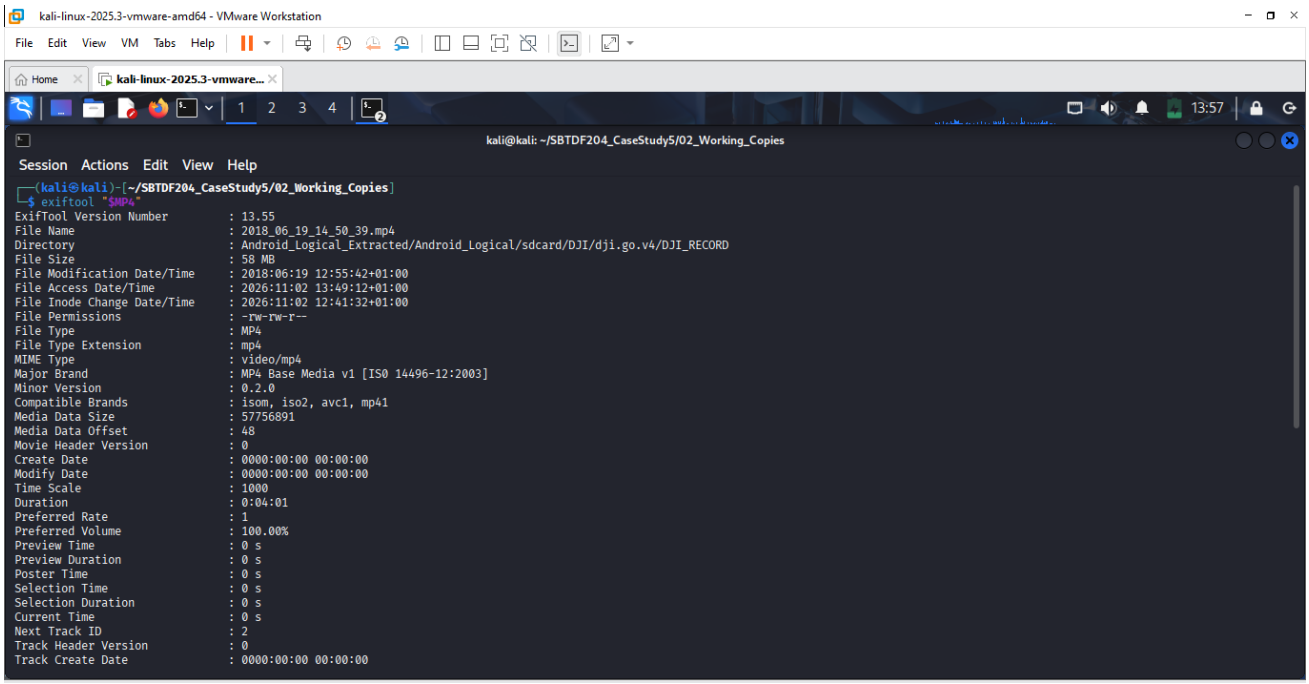


Fig 3.2 Cached Flight Video ExifTool Metadata.

Explanation

ExifTool independently recorded file and container metadata for the cached MP4. Using a second metadata tool provided another view of the file without altering it.

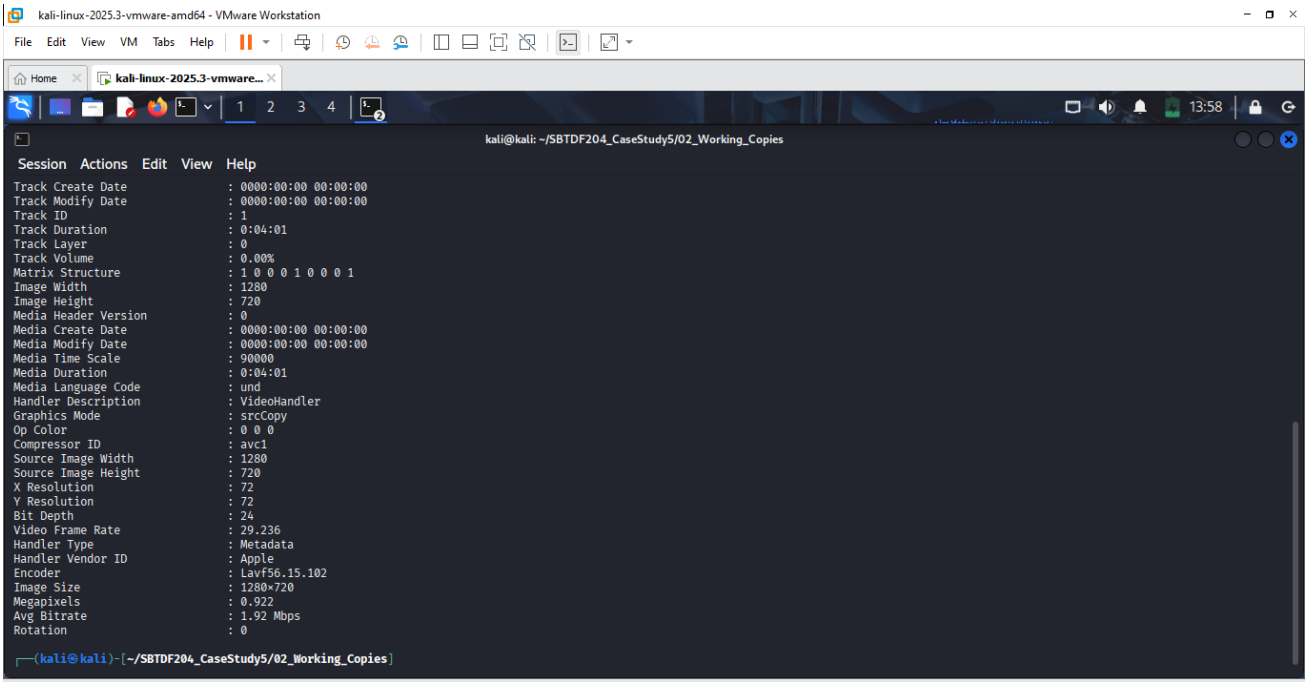


Fig 3.3 Cached Flight Video ExifTool Metadata page 1.

Explanation

The remaining ExifTool fields describe image dimensions, frame rate, encoder information, and other properties. These values were consistent with the earlier MediaInfo examination.

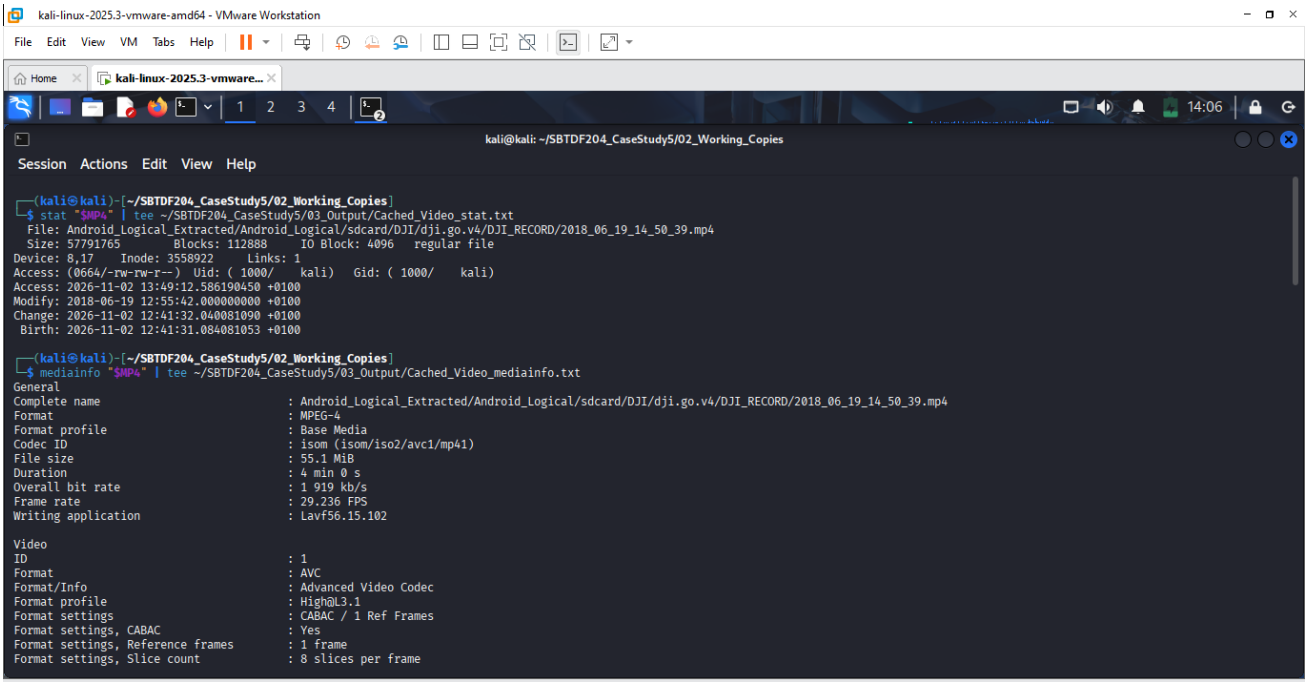


Fig 3.4 Cached Flight Video ZIP Timestamp Verification.

Explanation

The extracted filesystem information was compared with the preserved video metadata. This helped identify timestamp differences between extracted files and source archive information.

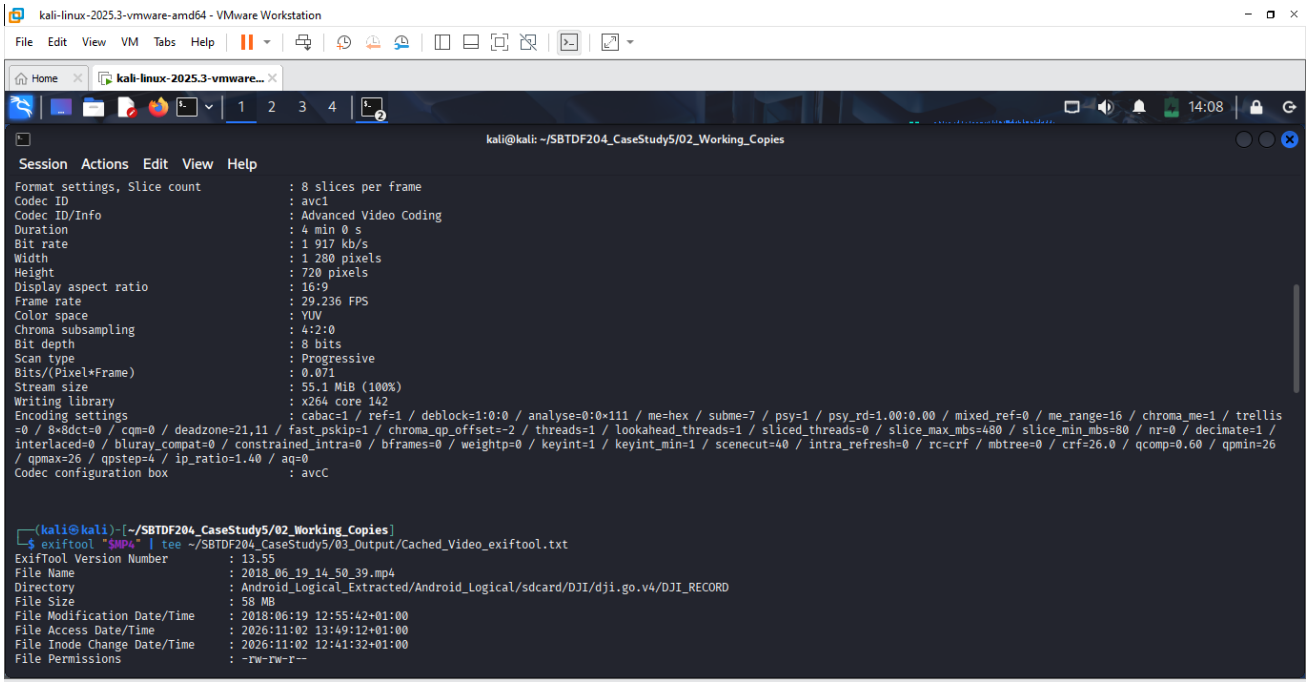


Fig 3.5 Cached Flight Video ZIP Timestamp Verification page 1.

Explanation

Further MediaInfo and ExifTool output was reviewed during the timestamp check. The comparison separated technical video properties from archive-level time information.

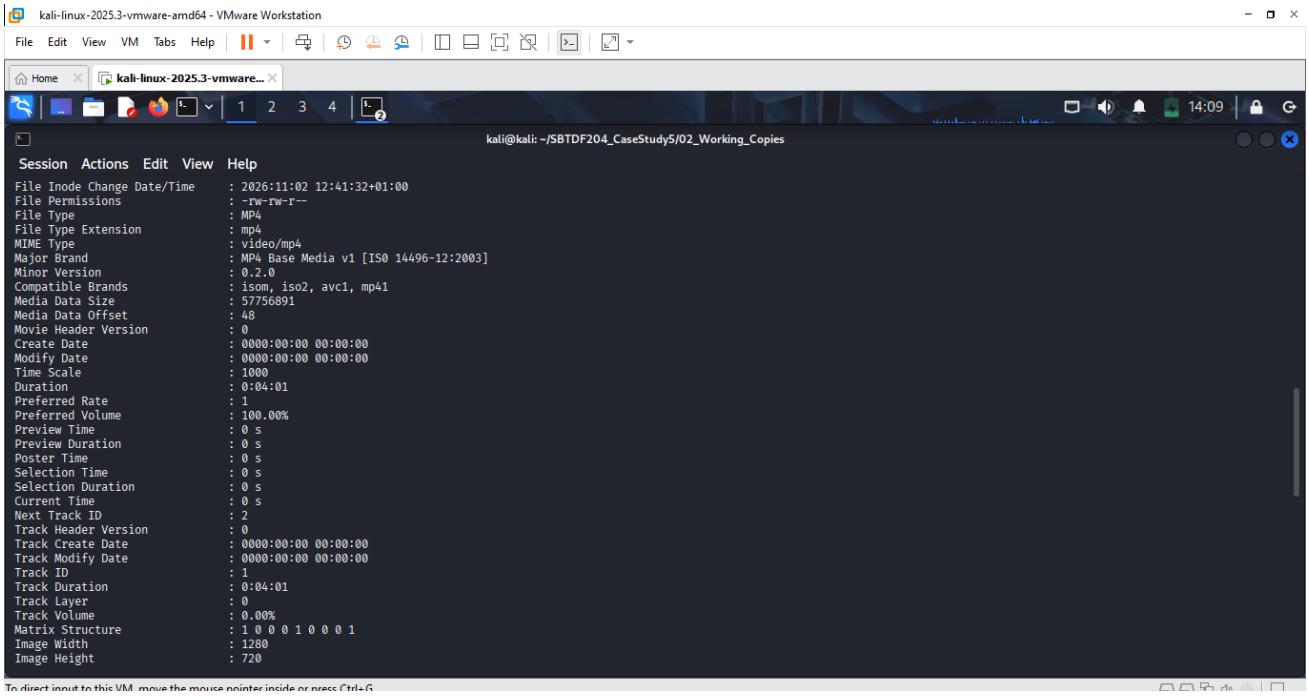


Fig 3.6 Cached Flight Video ZIP Timestamp Verification page 2.

Explanation

Several internal MP4 date fields were not useful for establishing the actual recording time. This reinforced the need to rely on supported source timestamps rather than zeroed or ambiguous embedded fields.

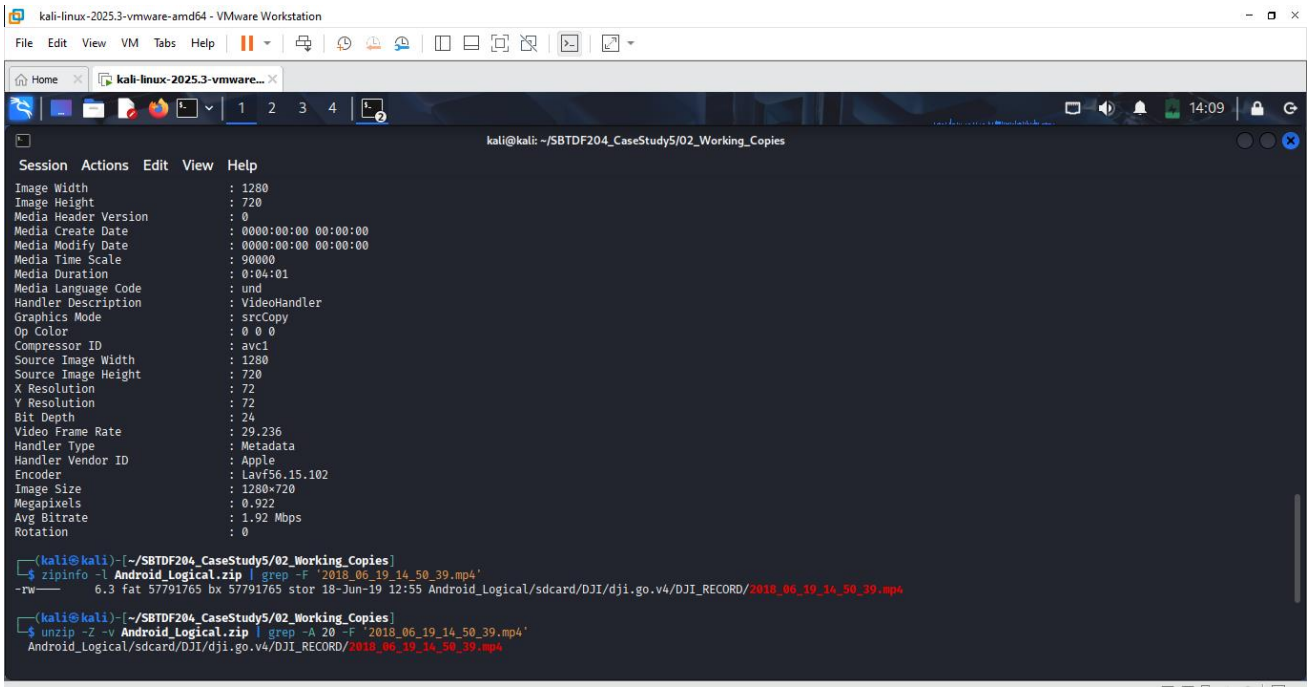


Fig 3.7 Cached Flight Video ZIP Timestamp Verification page 3.

Explanation

The final metadata view was checked together with the source archive lookup for the MP4. This confirmed that the ZIP information had been reviewed separately from the extracted filesystem time.

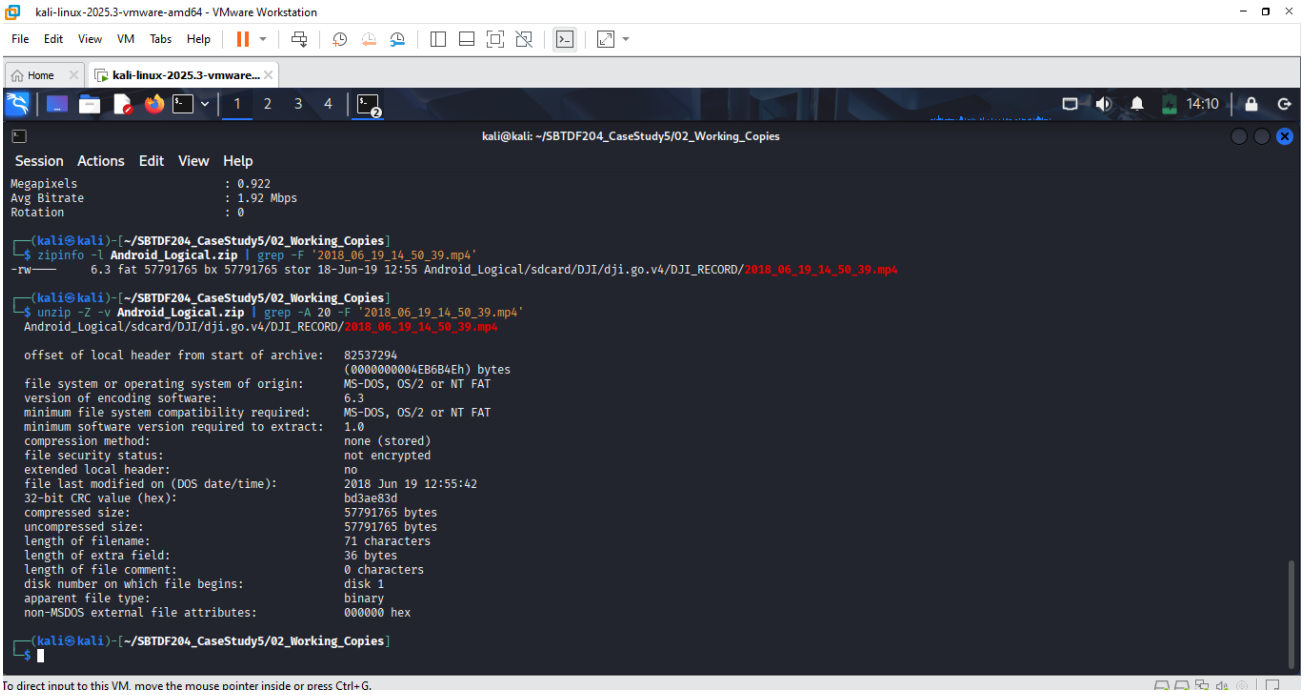


Fig 3.8 Cached Flight Video ZIP Timestamp Verification page 4.

Explanation

Detailed ZIP information records the cached video's source time and archive attributes. Because a trustworthy timezone was not stored, the time was retained as source evidence rather than assumed to be UTC.

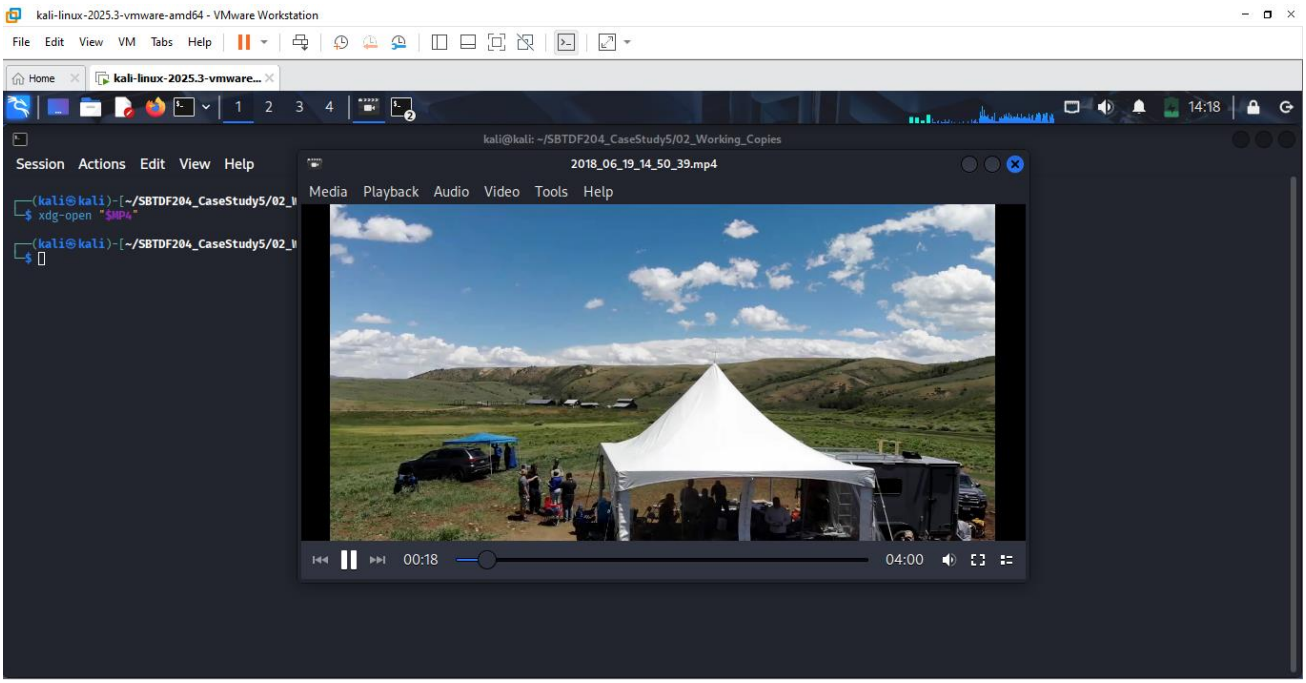


Fig 3.9 DJI GO 4 Cached Flight Video Preview.

Explanation

The controlled preview shows a rural outdoor area with a white tent, vehicles, people, and surrounding hills. This confirms the visible content of the preserved cached video.

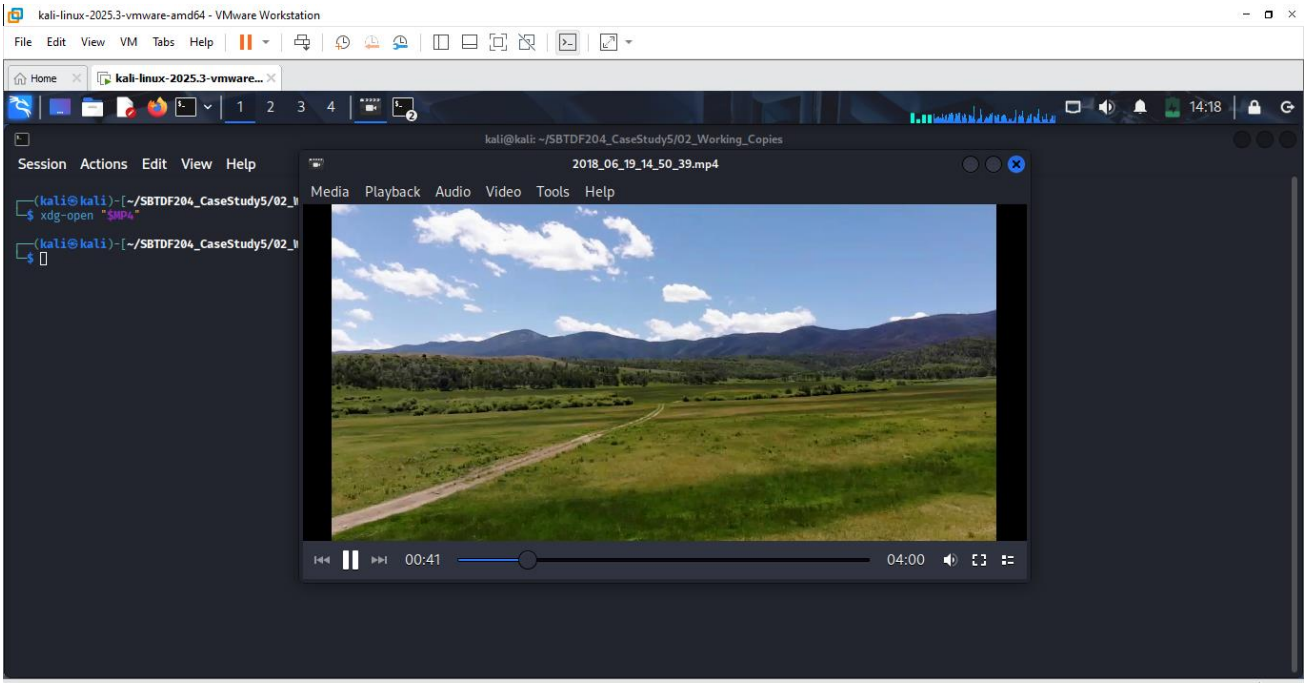


Fig 4.0 DJI GO 4 Cached Flight Video Preview page 1.

Explanation

This frame shows open fields, a track, hills, and vegetation. It is consistent with aerial movement across a rural landscape.

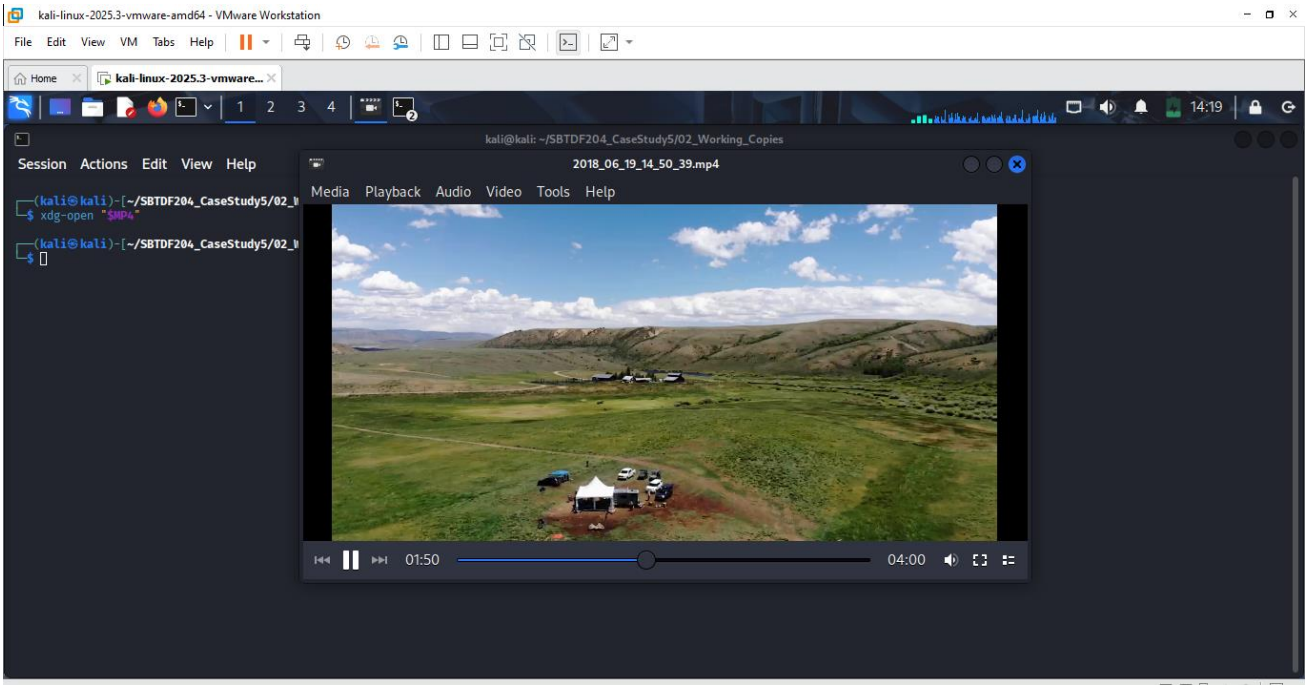


Fig 4.1 DJI GO 4 Cached Flight Video Preview page 2.

Explanation

The camera looks down toward the tent and nearby vehicles from a greater height. The view supports that the recording includes airborne footage.

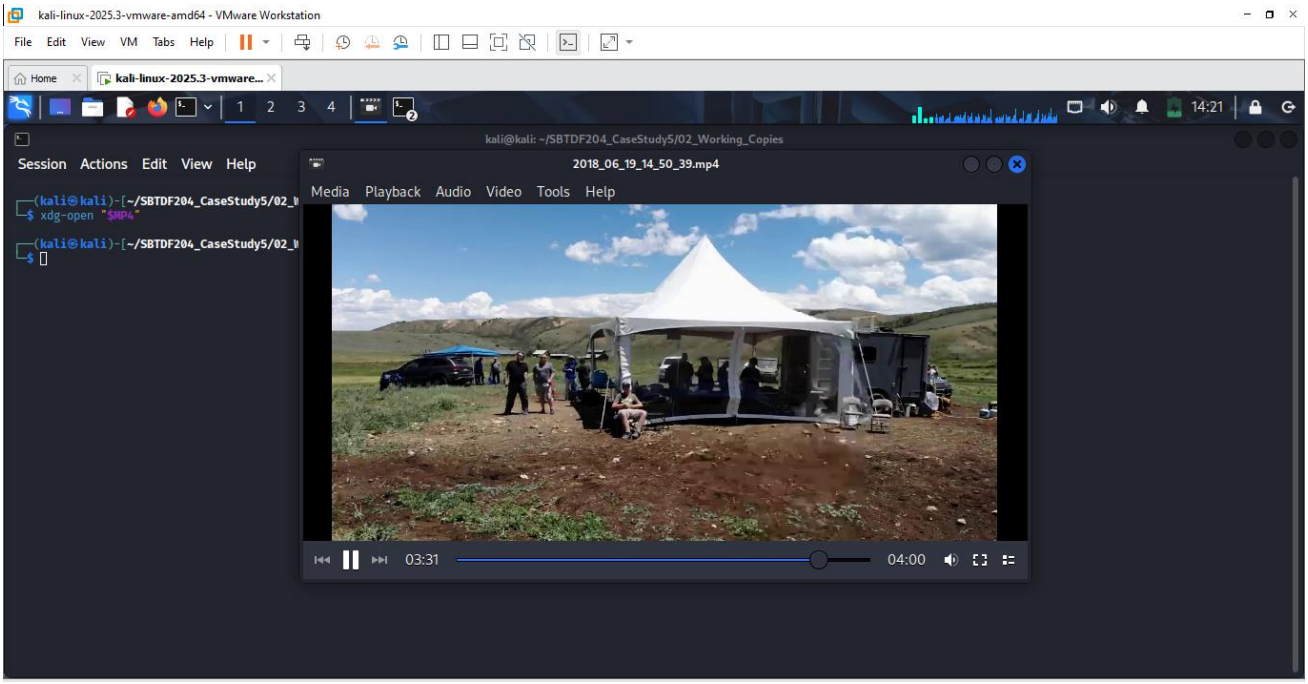


Fig 4.2 DJI GO 4 Cached Flight Video Preview page 3.

Explanation

The recording returns closer to the white tent where several people are visible. The scene confirms activity around the site but does not show who controlled the aircraft.

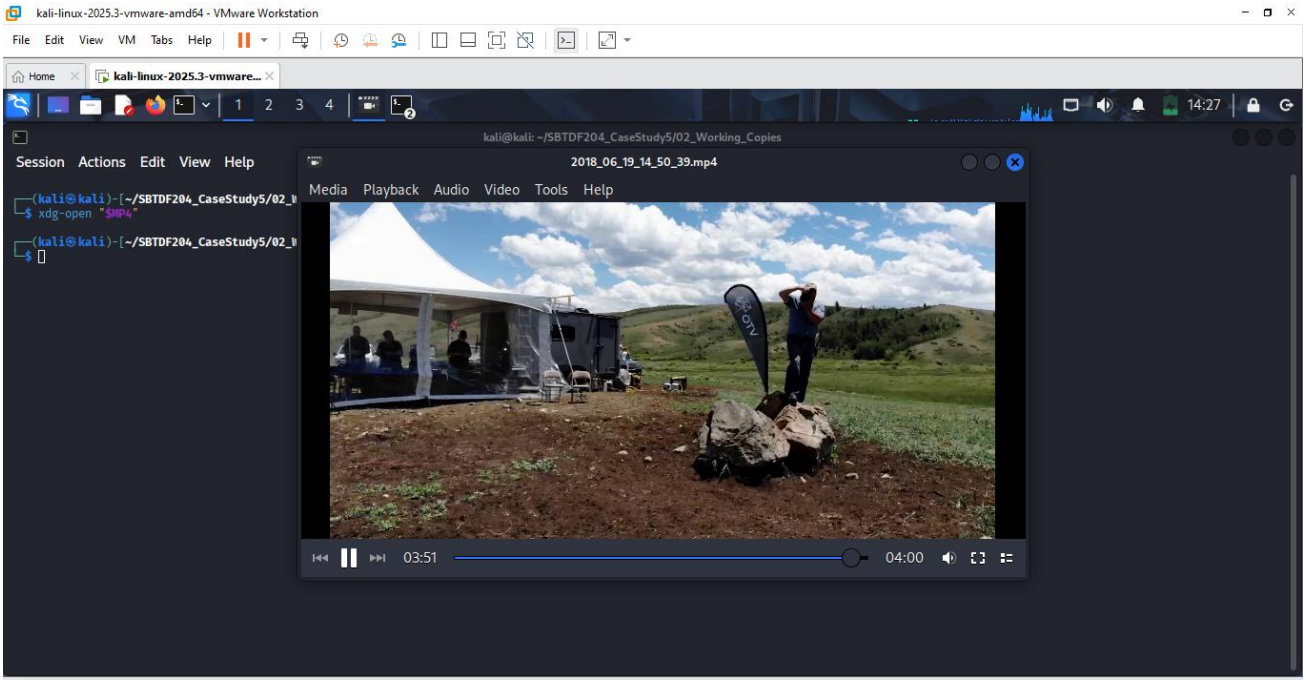


Fig 4.3 DJI GO 4 Cached Flight Video Preview page 4.

Explanation

A person is visible near the tent and equipment toward the end of the clip. Presence in the scene alone cannot identify the drone operator.

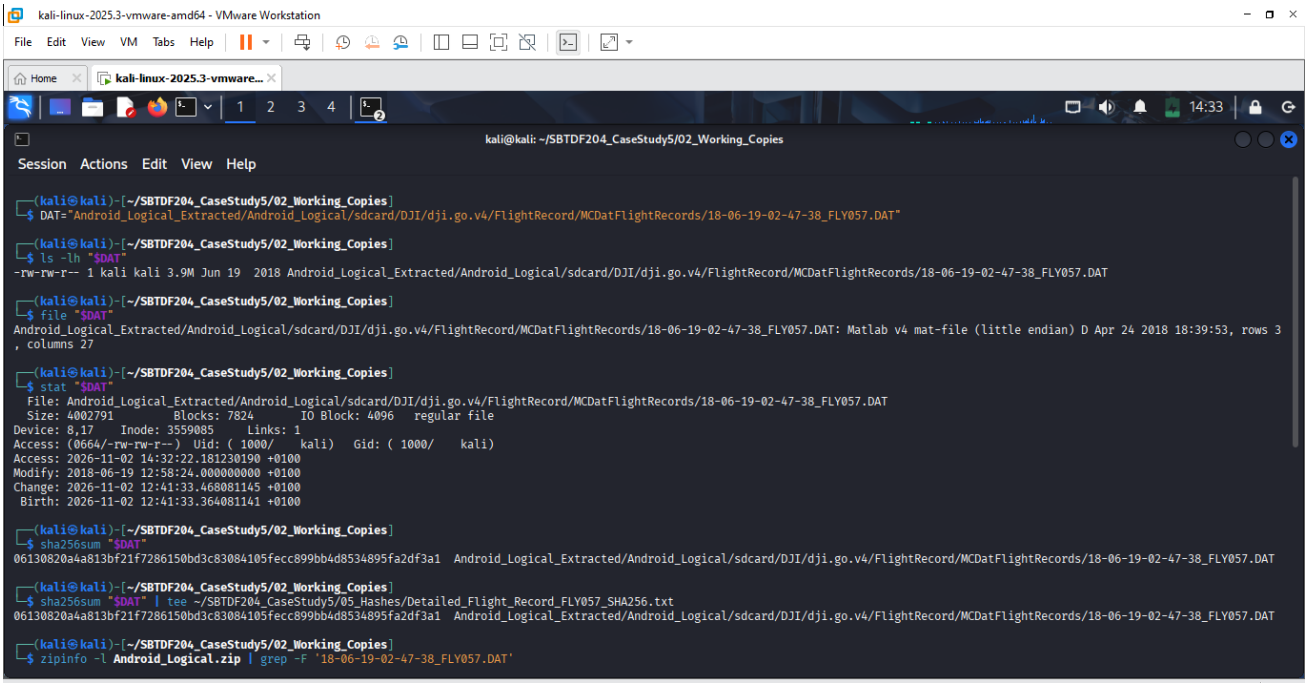


Fig 4.4 DJI GO 4 Detailed Flight Record Metadata and Integrity.

Explanation

FLY057.DAT was identified, sized, and hashed before analysis. Kali recognised it as a little-endian MATLAB v4 MAT-file, providing a technical description of the detailed record.

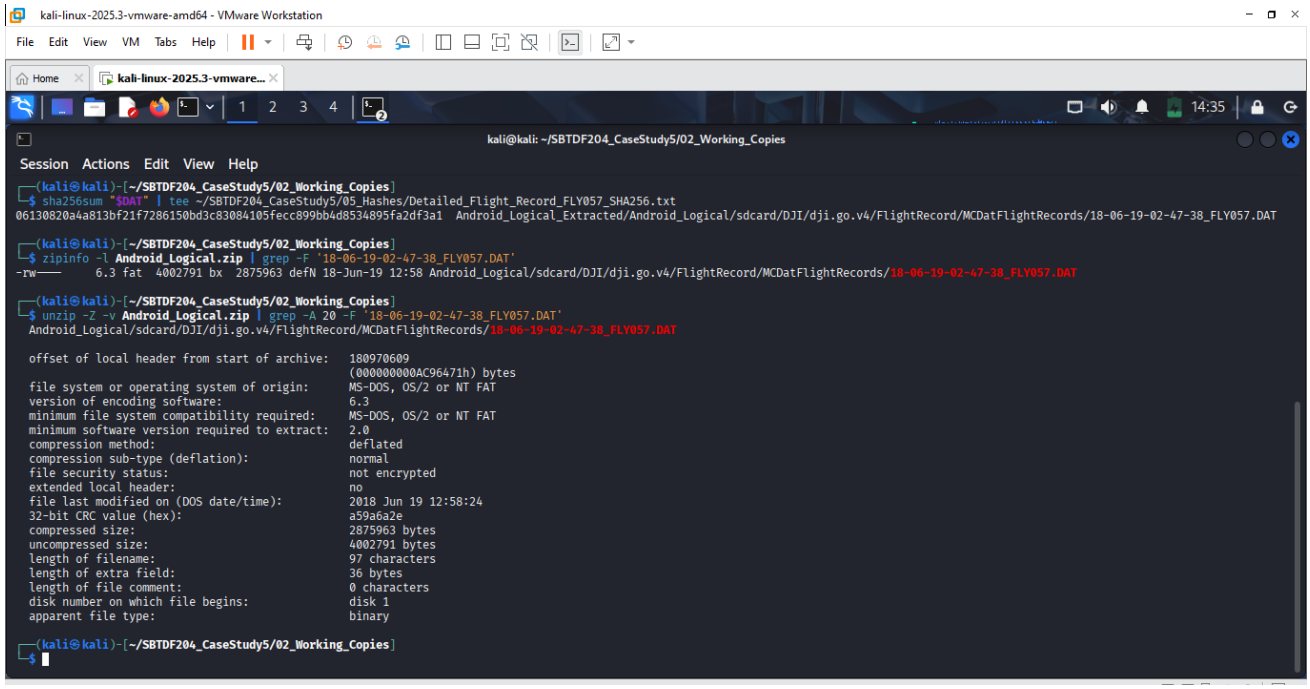


Fig 4.5 DJI GO 4 Detailed Flight Record Metadata and Integrity Page 1.

Explanation

The source ZIP metadata for the DAT record was examined separately. This preserved its archive timestamp and showed why extracted and archive times needed careful treatment.

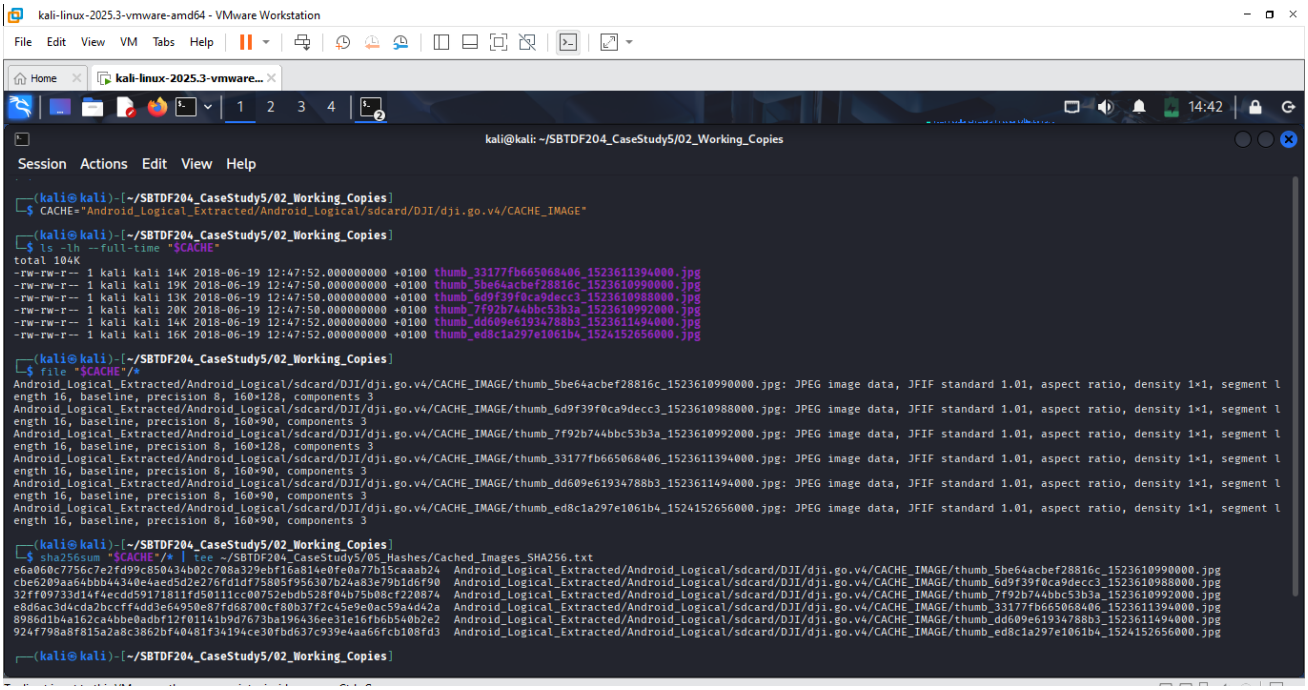


Fig 4.6 DJI GO 4 Cached Picture Metadata.

Explanation

The six cached images were inventoried, identified as JPEG files, and hashed. This established their format and integrity before visual inspection.

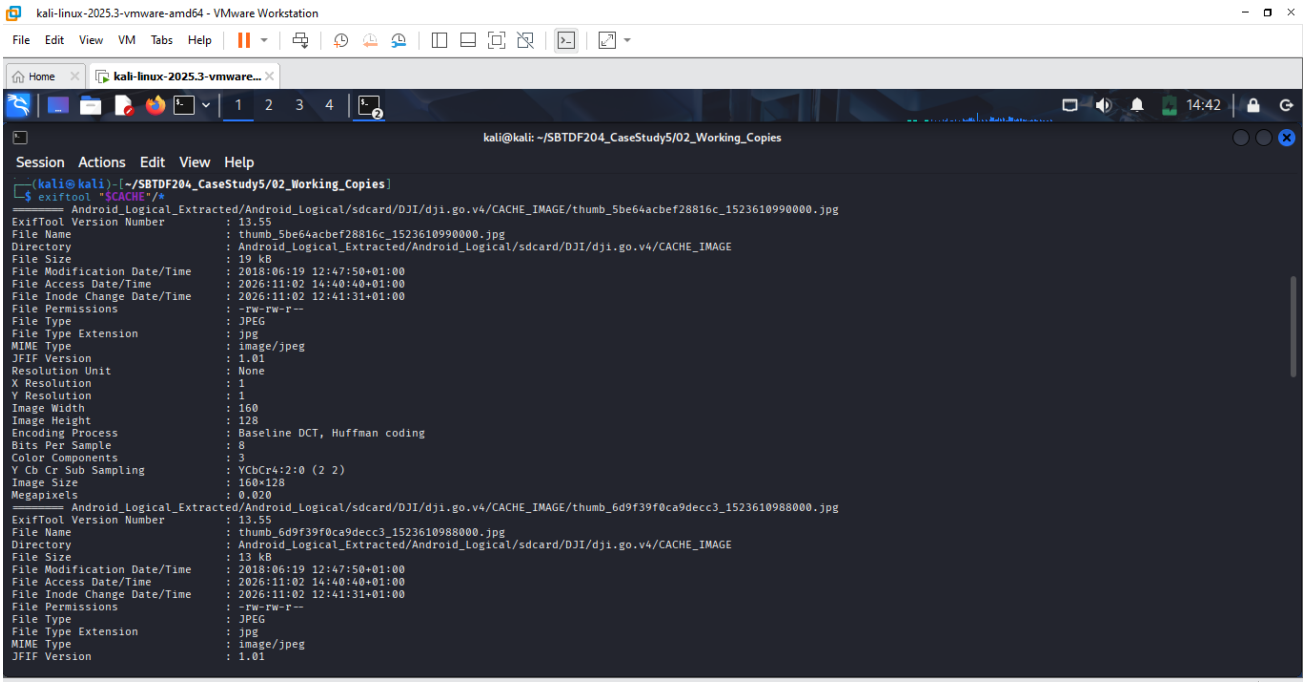


Fig 4.7 DJI GO 4 Cached Picture Metadata page 1.

Explanation

ExifTool shows that the cached files are small JPEG thumbnails with limited metadata. No useful original GPS information was available from the displayed fields.

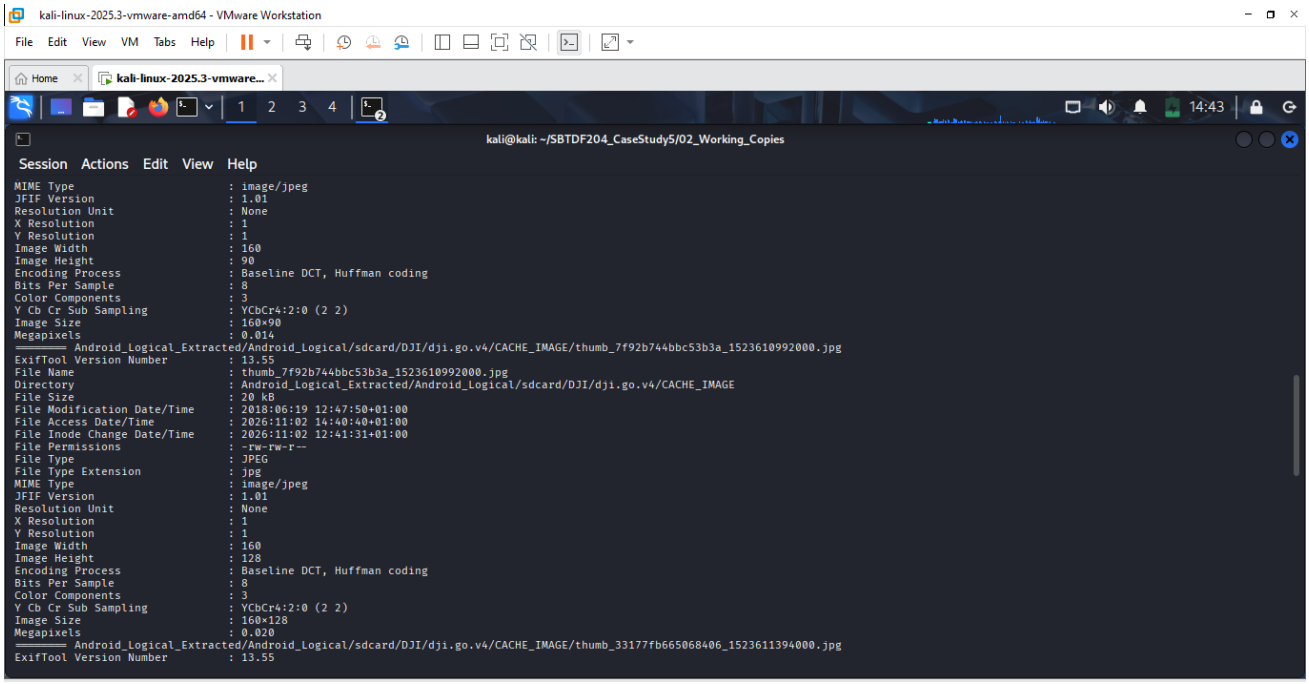


Fig 4.8 DJI GO 4 Cached Picture Metadata page 2.

Explanation

The metadata comparison continues across additional cached thumbnails. Their similar dimensions and JPEG properties support their interpretation as application cache images.

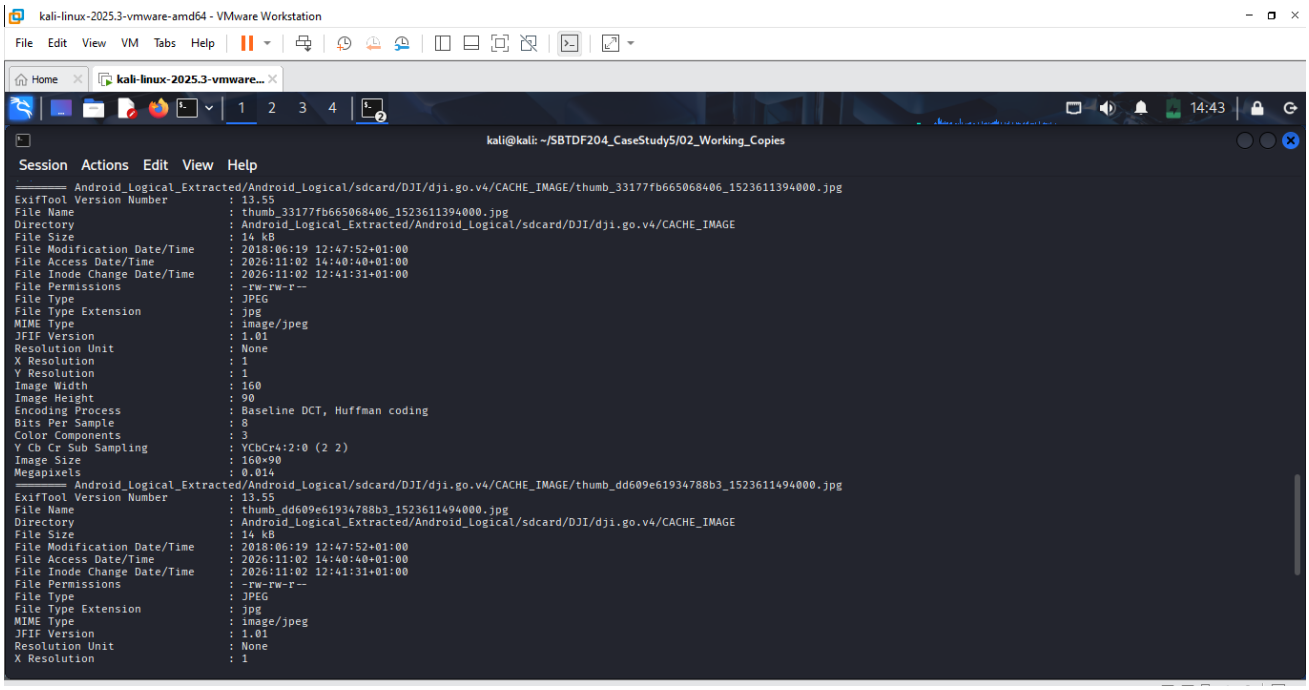


Fig 4.9 DJI GO 4 Cached Picture Metadata page 3.

Explanation

Additional thumbnail metadata was reviewed for consistency. The files still lacked reliable original camera or location information.

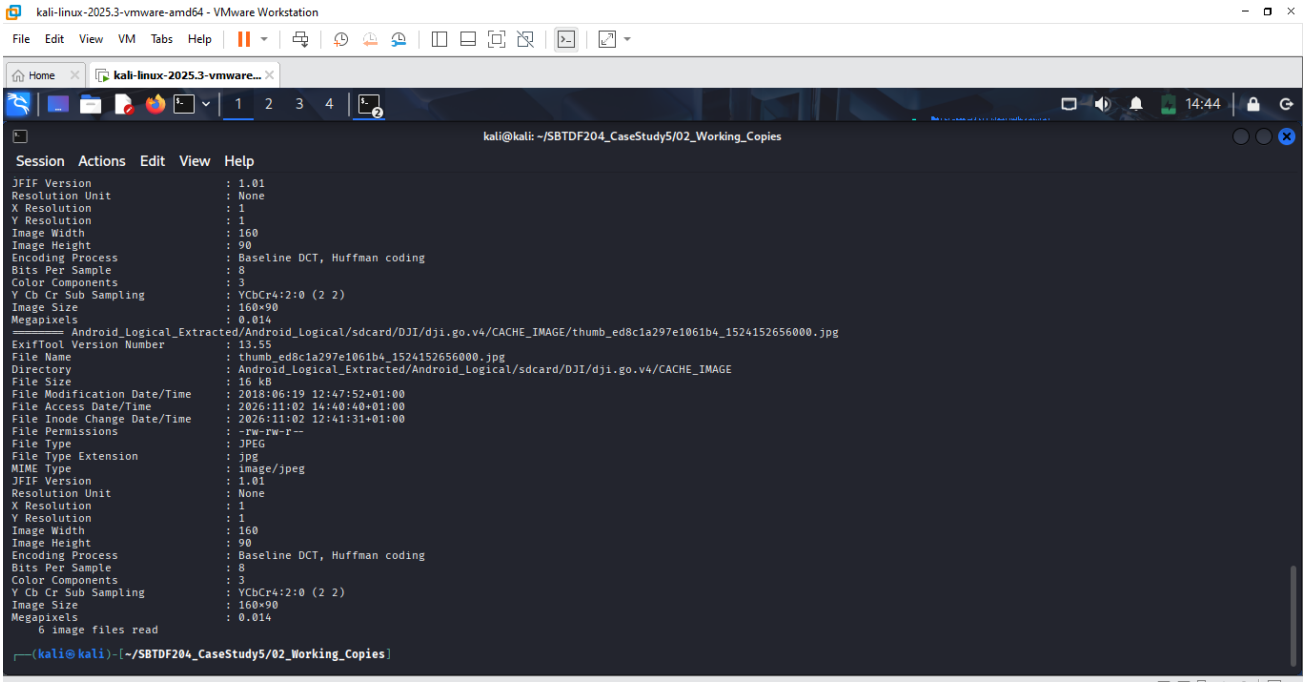


Fig 5.0 DJI GO 4 Cached Picture Metadata page 4.

Explanation

The last cached-picture entries show the same limited thumbnail characteristics. This completed the metadata review of all six cache images.

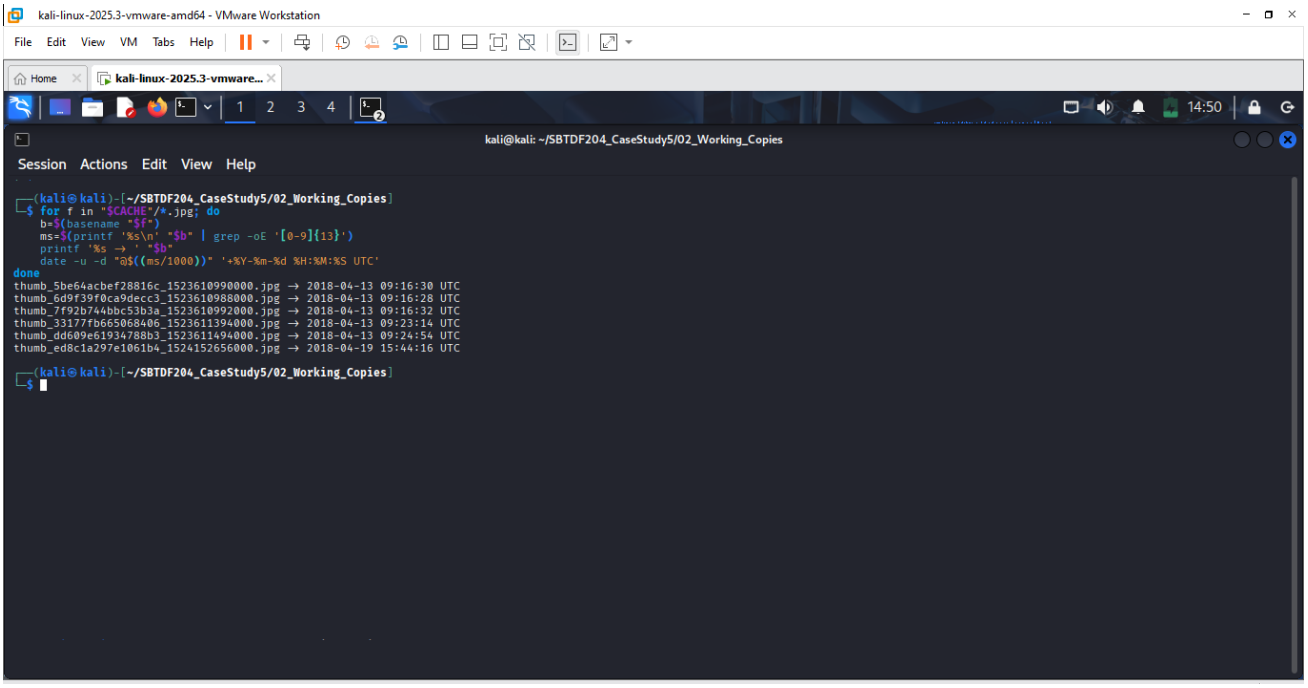


Fig 5.1 DJI GO 4 Cached Picture Filename Timestamp Conversion.

Explanation

The 13-digit values embedded in the cached-picture filenames were converted to UTC dates. The results fell between 13 and 19 April 2018, so they did not align directly with the June 19 flight evidence.

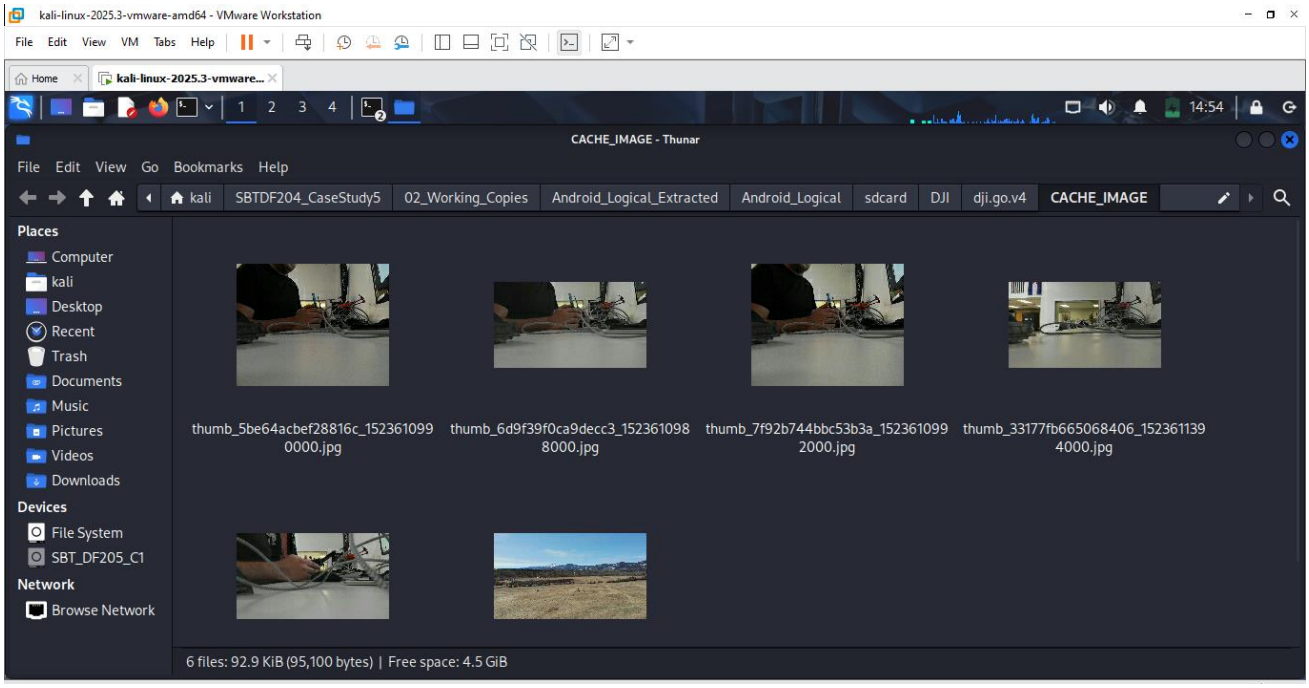


Fig 5.2 DJI GO 4 Cached Picture Visual Inventory.

Explanation

The six thumbnails were viewed together in Thunar. Five show indoor or workbench scenes and one shows an outdoor landscape, but the images alone do not establish when or by whom they were captured.

Android Evidence Log

Evidence ID	Source record, path or image	Tool / method used	Observed fact	Screenshot / appendix reference
E01	Android_Logical.zip	ls -lh, file, stat, sha256sum, cp --preserve=all, and cmp	The supplied Android logical acquisition was confirmed as a valid ZIP archive with an exact size of 415,154,789 bytes. Its SHA-256 value was 9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d. A separate working copy was created, and both copies produced the same hash. A byte-for-byte comparison also confirmed that the working copy matched the preserved original.	Figs 1.1-1.2
EO2	Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt	find, file, stat, sha256sum, xxd, strings, zipinfo, and unzip -Z -v	The main DJI GO 4 flight record was located in the expected FlightRecord directory. The file was 555,776 bytes with SHA-256 0a9f1816d52a98f840f6e67c08a14473ae5a0a2e75e9484d5de66368c4fd4c4a. Manual examination of the record exposed readable text including “Mavic Air”, supporting its association with a Mavic Air aircraft. ZIP metadata recorded a source modification time of 19 June 2018 at 12:55:40, although	Figs 2.1, 2.5-2.8

Evidence ID	Source record, path or image	Tool / method used	Observed fact	Screenshot / appendix reference
			the archive format does not preserve a timezone, so the time was treated cautiously during normalisation.	
E03	Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4	file, stat, sha256sum, mediainfo, exiftool, zipinfo, and controlled video preview	A cached flight video was recovered from the DJI GO 4 DJI_RECORD folder. The file measured 57,791,765 bytes and had SHA-256 2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00. MediaInfo identified it as an MPEG-4/H.264 video, approximately 4 minutes long, with a resolution of 1280 × 720. The video was previewed only after its hash and metadata had been preserved. It showed aerial footage of a rural outdoor location containing a white tent, vehicles, surrounding fields and hills, and several people. These visuals support flight-related recording but do not identify who operated the aircraft.	Figs 2.2, 2.9-4.3
E04	Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT	find, file, stat, sha256sum, zipinfo, and unzip -Z -v	The detailed flight record FLY057.DAT was located inside the MCDatFlightRecords directory. It was 4,002,791 bytes and produced SHA-256 06130820a4a813bf21f7286150bd3c83084105fecc899bb4d8534895fa2df3a1. Kali identified the file as a MATLAB v4 MAT-file, little-endian. ZIP metadata recorded a modification time of 19 June 2018 at 12:58:24. According to the course workflow, this record covers the wider power-on to power-off period and therefore extends beyond the narrower main flight and video records.	Figs 2.3, 4.4-4.5
E05	Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/	find, ls --full-time, file, sha256sum, exiftool, Unix-millisecond conversion, and controlled thumbnail preview.	Six cached JPEG thumbnails were recovered from the DJI GO 4 CACHE_IMAGE directory. Their SHA-256 values were preserved in Cached_Images_SHA256.txt. ExifTool showed that the files were small JPEG thumbnails without useful original camera date or GPS metadata. The 13-digit timestamps contained in the filenames converted to dates between 13 April and 19 April 2018, rather than the June 19 flight date. Visually, five thumbnails showed indoor/workbench scenes while one showed an outdoor landscape. For that reason, these cached thumbnails cannot reliably establish that photographs were captured during the June 19 flight or identify the person operating the drone.	Figs 2.4, 4.6-5.2

The original Android evidence package was preserved without alteration, while examination was carried out on a verified working copy. Hash values and file metadata were recorded before opening or previewing relevant media. Where extracted filesystem timestamps differed from source archive information, the original ZIP metadata was checked separately. Times without a reliable timezone were not treated as UTC automatically, and filename timestamps were not relied on as proof unless supported by additional evidence.

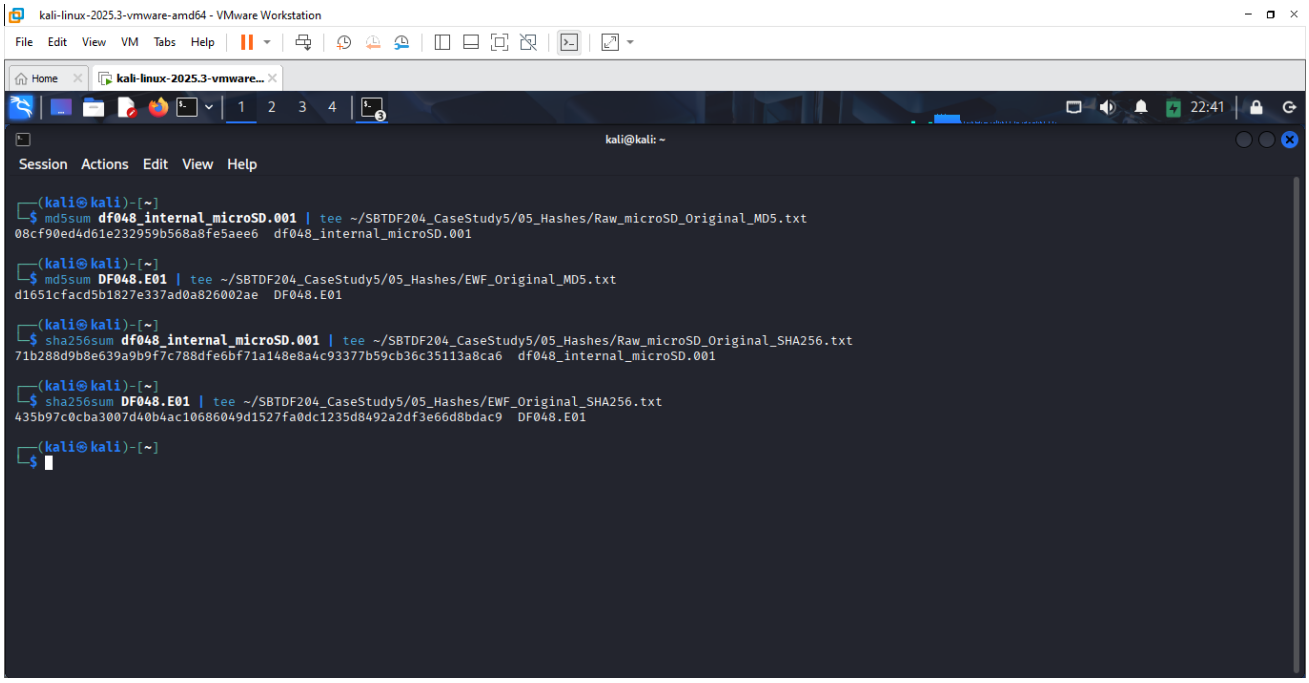


Fig 5.3 Raw microSD and EWF Original MD5 and SHA-256 Verification.

Explanation

MD5 and SHA-256 values were calculated for the original raw microSD and EWF files. The MD5 values matched the supplied reference checks, while SHA-256 was retained as the primary integrity record for the assignment.

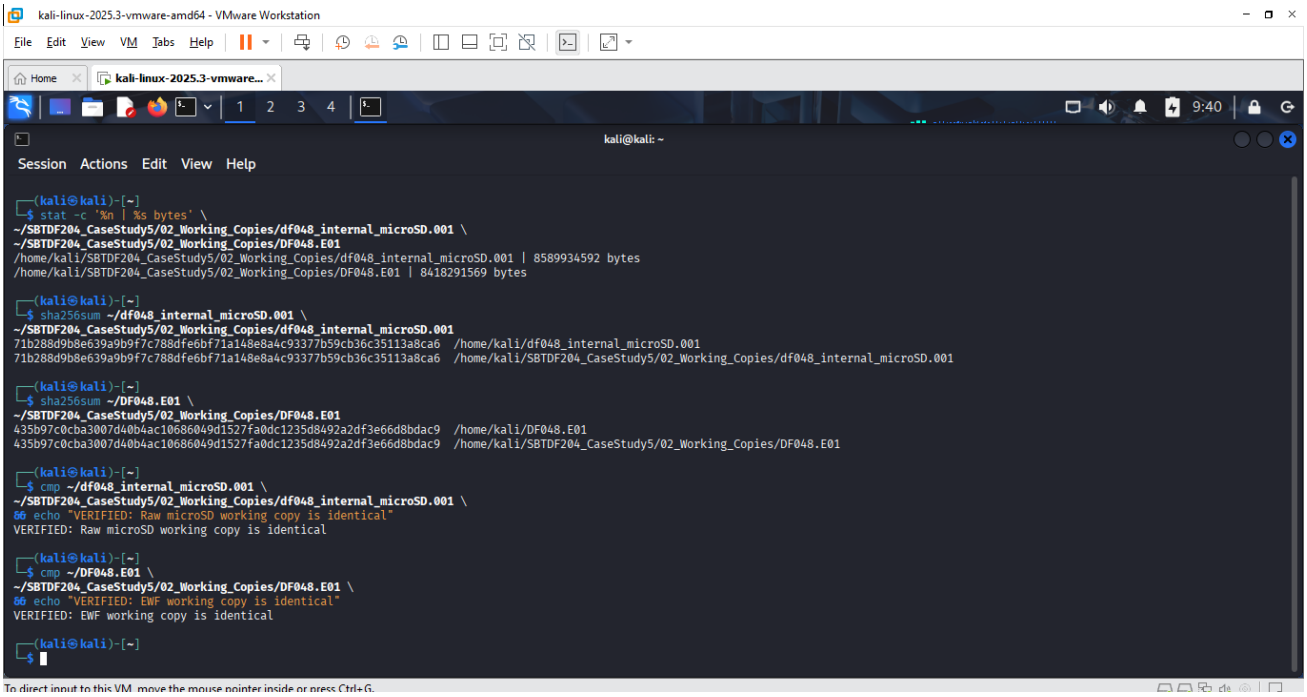


Fig 5.4 Raw microSD and EWF Working Copy Integrity Verification.

Explanation

The original and working microSD evidence files were compared by size, hashes, and byte-for-byte verification. Both working copies were confirmed as faithful copies before analysis began.

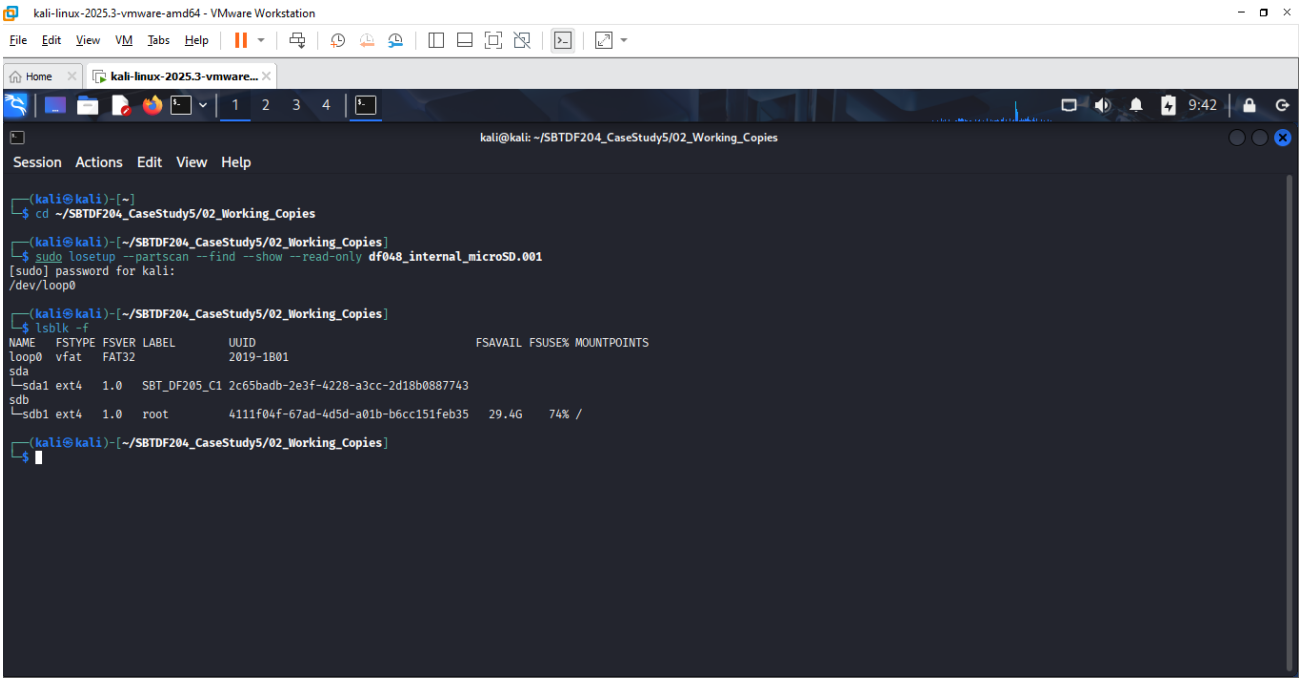


Fig 5.5 Raw microSD Read-Only Loop Device Identification.

Explanation

The raw .001 image was attached as /dev/loop0 using a read-only loop-device command. lsblk identified the exposed volume as FAT32 while the read-only setting protected the evidence from writes.

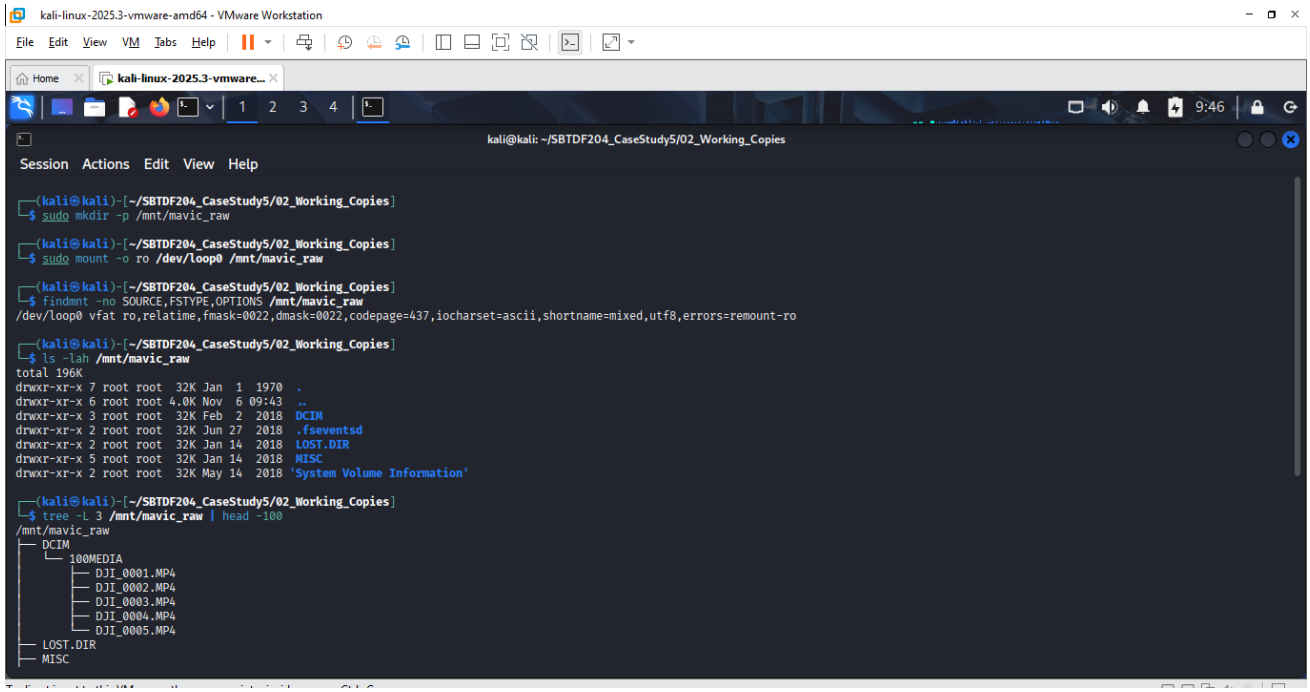


Fig 5.6 Raw microSD Read-Only Mount and Filesystem Structure.

Explanation

The FAT32 volume was mounted read-only and the mount options were verified. The root directory included

DCIM, MISC, LOST.DIR, and System Volume Information.

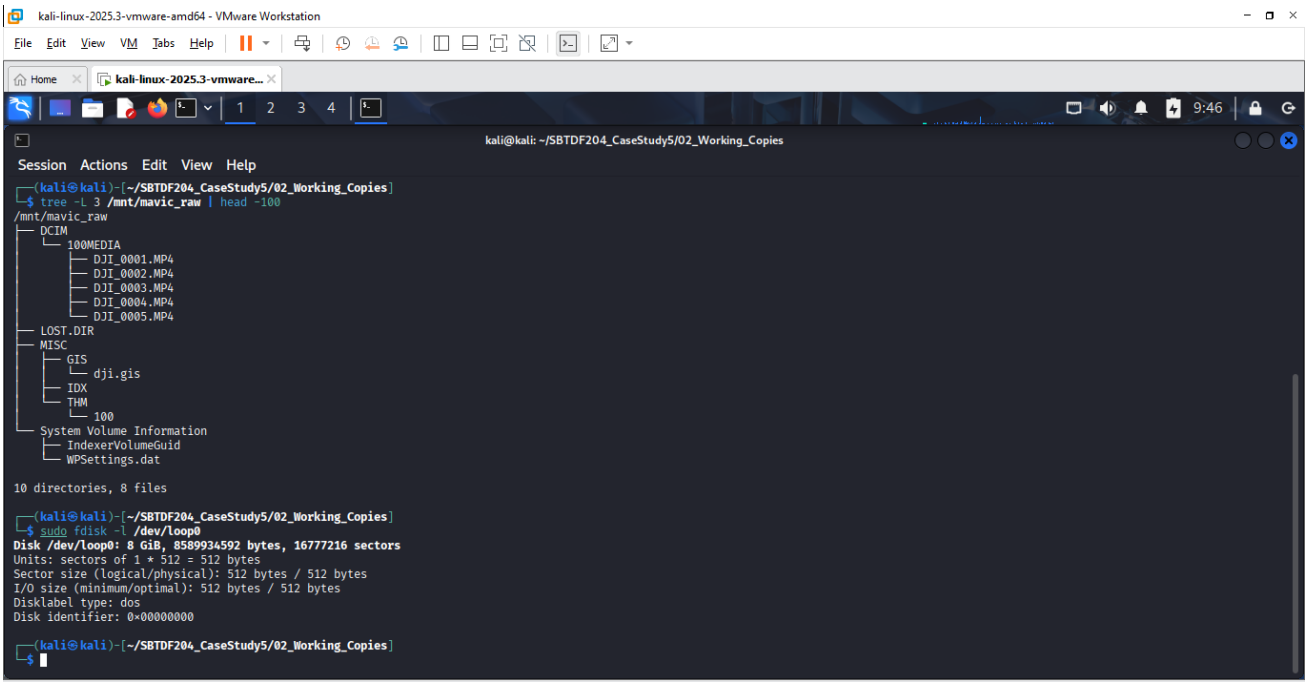


Fig 5.7 Raw microSD Read-Only Mount and Filesystem Structure Page 1.

Explanation

The directory tree shows five MP4 files in DCIM/100MEDIA together with thumbnail and system-volume artefacts. fdisk also records the 8 GiB media size and 512-byte sector size.

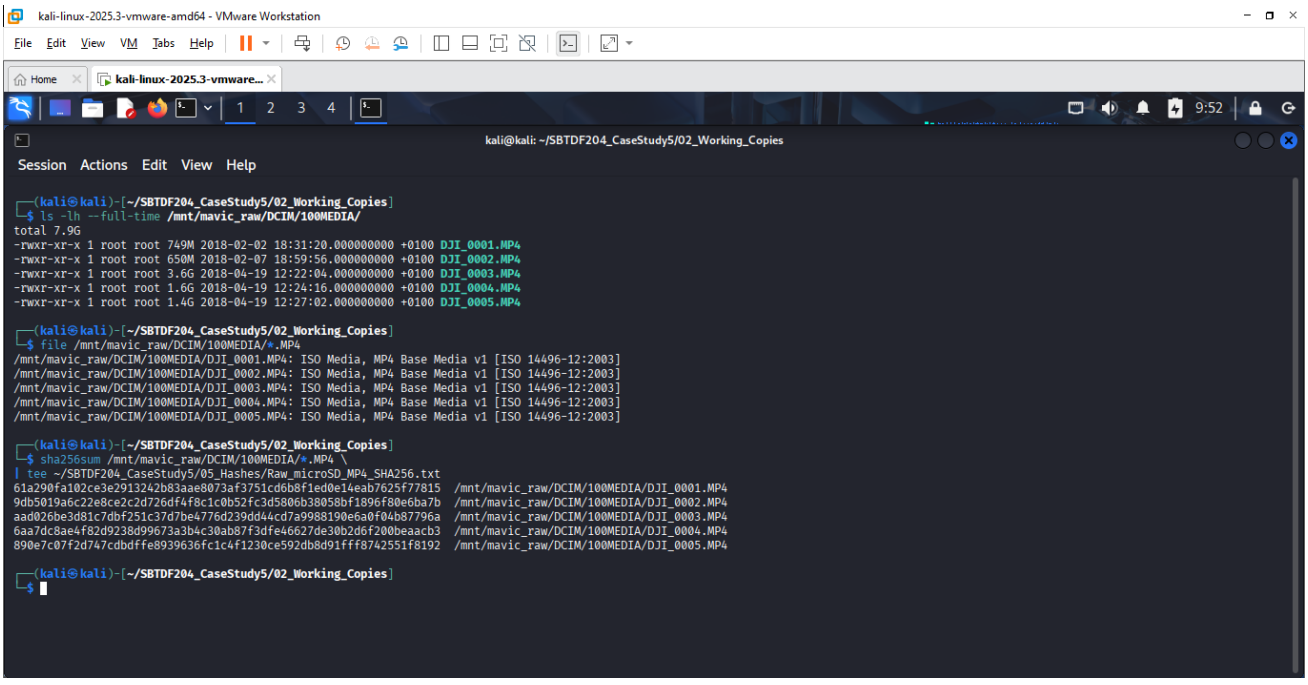


Fig 5.8 Raw microSD Video File Inventory and Timestamps.

Explanation

The five DJI MP4 recordings were listed with their sizes and filesystem timestamps. Their formats were

verified before media hashes were calculated.

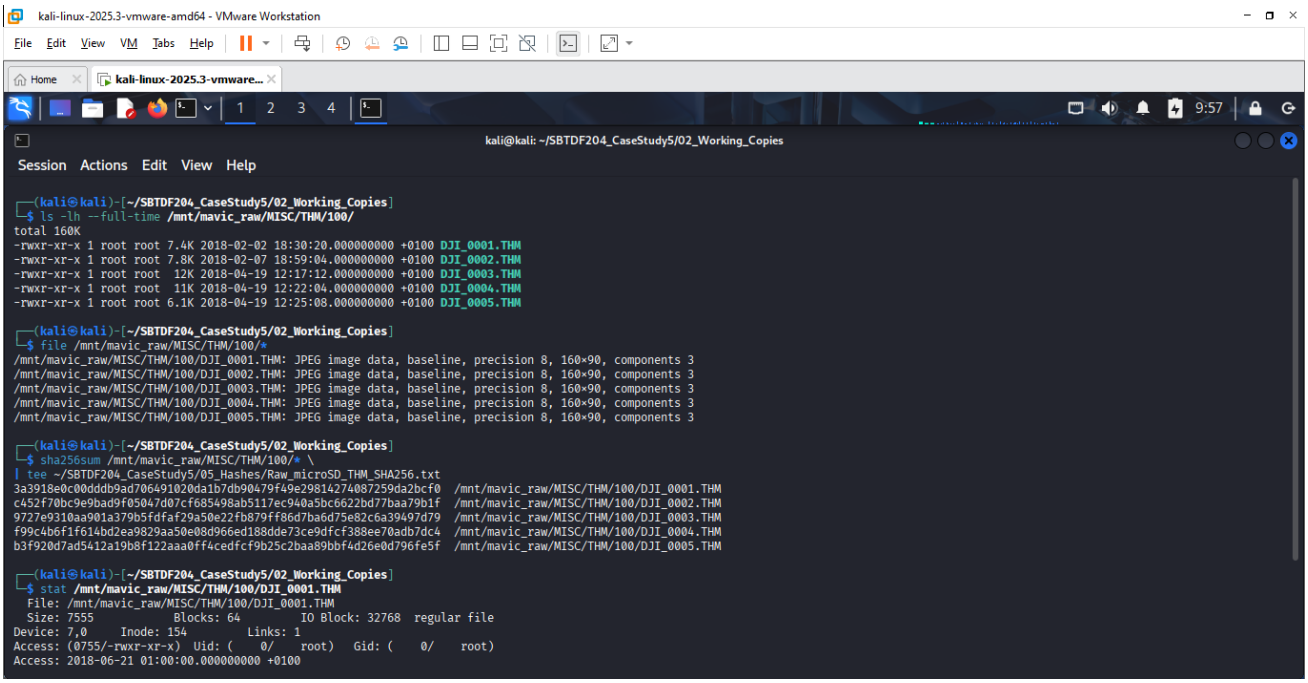


Fig 5.9 Raw microSD Video SHA-256.

Explanation

SHA-256 values were preserved for the raw microSD videos. These values later became the main file-level comparison against the EWF evidence.

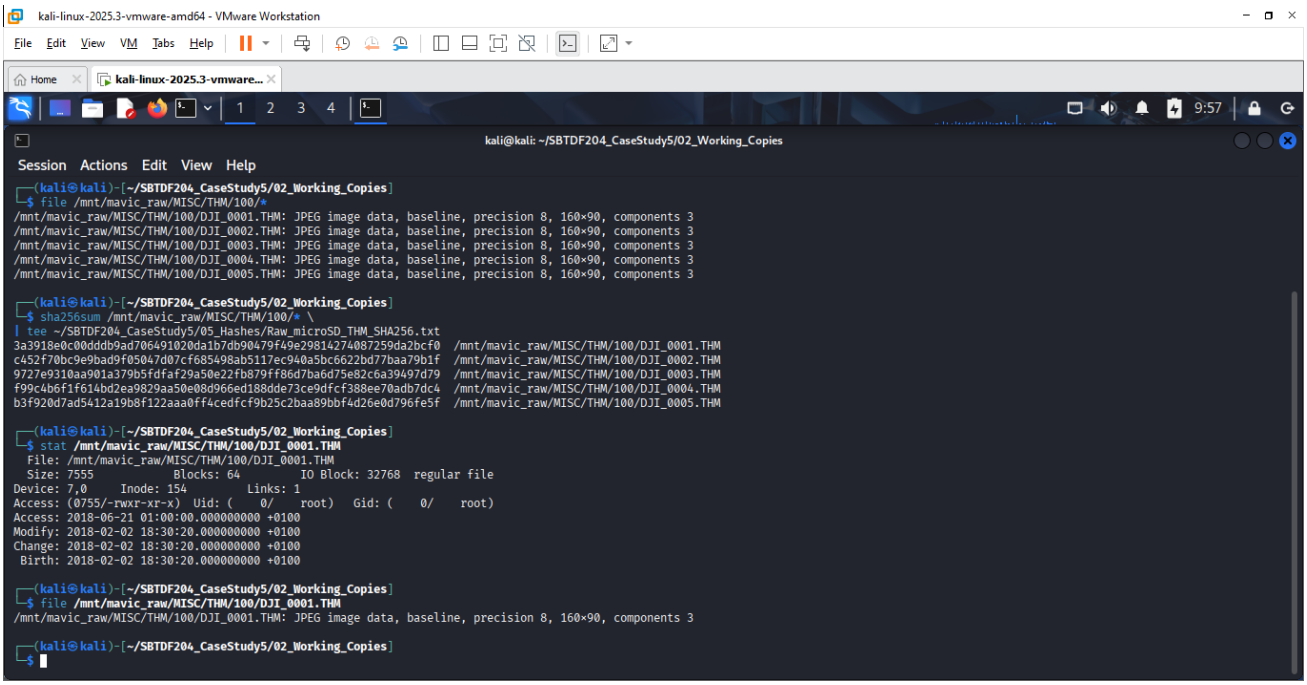


Fig 6.0 Raw microSD Thumbnail Evidence Inventory.

Explanation

The THM files were identified as JPEG image data rather than unknown binary files. Their hashes and

metadata were preserved so the thumbnail artefacts could be independently verified.

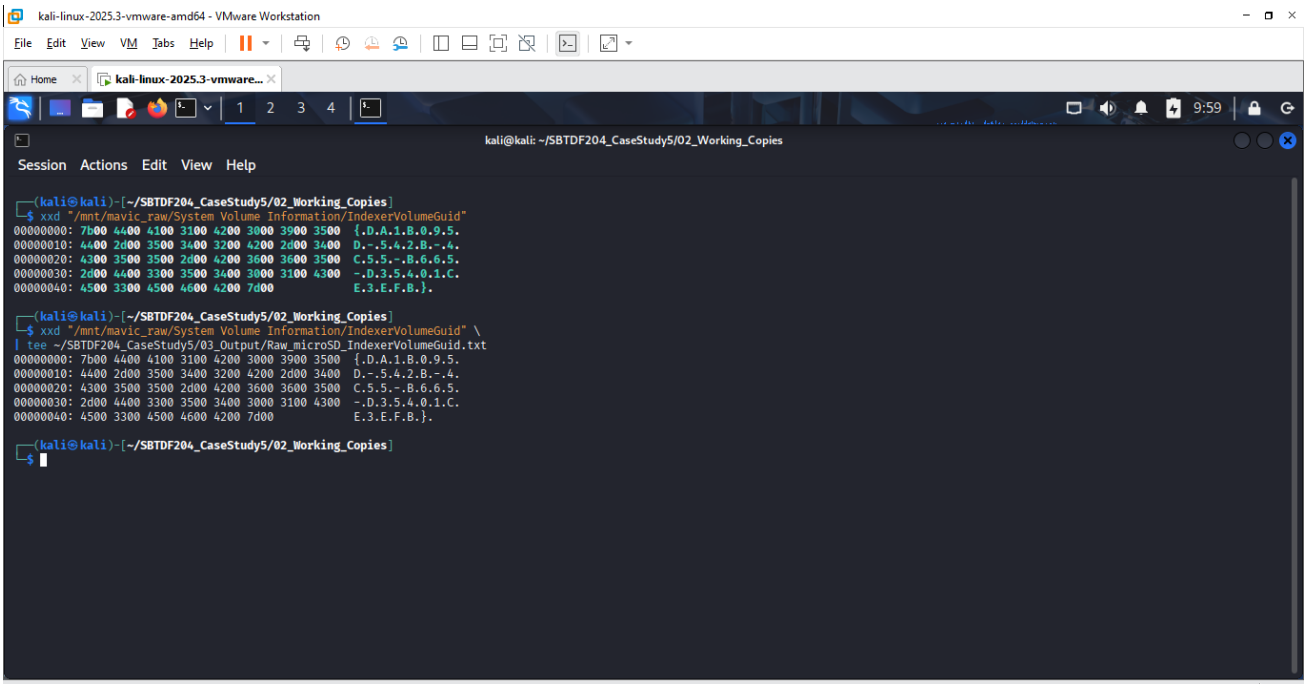


Fig 6.1 Raw microSD Volume GUID Identification.

Explanation

IndexerVolumeGuid was read with xxd to expose the microSD volume GUID. This identifier later provided an independent comparison point with the EWF representation.

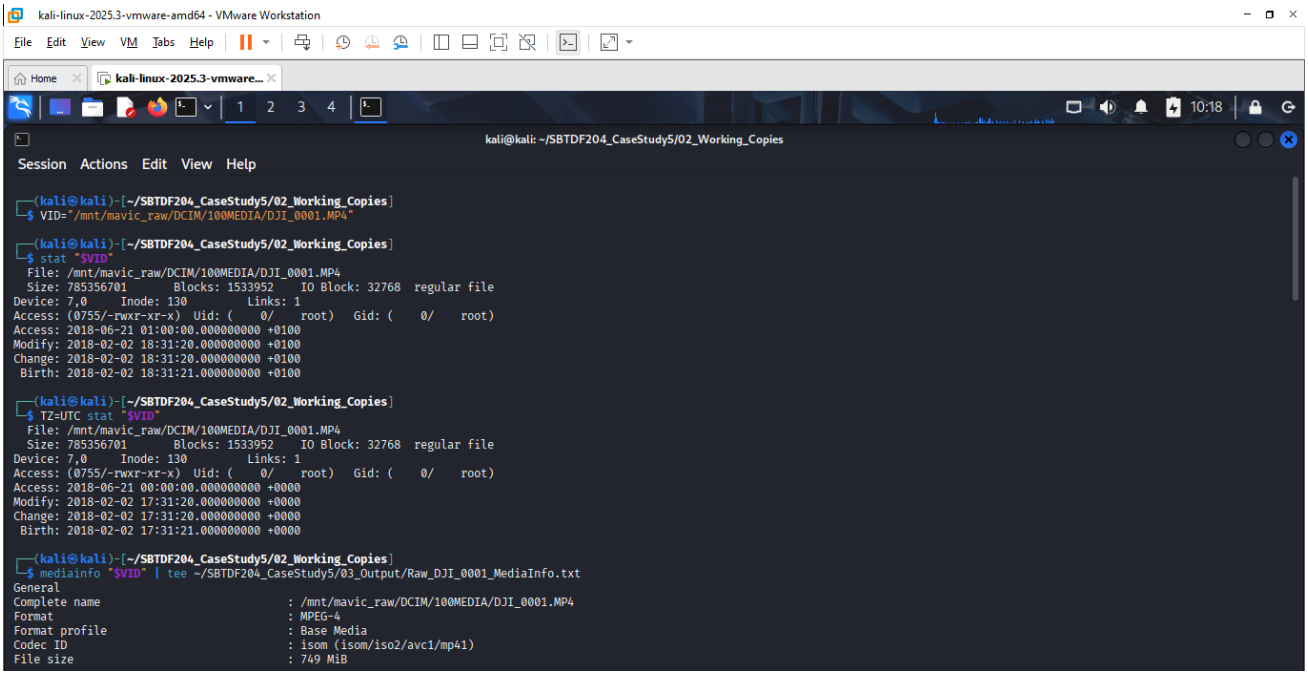


Fig 6.2 Raw microSD DJI 0001 Video Metadata and UTC Time.

Explanation

DJI_0001.MP4 was examined with stat and its filesystem time was also displayed explicitly in UTC. MediaInfo

examination began only after the basic metadata had been preserved.

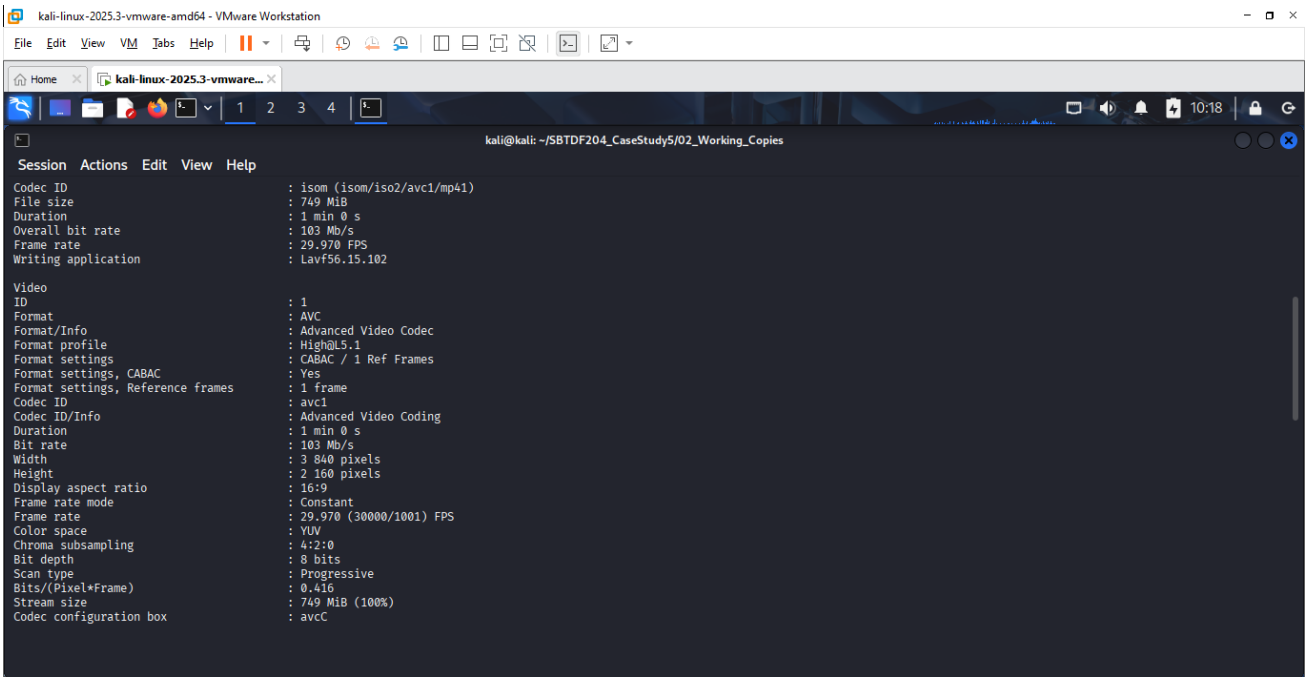


Fig 6.3 Raw microSD DJI 0001 Video Metadata and UTC Time Page 1.

Explanation

MediaInfo identifies the recording as a one-minute AVC/H.264 video at 3840 x 2160. The output also records frame rate, bitrate, and other technical details.

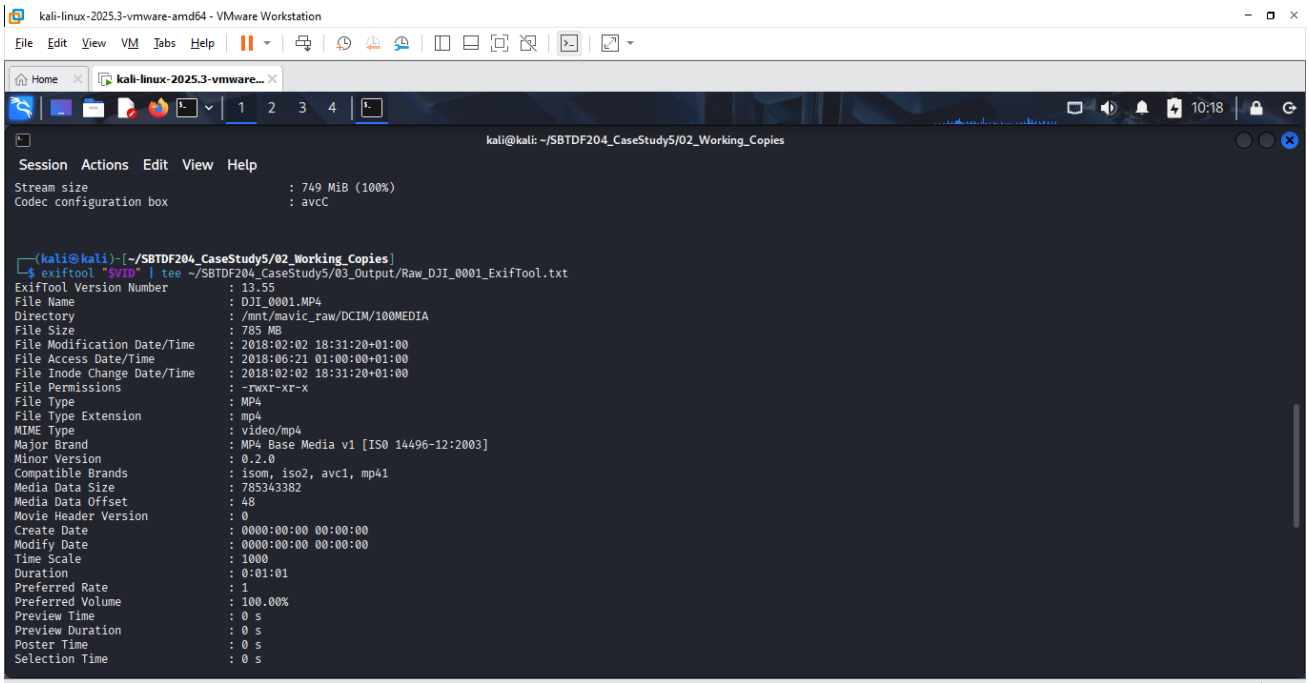


Fig 6.4 Raw microSD DJI 0001 Video Metadata and UTC Time Page 2.

Explanation

ExifTool independently examined the MP4 container and filesystem information. Several embedded creation-

date fields were zeroed, making the verified filesystem time more useful for chronology.

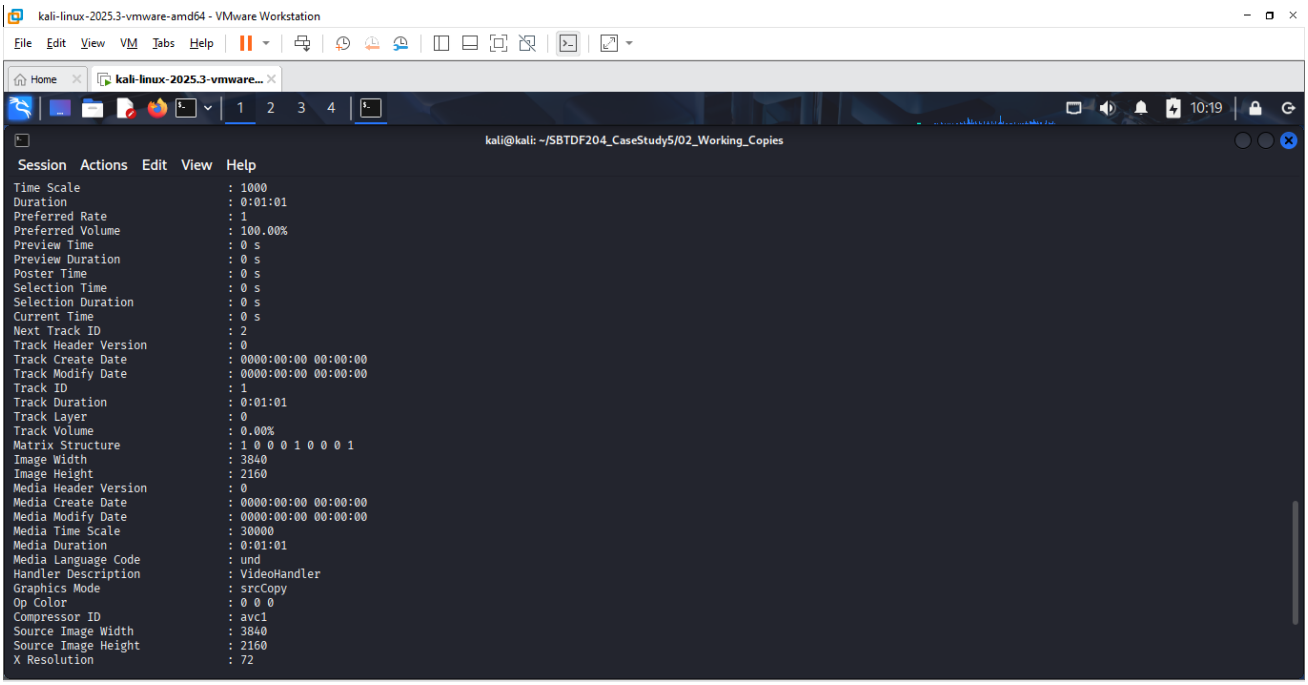


Fig 6.5 Raw microSD DJI 0001 Video Metadata and UTC Time Page 3.

Explanation

The ExifTool output continues through track and media-level properties. These fields document the internal video structure without changing the file.

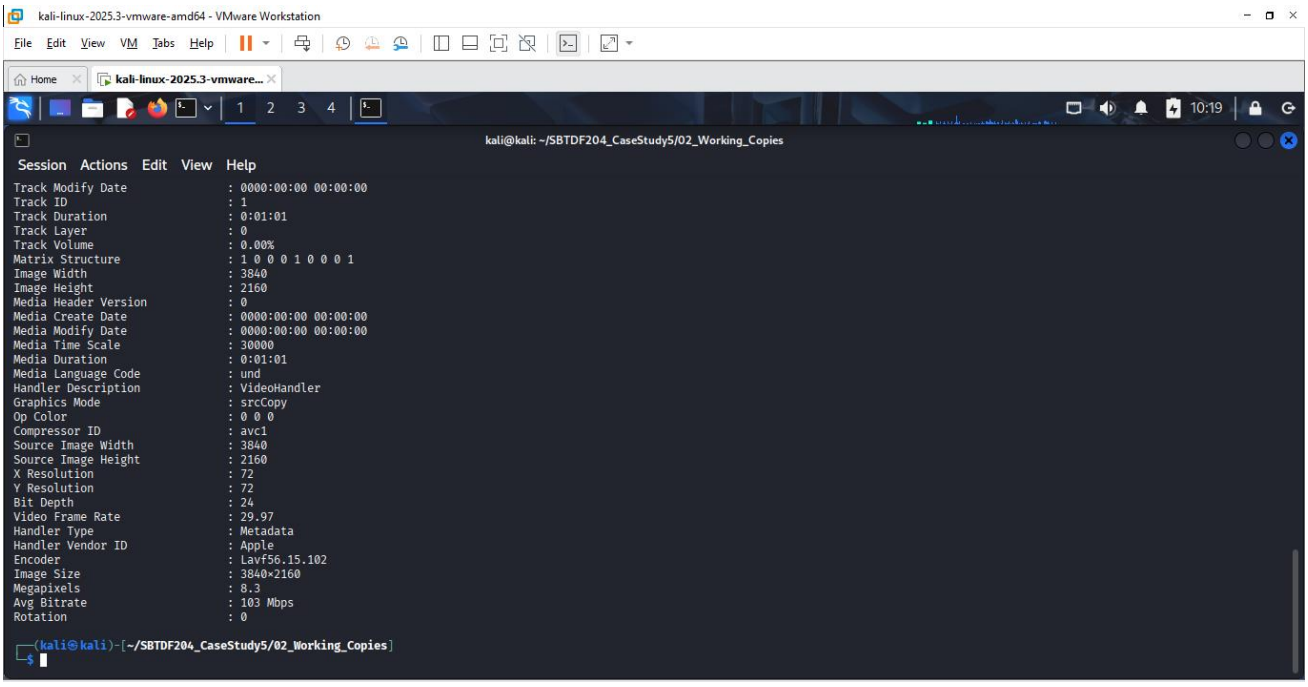


Fig 6.6 Raw microSD DJI 0001 Video Metadata and UTC Time Page 4.

Explanation

The final metadata fields confirm the 3840 x 2160 image size, approximate 29.97 fps frame rate, and AVC

encoding. This completed the non-visual examination before preview.

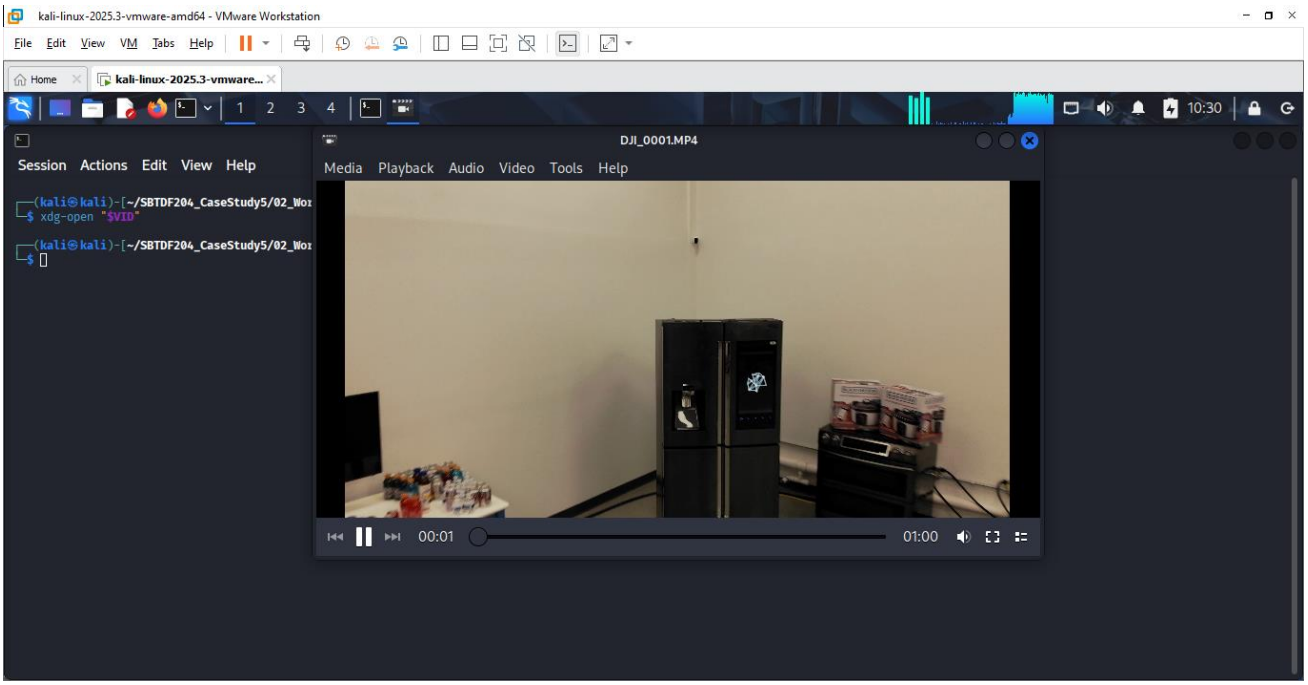


Fig 6.7 Raw microSD DJI 0001 Video Preview.

Explanation

The beginning of DJI_0001.MP4 shows an indoor laboratory or workshop containing equipment. Its visible setting is clearly different from the outdoor Android cached flight video.

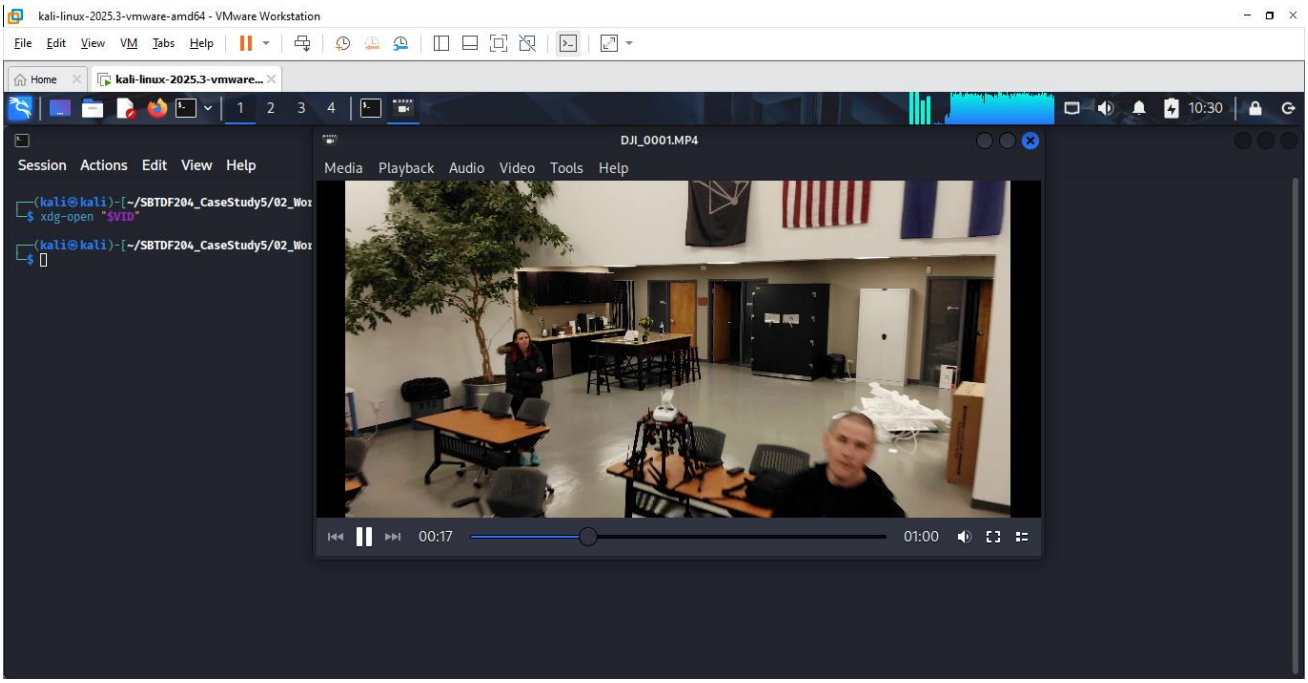


Fig 6.8 Raw microSD DJI 0001 Video Preview page 1.

Explanation

Two people are clearly visible in the indoor drone-laboratory scene alongside equipment and worktables.

Their presence was recorded as a visual observation without assigning an operator identity.

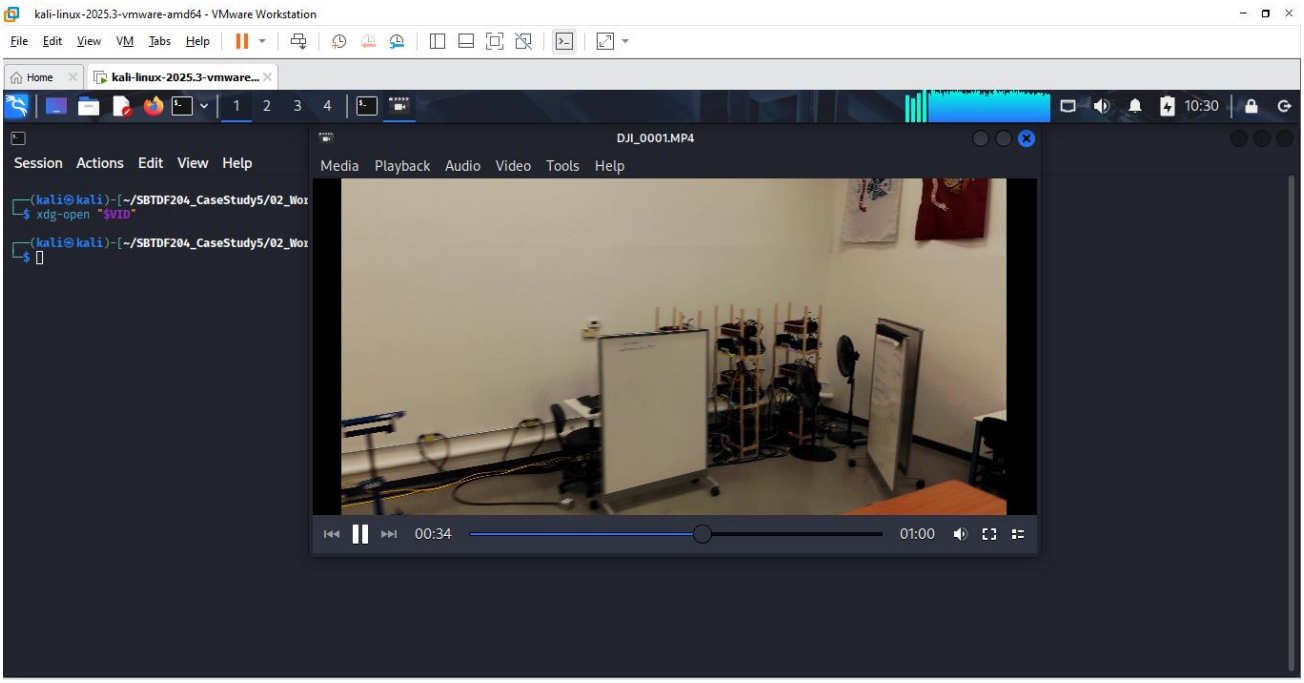


Fig 6.9 Raw microSD DJI 0001 Video Preview page 2.

Explanation

The video continues through another area of the room containing equipment and boards. This confirms that the clip mainly records an indoor technical environment.

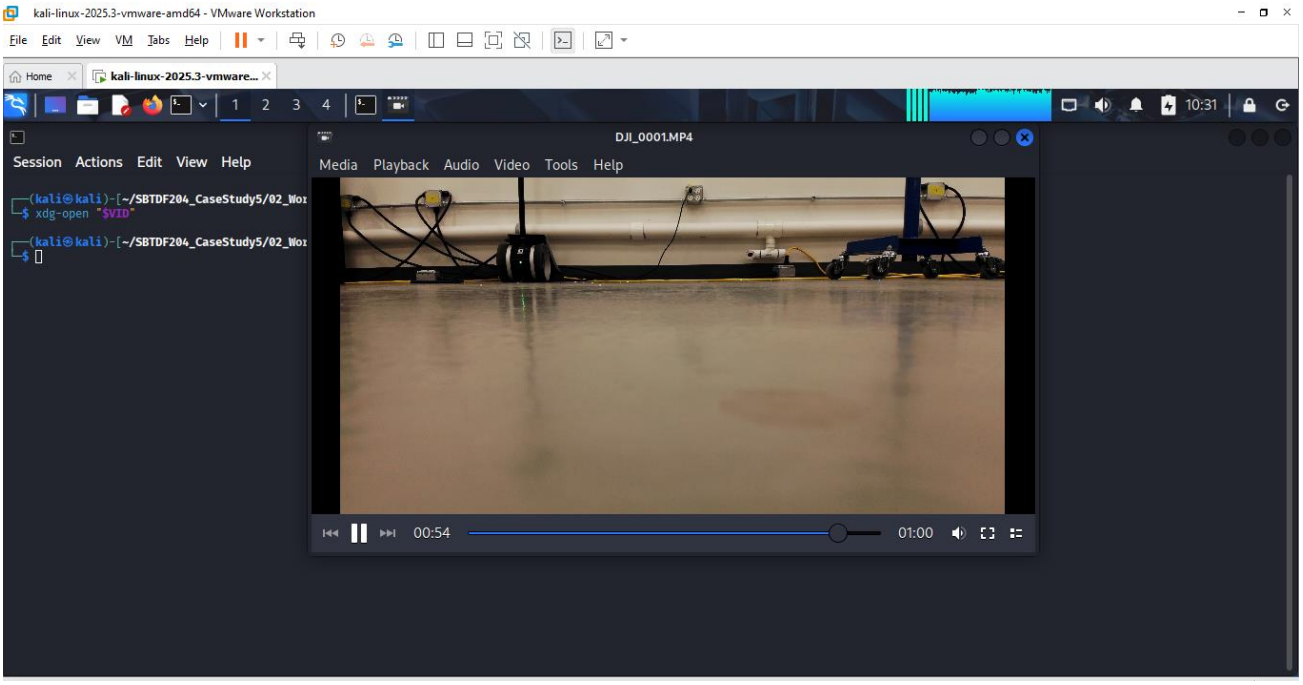


Fig 7.0 Raw microSD DJI 0001 Video Preview page 3.

Explanation

Near the end of the clip, the camera faces the floor and nearby equipment. This completes the preserved visual sample from the one-minute recording.

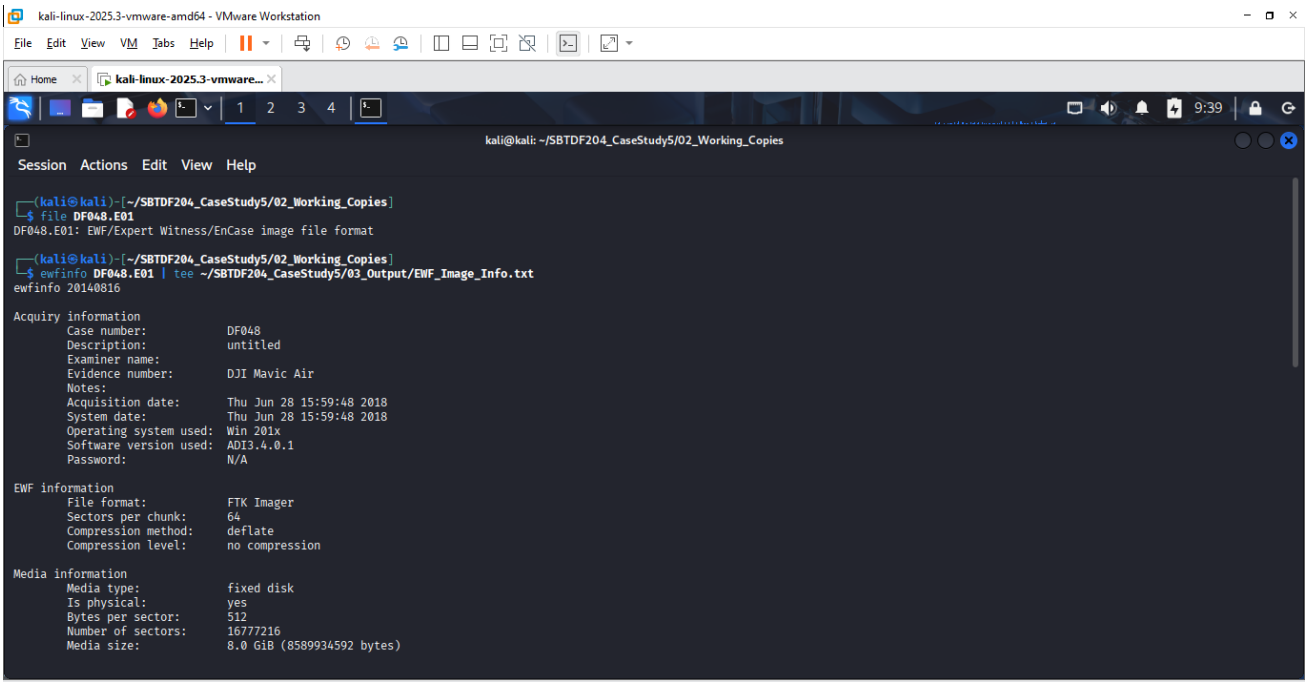


Fig 7.1 EWF Image Format and Acquisition Information.

Explanation

DF048.E01 was identified as an EWF/EnCase image and ewfinfo recorded its acquisition details. The image represents 8 GiB of media with 512-byte sectors and was created using FTK Imager.

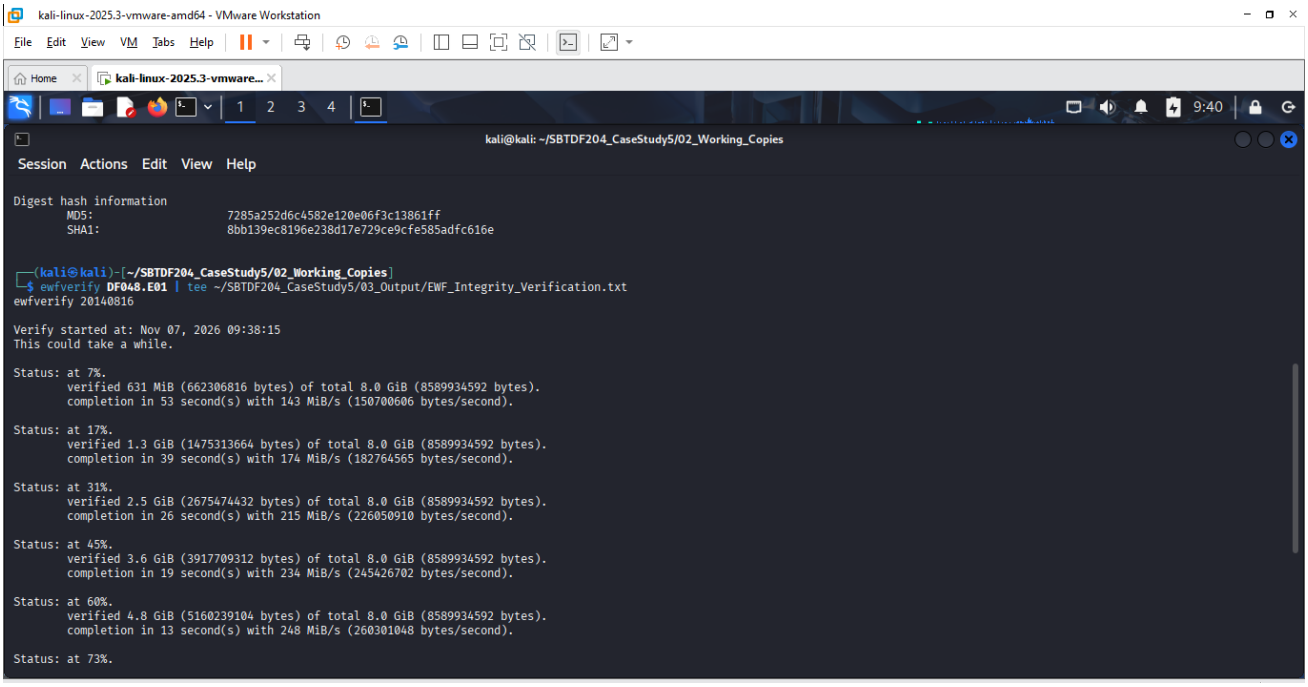


Fig 7.2 EWF Image Integrity Verification.

Explanation

ewfverify began reading and checking the complete EWF evidence data. The progress output documents that verification covered the full 8 GiB media.

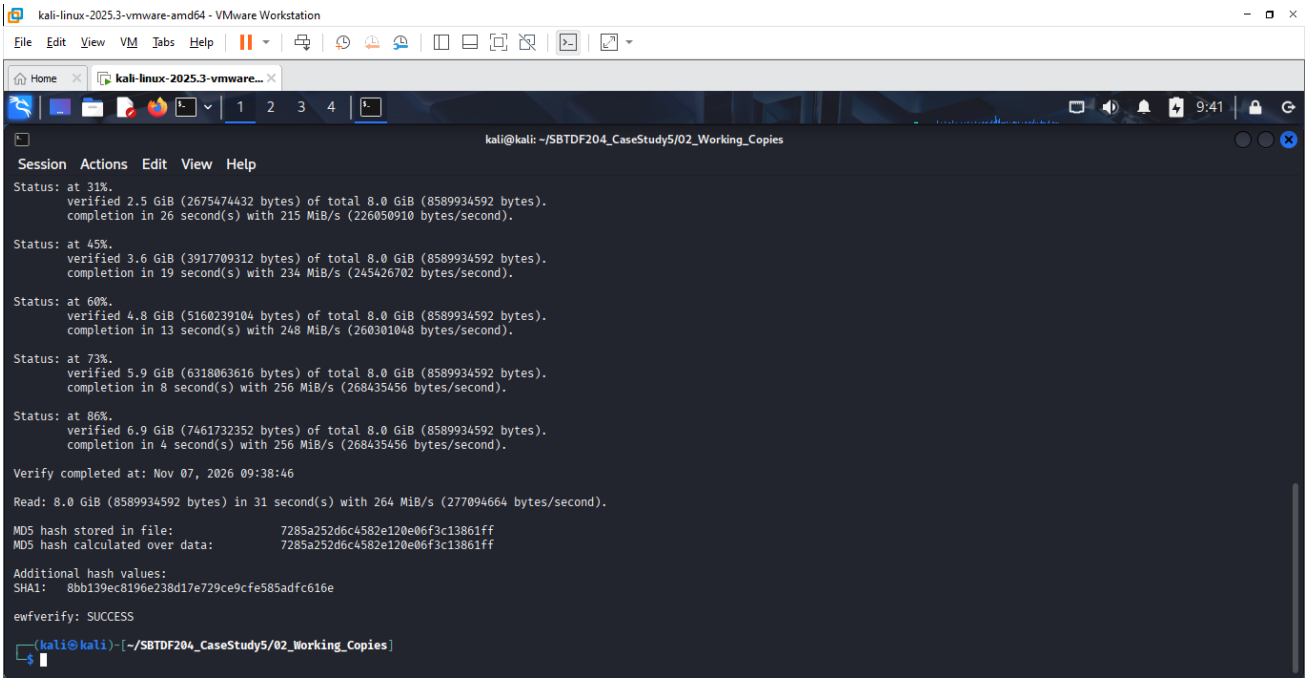


Fig 7.3 EWF Image Integrity Verification page 1.

Explanation

The verification completed successfully and the internally stored and calculated integrity values agreed. This confirms that the EWF passed its own integrity check.

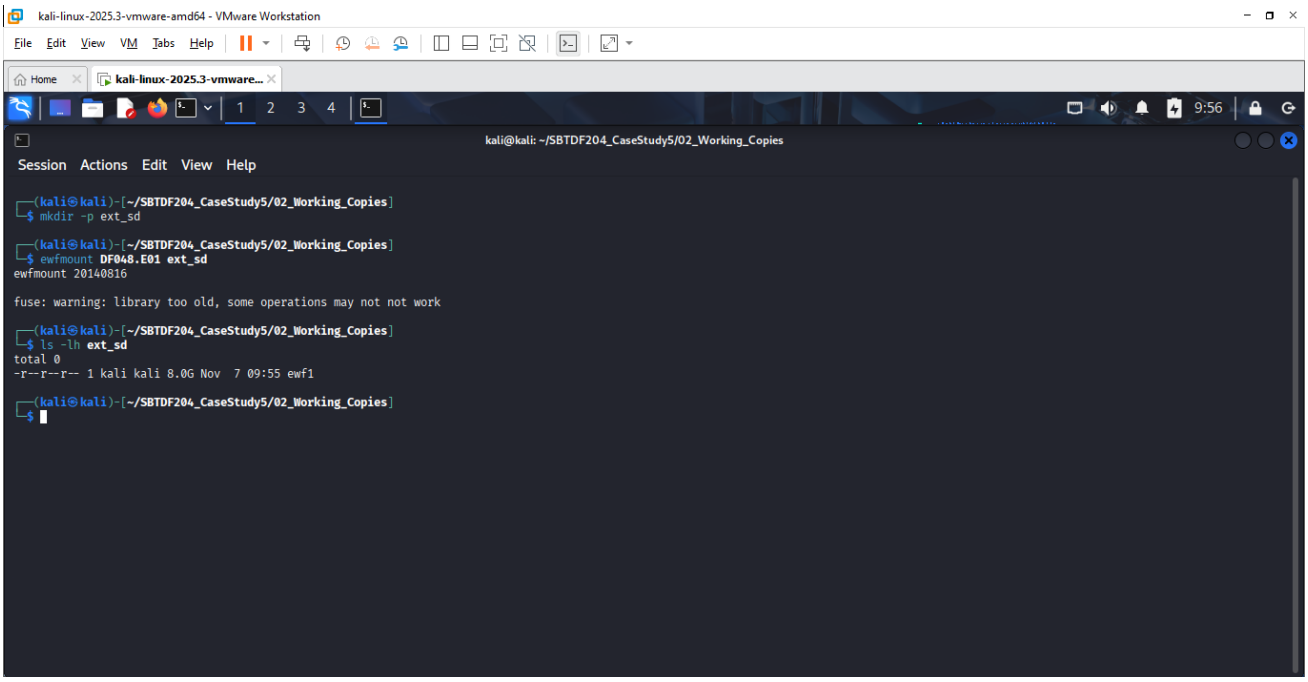


Fig 7.4 EWF Exposed Media Image Identification.

Explanation

ewfmount exposed the EWF evidence through the ext_sd directory as the 8 GiB ewf1 media file. This allowed the filesystem to be examined with forensic tools without modifying the EWF container.

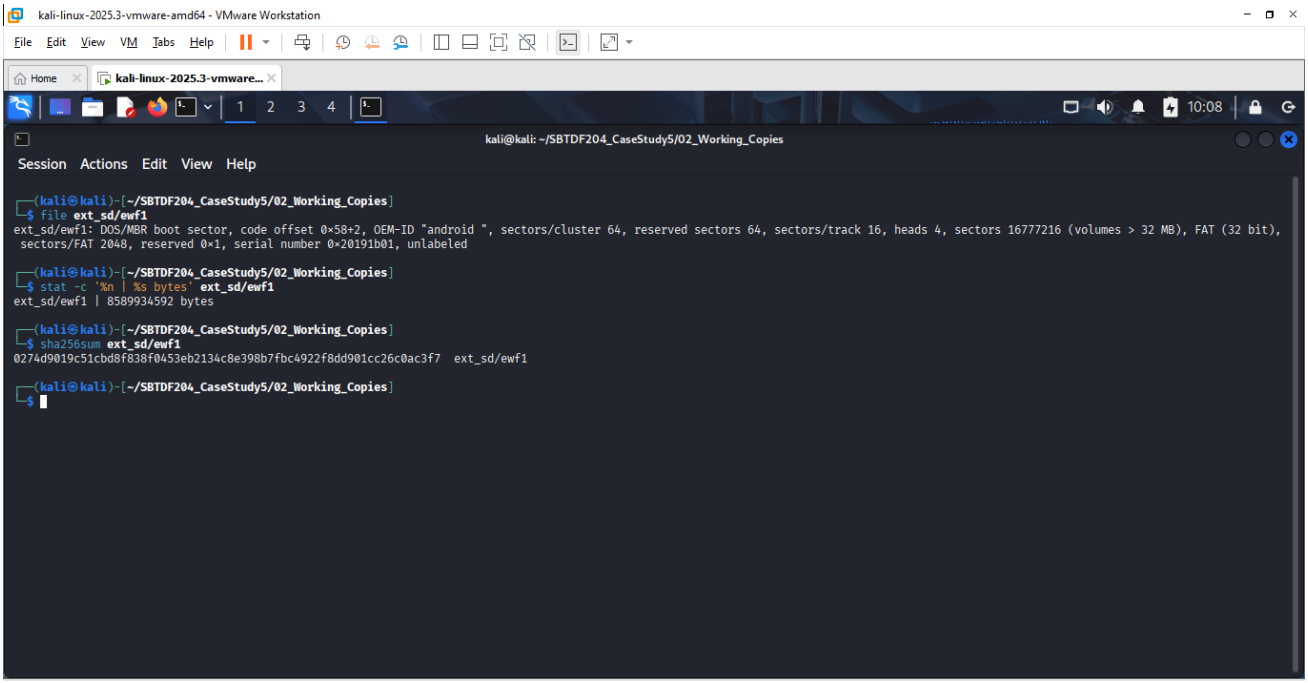


Fig 7.5 EWF Exposed Media Size Serial and SHA-256.

Explanation

The exposed ewf1 data was identified as FAT32, measured exactly 8,589,934,592 bytes, and had its SHA-256 recorded. The FAT volume serial was also visible as 0x20191b01.

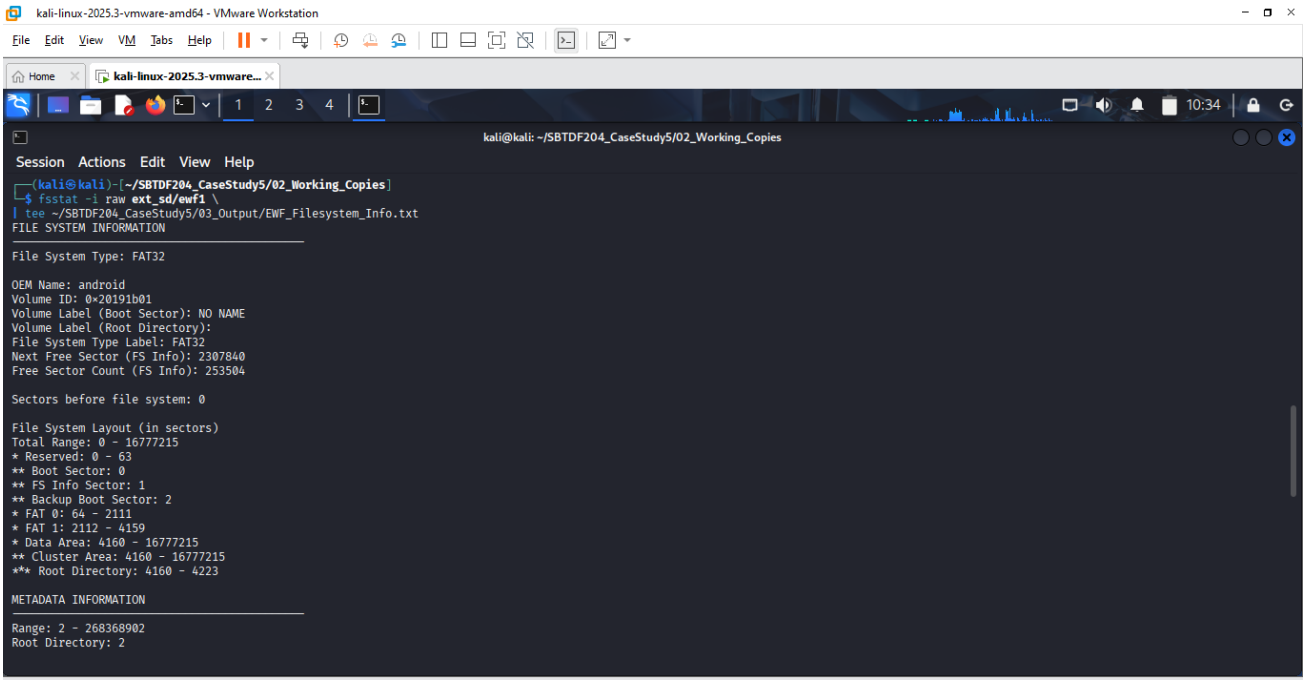


Fig 7.6 EWF FAT32 Filesystem Information.

Explanation

fsstat identified the exposed media as FAT32 and recorded its volume ID, sector layout, FAT locations, and root-directory information. This established the filesystem structure independently of the raw-image mount.

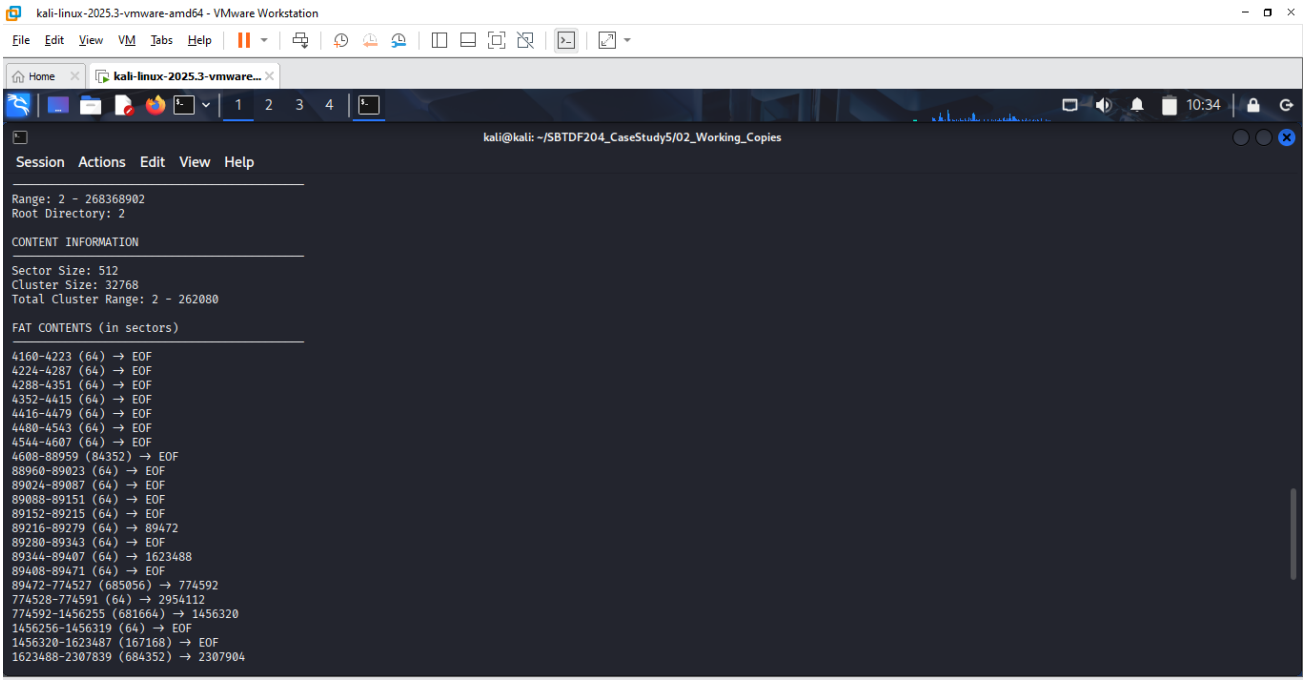


Fig 7.7 EWF FAT32 Filesystem Information page 1.

Explanation

The continuation shows FAT allocation chains and cluster relationships within the EWF filesystem. These details document how the media is organised at filesystem level.

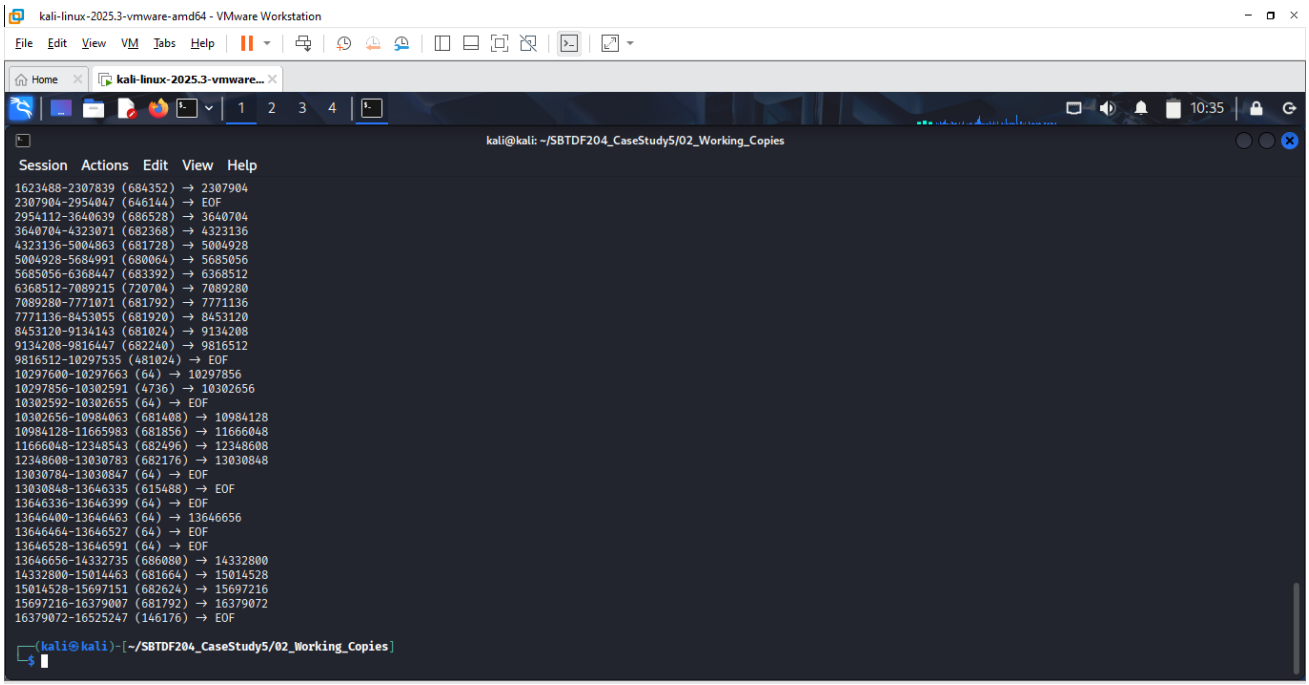


Fig 7.8 EWF FAT32 Filesystem Information page 2.

Explanation

Additional FAT chain information continues through the allocation map. This forms part of the technical record of the EWF media structure.

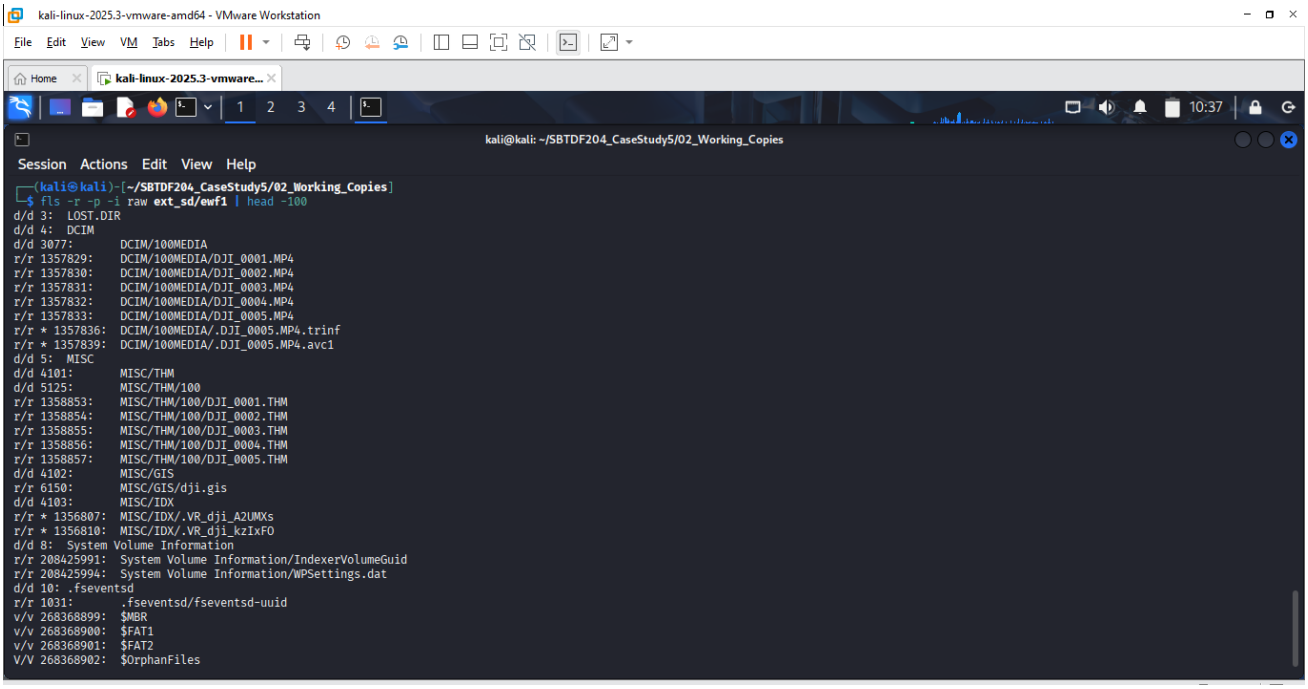


Fig 7.9 EWF FAT32 Filesystem Information.

Explanation

fls produced a recursive forensic listing of the EWF filesystem. The listing includes the same five DJI MP4 files, THM images, and system-volume artefacts seen in the raw microSD image.

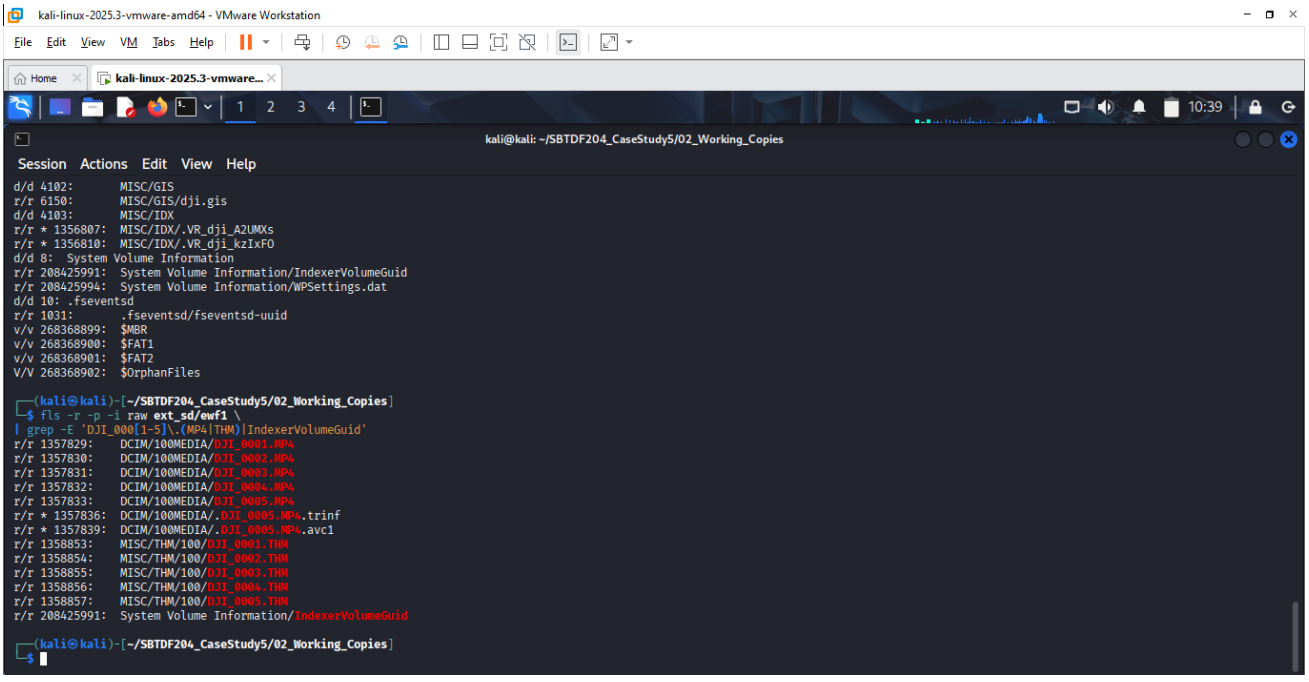


Fig 8.0 EWF Media File Inventory with Inode References.

Explanation

The fls results were filtered to show the five DJI videos, five THM files, and IndexerVolumeGuid with their

metadata addresses. These inode references allowed the files to be examined directly without a normal filesystem mount.

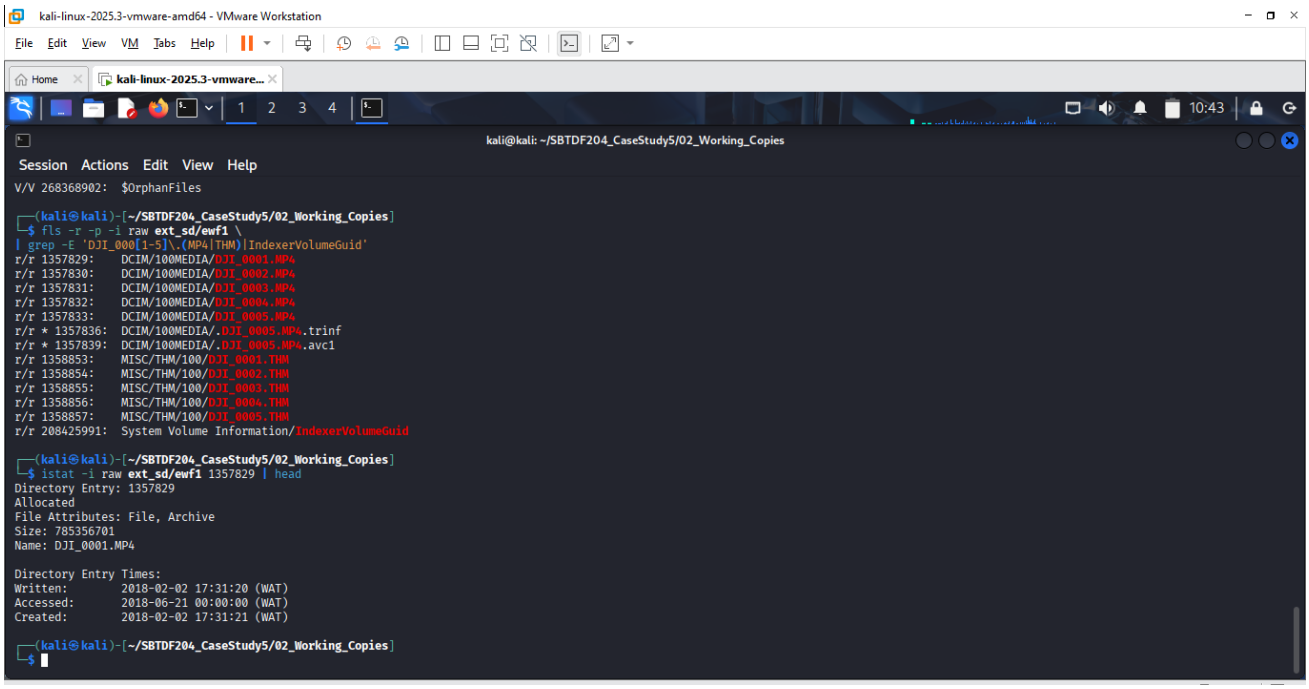


Fig 8.1 EWF DJI 0001 Video Timestamp Metadata.

Explanation

istat examined inode 1357829, identified as DJI_0001.MP4, and displayed its filesystem times. This preserved the file-level timestamp evidence from the EWF representation.

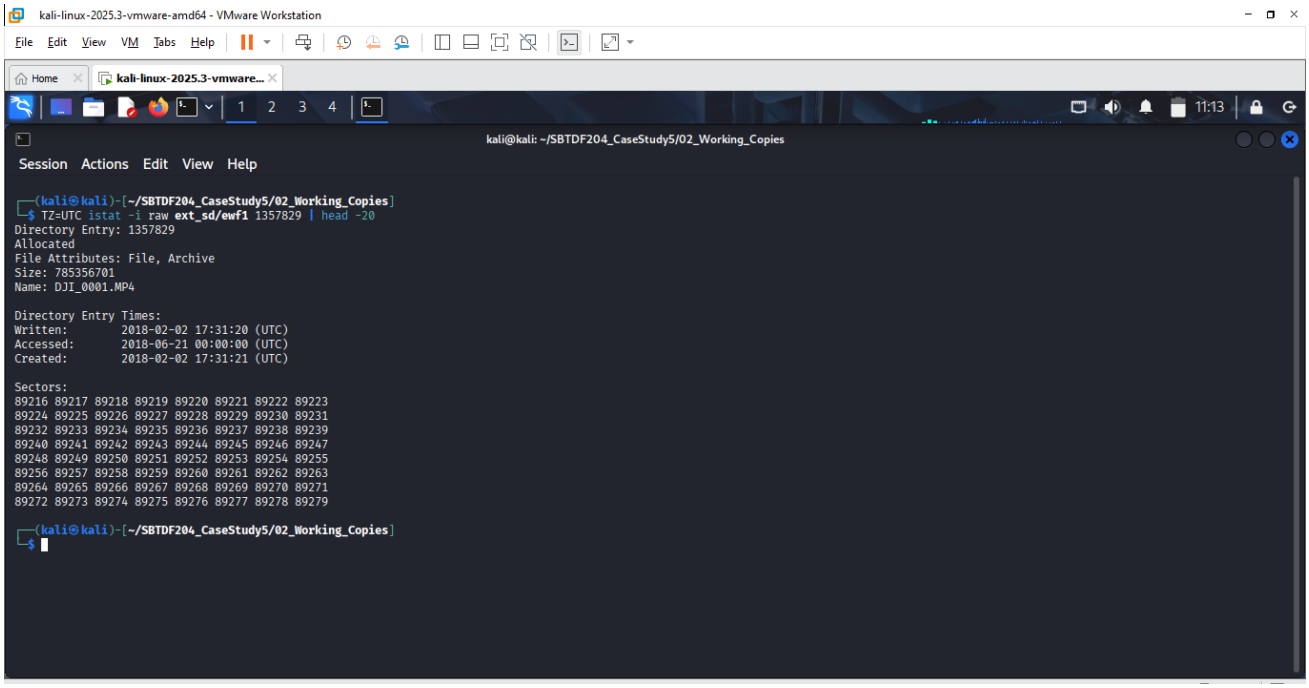


Fig 8.2 EWF DJI 0001 Video UTC Timestamp.

Explanation

The same istat result was displayed using UTC, giving a written time of 02 February 2018 at 17:31:20 UTC. This matched the value established from the raw microSD examination.

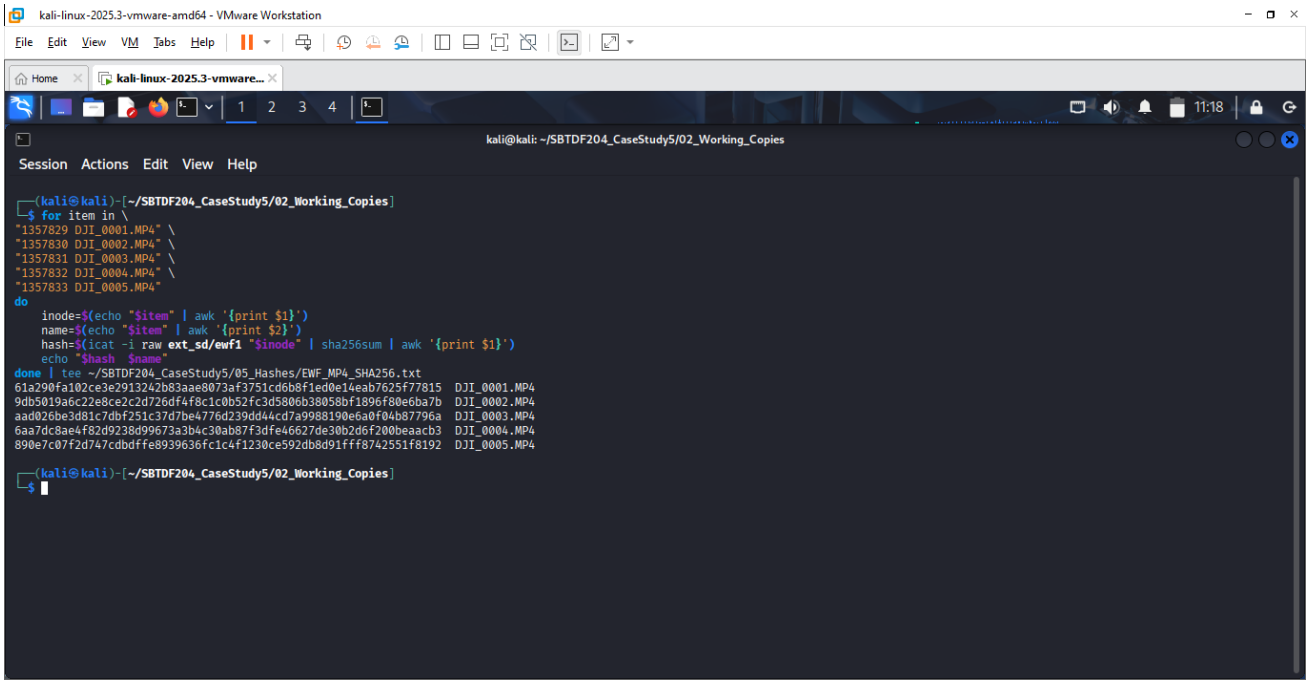


Fig 8.3 EWF Video SHA-256 Hashes.

Explanation

Each of the five MP4 files was read directly from the EWF with icat and hashed with SHA-256. All five values matched their corresponding videos from the raw microSD image, providing strong file-level correlation.

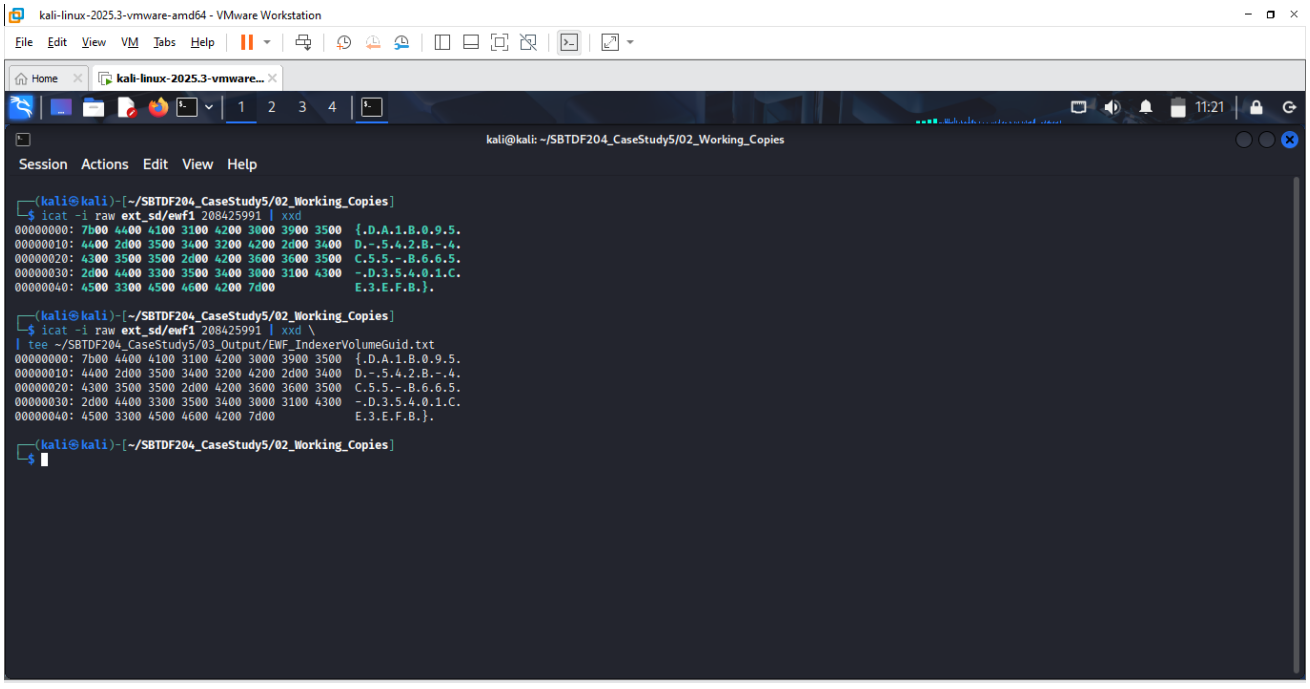


Fig 8.4 EWF Volume GUID Identification.

Explanation

The EWF copy of IndexerVolumeGuid was read directly with icat and displayed using xxd. The recovered GUID matched the raw image, providing another independent link between the two media representations.

Investigation Question 9 - Final UTC Timeline and Cross-Source Correlation

The evidence forms two distinct time groups. The raw microSD and EWF media contain recordings mainly from February and April 2018, while the Android DJI GO 4 artefacts relate to activity on 19 June 2018. These dates were not merged into one flight event, and June values without a trustworthy timezone are shown as source times rather than forced into UTC.

Time	Source	Artefact / Event	Evidential treatment
2018-02-02 17:31:20 UTC	Raw microSD + EWF	DJI_0001.MP4 written time	Direct UTC-normalised filesystem time. The raw and EWF values matched.
2018-02-07 17:59:56 UTC	Raw microSD	DJI_0002.MP4	Converted from the displayed +0100 filesystem time.
2018-04-13 09:16:28-09:24:54 UTC	Android cache	Five cached-picture filename times	Filename-derived cache times only; not proof of a flight or capture event.
2018-04-19 11:22:04 UTC	Raw microSD	DJI_0003.MP4	Converted from the displayed +0100 filesystem time.
2018-04-19 11:24:16 UTC	Raw microSD	DJI_0004.MP4	Converted from the displayed +0100 filesystem time.
2018-04-19 11:27:02 UTC	Raw microSD	DJI_0005.MP4	Converted from the displayed +0100 filesystem time.
2018-04-19 15:44:16 UTC	Android cache	Final cached-picture filename time	Filename-derived cache time only; no reliable link to the June flight.
2018-06-19 12:55:40 source time	Android ZIP	Main flight record modification time	Timezone was not preserved reliably, so this was not labelled UTC.
2018-06-19 12:58:24 source time	Android ZIP	FLY057.DAT modification time	Timezone unresolved. The DAT represents a wider power-on to power-off scope.
2018-06-19 14:50:34 source filename time	Android DJI GO 4	DJIFlightRecord_2018-06-19_[14-50-34].txt	Useful source time, but the timezone was not established.
2018-06-19 14:50:39 source filename time	Android DJI GO 4	2018_06_19_14_50_39.mp4	Useful source time for the cached video, but not safely convertible to UTC from the available evidence.

Cross-Source Correlation

Raw microSD and EWF: This is the strongest correlation in the case. Both represented an 8 GiB FAT32 volume with volume ID 0x20191b01, the same five DJI MP4 filenames, the same THM files, and the same IndexerVolumeGuid. All five MP4 SHA-256 values matched across the raw and EWF sources, and DJI_0001.MP4 showed the same 2018-02-02 17:31:20 UTC written time.

Whole-media difference: The full raw .001 image and the exposed EWF media produced different whole-image SHA-256 values, so they were not described as byte-for-byte identical. Their relationship was established instead through matching filesystem identity, GUID, filenames, individual MP4 hashes, and timestamps.

Android evidence: The Android logical extraction contained a Mavic Air-associated main flight record, a roughly four-minute 1280 x 720 cached H.264 video, a detailed DAT record, and six cached JPEG thumbnails. The cached video showed an outdoor rural scene with a white tent, vehicles, fields, hills, and people.

Android versus microSD: No direct file-level match was established between the Android cached video and raw/EWF DJI_0001.MP4. The Android video was an outdoor June artefact, whereas DJI_0001.MP4 was a one-minute 4K indoor recording dated 2 February 2018; their dates, duration, resolution, visible content, and hashes were different.

Record scope: The main flight record, cached video, and detailed DAT file were not treated as if they described identical start and end points. The DAT record covers the broader power-on to power-off period and therefore can extend beyond the narrower flight or video activity.

Confidence Assessment

Overall confidence: Moderate.

Confidence is high in evidence integrity because original files were preserved, verified working copies were used, and hashes were recorded before analysis. Confidence is also high that the raw microSD and EWF contain the same allocated DJI video material because all five MP4 hashes, the volume identity, GUID, and key timestamp evidence matched.

Confidence is moderate when reconstructing the exact June 19 sequence because the source ZIP and filename times do not carry a trustworthy timezone. The evidence supports DJI GO 4 activity on that date, but not every event can be placed precisely on a UTC timeline.

Confidence is low for identifying the drone owner or operator. People are visible in some recordings, but appearance in the footage does not prove who owned, launched, piloted, or controlled the aircraft.

Limitations

- Timestamp uncertainty: Several Android ZIP and FAT-derived timestamps did not contain a reliable embedded timezone. Converting them blindly to UTC could create a false chronology, so uncertain values were retained as source times.
- Logical extraction scope: The Android evidence was a logical acquisition rather than a complete physical image. Material outside the extracted application and storage areas, including potentially deleted or inaccessible data, may not have been available.
- Cached-picture limitations: The cached JPEG files were small thumbnails without useful original GPS or camera-date metadata. Their filename-derived dates fell mainly in April, so they could not reliably be tied to the June 19 flight.
- Operator identification: Videos showed people and locations, but the visual evidence did not establish which person, if any, controlled the drone. Presence near the aircraft or recording site cannot be treated as proof of operation.
- Different record scopes: The main flight record, cached video, and detailed DAT record do not necessarily begin and end at the same event. The DAT covers a wider power-on to power-off period and cannot automatically define the narrower flight or video interval.
- Raw/EWF whole-image difference: Although the individual MP4 hashes and media identifiers matched, the complete raw and EWF-exposed media SHA-256 values differed. The investigation established matching allocated evidence but did not determine the precise cause of differences elsewhere in the full media representation.
- Limited direct Android-to-microSD correlation: No SHA-256 match was established between the Android cached flight video and DJI_0001.MP4 from the microSD. The evidence supports their presence within the wider DJI case but not equality of those two files.

Conclusion

The investigation successfully preserved and examined the Android logical acquisition, raw internal microSD image, and EWF image using verified working copies and read-only or forensic-tool workflows. The Android extraction contained a DJI GO 4 main flight record associated with a Mavic Air, a cached flight video, a detailed DAT record, and cached picture artefacts. Each artefact was treated separately because the records have different evidential scopes.

The raw microSD examination identified an 8 GiB FAT32 filesystem containing five DJI MP4 recordings and five THM thumbnails. DJI_0001.MP4 was independently examined as a one-minute 3840 x 2160 H.264 recording with a verified written time of 02 February 2018 at 17:31:20 UTC, and its controlled preview showed an indoor drone-laboratory setting.

The EWF image passed its internal integrity verification and exposed an equivalent FAT32 media structure. The same five DJI MP4 filenames, matching SHA-256 values for all five videos, matching filesystem identity, and matching volume GUID strongly corroborate the relationship between the EWF and raw microSD evidence. Their complete media SHA-256 values were different, so the two full image representations were not described as byte-for-byte identical.

The Android evidence documents separate DJI GO 4 activity on 19 June 2018. Its cached video showed an outdoor rural environment, whereas the examined February microSD video showed an indoor laboratory. Their technical characteristics and timestamps differed, so no unsupported claim was made that they were the same recording or the same flight.

Overall, the evidence provides strong technical correlation between the raw and EWF microSD sources and moderate support for reconstructing the June 19 DJI GO 4 activity. It does not establish the identity of the drone operator, prove ownership, or allow every source timestamp to be placed precisely in UTC.

Screenshots

Fig 1.0 Forensic Investigation Directory Structure.
Fig 1.1 Android Logical Original File Integrity Verification.
Fig 1.2 Android Logical Working Copy Integrity Verification.
Fig 1.3 Android Logical Working Copy Extraction.
Fig 1.4 Android Logical Working Copy Extraction page 1.
Fig 1.5 Android Logical Working Copy Extraction page 2.
Fig 1.6 Android Logical Working Copy Extraction page 3.
Fig 1.7 Android Logical Working Copy Extraction page 4.
Fig 1.8 DJI GO 4 Application Directory Structure.
Fig 1.9 DJI GO 4 Application Directory Structure page 1.
Fig 2.0 DJI GO 4 Application Directory Structure page 2.
Fig 2.1 DJI GO 4 Main Flight Record Located.
Fig 2.2 DJI GO 4 Cached Flight Video Located.
Fig 2.3 DJI GO 4 Detailed DAT Flight Records Located.
Fig 2.4 DJI GO 4 Cached Picture Artefacts Located.
Fig 2.5 DJI GO 4 Main Flight Record Metadata and Integrity.
Fig 2.6 Main Flight Record Aircraft Identification.
Fig 2.7 Main Flight Record Aircraft Identification page 1.
Fig 2.8 Main Flight Record ZIP Timestamp Verification.
Fig 2.9 DJI GO 4 Cached Flight Video Metadata and Integrity.
Fig 3.0 Cached Flight Video MediaInfo Metadata.
Fig 3.1 Cached Flight Video MediaInfo Metadata page 1.
Fig 3.2 Cached Flight Video ExifTool Metadata.
Fig 3.3 Cached Flight Video ExifTool Metadata page 1.
Fig 3.4 Cached Flight Video ZIP Timestamp Verification.
Fig 3.5 Cached Flight Video ZIP Timestamp Verification page 1.
Fig 3.6 Cached Flight Video ZIP Timestamp Verification page 2.
Fig 3.7 Cached Flight Video ZIP Timestamp Verification page 3.
Fig 3.8 Cached Flight Video ZIP Timestamp Verification page 4.
Fig 3.9 DJI GO 4 Cached Flight Video Preview.
Fig 4.0 DJI GO 4 Cached Flight Video Preview page 1.
Fig 4.1 DJI GO 4 Cached Flight Video Preview page 2.
Fig 4.2 DJI GO 4 Cached Flight Video Preview page 3.
Fig 4.3 DJI GO 4 Cached Flight Video Preview page 4.
Fig 4.4 DJI GO 4 Detailed Flight Record Metadata and Integrity.
Fig 4.5 DJI GO 4 Detailed Flight Record Metadata and Integrity Page 1.
Fig 4.6 DJI GO 4 Cached Picture Metadata.
Fig 4.7 DJI GO 4 Cached Picture Metadata page 1.
Fig 4.8 DJI GO 4 Cached Picture Metadata page 2.
Fig 4.9 DJI GO 4 Cached Picture Metadata page 3.
Fig 5.0 DJI GO 4 Cached Picture Metadata page 4.
Fig 5.1 DJI GO 4 Cached Picture Filename Timestamp Conversion.
Fig 5.2 DJI GO 4 Cached Picture Visual Inventory.
Fig 5.3 Raw microSD and EWF Original MD5 and SHA-256 Verification.
Fig 5.4 Raw microSD and EWF Working Copy Integrity Verification.
Fig 5.5 Raw microSD Read-Only Loop Device Identification.

Fig 5.6 Raw microSD Read-Only Mount and Filesystem Structure.
Fig 5.7 Raw microSD Read-Only Mount and Filesystem Structure Page 1.
Fig 5.8 Raw microSD Video File Inventory and Timestamps.
Fig 5.9 Raw microSD Video SHA-256.
Fig 6.0 Raw microSD Thumbnail Evidence Inventory.
Fig 6.1 Raw microSD Volume GUID Identification.
Fig 6.2 Raw microSD DJI 0001 Video Metadata and UTC Time.
Fig 6.3 Raw microSD DJI 0001 Video Metadata and UTC Time Page 1.
Fig 6.4 Raw microSD DJI 0001 Video Metadata and UTC Time Page 2.
Fig 6.5 Raw microSD DJI 0001 Video Metadata and UTC Time Page 3.
Fig 6.6 Raw microSD DJI 0001 Video Metadata and UTC Time Page 4.
Fig 6.7 Raw microSD DJI 0001 Video Preview.
Fig 6.8 Raw microSD DJI 0001 Video Preview page 1.
Fig 6.9 Raw microSD DJI 0001 Video Preview page 2.
Fig 7.0 Raw microSD DJI 0001 Video Preview page 3.
Fig 7.1 EWF Image Format and Acquisition Information.
Fig 7.2 EWF Image Integrity Verification.
Fig 7.3 EWF Image Integrity Verification page 1.
Fig 7.4 EWF Exposed Media Image Identification.
Fig 7.5 EWF Exposed Media Size Serial and SHA-256.
Fig 7.6 EWF FAT32 Filesystem Information.
Fig 7.7 EWF FAT32 Filesystem Information page 1.
Fig 7.8 EWF FAT32 Filesystem Information page 2.
Fig 7.9 EWF FAT32 Filesystem Information.
Fig 8.0 EWF Media File Inventory with Inode References.
Fig 8.1 EWF DJI 0001 Video Timestamp Metadata.
Fig 8.2 EWF DJI 0001 Video UTC Timestamp.
Fig 8.3 EWF Video SHA-256 Hashes.
Fig 8.4 EWF Volume GUID Identification.

References

ICDFA Materials.